MA205

Calculus II

MA205 CALCULUS II

ABET eCOURSEBOOK AY2025-2026

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UNITED STATES MILITARY ACADEMY WEST POINT

TAB A AY2026 COURSE ASSESSMENT

DEPARTMENT OF THE ARMY
DEPARTMENT OF MATHEMATICAL SCIENCES
UNITED STATES MILITARY ACADEMY
601 THAYER ROAD
WEST POINT, NEW YORK 10996-1704

MADN-MATH 30 January 2026

MEMORANDUM THRU

Dr. Rachelle DeCoste, Professor and Director, Mathematical Sciences Program, Department of Mathematical Sciences, USMA, West Point, NY 10996

COL Michael Scioletti, Professor and Head, Department of Mathematical Sciences, USMA, West Point, NY 10996

FOR Office of the Dean, AARS, United States Military Academy, West Point, NY 10996

SUBJECT: MA205 Multivariable Calculus Executive Summary (AY26-1)

1. **Purpose.** The purpose of this memorandum is to assess the MA205 Multivariable Calculus curriculum for Academic Year 26-1 (AY26-1).

2. **Assessment Summary.** In accordance with the Department's three-year cyclical assessment schedule (Enclosure 1) and the AY26-1 course assessment plan, this year's assessment effort focused on two course Student Learning Outcomes (SLOs): SLO 3 (Construct and Evaluate Multiple Integrals) and SLO 5 (Use Technology to Visualize and Analyze). The evaluations are summarized in Table 1.

Course SLO	Strength of Attainment	Strength of Evidence	Overall Evaluation
SLO 3: Construct and Evaluate Multiple Integrals	D	B	C
SLO 5: Use Technology to Visualize and Analyze	C	C	C

Figure 1: SLO Evaluations. Key: A = exceeds the standard; B = meets the standard; C = meets the standard with some weakness; D = fails the standard (can be addressed in the short-term); F = catastrophically fails the standard (requires immediate remediation).

Note: SLO 5 is supported by a single direct measure (the TEE visualization item, Problem 6a), which limits the strength of evidence. Establishing additional direct technology measures (e.g., a scored Mathematica lab rubric) is recommended for future cycles (see Enclosures 2 and 4).

3. **Assessment Discussion.** This section presents an overview of assessment findings for AY26-1. A detailed discussion of direct and indirect indicators can be found in the Course Design and Analysis Report (CDAR) at Enclosure 4.

a. Status of significant changes. The course maintained the 55-minute format (56 lessons) implemented in AY24-2 after transitioning from 40 lessons with 75-minute class periods; this structure continued to provide adequate instructional time without negatively impacting student performance. For AY26-1, the assessment effort focused on SLO 3 and SLO 5 in accordance with the cyclical schedule and the course assessment plan. Consistent with that plan, high-difficulty integral problems—particularly iterated integrals on the third Written Partial Review (WPR 3)—were modified to assess whether cadets can correctly establish limits and coordinate transformations (integral setup) rather than simply compute a final numerical result, directly targeting SLO 3. In addition, long-term Mathematica projects were eliminated to recover in-class instructional time, shifting the assessment of technology and visualization (SLO 5) toward in-class technology labs.

b. Strengths in the course. The course maintains a robust structure with daily WebAssign homework, quizzes, and WPRs reinforcing the learning objectives for each block. The four-block structure (Vectors and Geometry of Space, Multivariable Differentiation, Multivariable Integration, and Vector Calculus) provides logical progression from foundational concepts to advanced applications. Within SLO 3, cadets demonstrated strong proficiency constructing and evaluating double integrals: the double-integral mechanism (TEE Problem 6b) was met by 94.2% of cadets, well above the 73% criterion. WPR performance was consistent throughout the semester: WPR 1 (85.85%), WPR 2 (86.49%), WPR 3 (78.95%), and WPR 4 (85.08%), with a TEE average of 78.20% and a final course average of 84.66%. The course continues to serve as a prerequisite for 38 follow-on courses across seven departments.

c. Areas for improvement in the course. For SLO 3, two of the three direct mechanisms fell below the 73% success criterion: the surface-area mechanism (TEE Problem 3) was met by only 57.2% of cadets, and the surface-integral mechanism (TEE Problem 8) by 67.0%. Cadets struggle to set up surface-area and flux integrals in non-rectangular coordinate systems and to evaluate parametric surface integrals. For SLO 5, the single direct visualization measure (TEE Problem 6a) was met by 71.0% of cadets—just below threshold—and the evidence base is thin. Student feedback indicates that the Mathematica learning curve and the time-constrained environment limit the software's effectiveness as a learning tool.

d. Changes to be implemented in AY27. Based on assessment findings, we recommend: (1) reinforcing surface-area, flux, and parametric-surface-integral setup for SLO 3 through additional formative assessment across coordinate systems earlier in the integration block; (2) establishing a scored technology/visualization rubric (e.g., a Mathematica lab) to provide robust direct evidence for SLO 5, which currently rests on a single TEE measure; (3) expanding the use of visualization software (Mathematica and GeoGebra) in daily instruction rather than relegating technology to standalone lab days; (4) continuing to refine setup-focused conceptual assessment of multiple integrals; and (5) developing the TEE prior to drafting the WPRs to align SLO 3 and SLO 5 indicators with assessment questions.

e. Proposed changes to the SLOs. No changes to the Student Learning Outcomes are proposed at this time. The current six SLOs adequately capture the course objectives, and the assessment mechanisms effectively measure student attainment. The full list of current SLOs can be found in Enclosure 1.

4. The point of contact for this memorandum is MAJ Javier Sustaita, MA205 Course Director, at (845) 938-8137 or javier.sustaita@westpoint.edu.

JAMES C. GRYMES
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4 Encls
1. Full Set of Student Learning Outcomes (SLOs)
2. Assessment Data Summary
3. Grading Standards
4. Course Design and Analysis Report (CDAR)

Enclosure 1: Full Set of Student Learning Outcomes (SLOs)

MA205 maintains six Student Learning Outcomes. At the end of the course, cadets will be able to:

SLO 1. Use vectors, vector operators, and vector functions to model motion and represent surfaces in three-dimensional space.

SLO 2. Calculate partial derivatives and apply them to solve problems associated with instantaneous rates of change, extreme values, and constrained optimization.

SLO 3. Construct and evaluate multiple integrals in appropriate coordinate systems such as rectangular, polar, cylindrical, and spherical coordinates and apply them to solve problems involving volume, surface area, flux, density, and/or center of mass.

SLO 4. Identify and use key theorems from vector calculus, such as the Fundamental Theorem for Line Integrals, Green's Theorem, Stokes' Theorem, and/or the Divergence Theorem, as appropriate to solve applied problems.

SLO 5. Using technology, visualize and analyze functions and models and solve problems in multiple dimensions. This demonstrates a clear understanding of multivariable and vector calculus concepts, including partial derivatives, gradients, surfaces, integrals, and vector fields.

SLO 6. Improve the quality of both their written and verbal communication of mathematics.

AY26-1 cyclical assessment focus: SLO 3 and SLO 5.

Enclosure 2: Assessment Data Summary

Written Partial Review (WPR) Performance.

Assessment	Average	Median	Std Dev	Range
WPR 1	85.85%	88.00%	10.19	47.3% – 100.0%
WPR 2	86.49%	88.16%	8.00	53.5% – 99.2%
WPR 3	78.95%	79.75%	12.43	40.5% – 99.8%
WPR 4	85.08%	87.65%	11.71	46.7% – 100.0%

Term End Exam (TEE) Performance.

Assessment	Average	Median	Std Dev	Range
TEE	78.20%	79.39%	10.39	39.1% – 97.8%

Final Course Performance.

Metric	Value
Final Course Average	84.66%
Final Course Median	84.78%

SLO 3 Direct Assessment Mechanisms (TEE).

Mechanism	Assessment Item	% Achieving C	Meets Criterion?
Surface-area integral	TEE Problem 3	57.2%	No
Double integral	TEE Problem 6b	94.2%	Yes
Surface integral	TEE Problem 8	67.0%	No

SLO 3 result: one of three direct mechanisms met the 73% success criterion (double integral, 94.2%) — Overall Evaluation C (meets the standard with some weakness).

SLO 5 Direct Assessment Mechanisms.

Mechanism	Assessment Item	% Achieving C	Meets Criterion?
Visualization (sketch surface, draw/label)	TEE Problem 6a	71.0%	No

SLO 5 result: the single direct measure (TEE visualization item) was met by 71.0% of cadets, just below the 73% criterion; evidence is limited to one mechanism — Overall Evaluation C (meets the standard with some weakness).

Instructor Section Performance (Section Averages).

Instructor	WPR 1	WPR 2	WPR 3	WPR 4	TEE	n
BLUMAN	87.71%	89.18%	84.10%	87.30%	82.55%	16
EVANS	86.21%	86.66%	79.49%	86.53%	79.60%	50
FISHER	87.21%	88.75%	79.26%	88.23%	80.92%	52
FREENY	84.38%	84.27%	77.92%	83.08%	77.10%	55

GRYMES	85.10%	87.47%	73.50%	84.65%	77.81%	18
SUSTAITA	86.75%	84.07%	80.31%	83.69%	76.78%	35
WHITAKER	84.63%	86.73%	78.27%	83.08%	74.93%	50

Final Course Grade Distribution (n = 276).

Letter Grade	Count	Percentage
A+	5	1.8%
A	18	6.5%
A-	35	12.7%
B+	25	9.1%
B	49	17.8%
B-	50	18.1%
C+	34	12.3%
C	38	13.8%
C-	13	4.7%
D	6	2.2%
F	3	1.1%

Summary by band: A range 58 (21.0%); B range 124 (44.9%); C range 85 (30.8%); D 6 (2.2%); F 3 (1.1%).

Enclosure 3: Grading Standards

MA205 uses a holistic grading scheme to tie grades to qualitative performance feedback. Performance standards are criterion-referenced and defined independently of peer comparison.

Letter Grade	Numerical Grade (%)	Description
A+	97 – 100	Beyond Expectations
A	93 – 96.99	Dominates the material
A-	90 – 92.99	Mastery
B+	87 – 89.99	Excellent performance
B	83 – 86.99	Good understanding
B-	80 – 82.99	Proficient; aptitude for the subject
C+	77 – 79.99	Can build on foundations
C	73 – 76.99	Passing; Proficient
C-	70 – 72.99	Short-range understanding
D	65 – 69.99	Marginal performance / some learning
F	< 65	Failed to demonstrate understanding

Course Evaluation Plan (AY26-1).

Graded Event	Points
WPR 1	75
WPR 2	100
WPR 3	100
WPR 4	100
WebAssign	100
Labs	90
Application Paper/Video	20
Quizzes	75
TEE	300
Instructor Points	40
Total	1000

SLO Success Criteria.

- Individual Assessment Level: success requires at least 70% of cadets achieving C- or greater on each WPR question and at least 73% of cadets achieving C or greater on each TEE question.
- Course Level: an SLO meets the standard when at least two-thirds of aligned assessment mechanisms achieve their respective success criteria.

Enclosure 4: Course Design and Analysis Report (CDAR)

The complete Course Design and Analysis Report (CDAR) is provided as a separate attachment. Key findings include:

- The 55-minute class format (56 lessons) continues to provide adequate instructional time for course material coverage without negatively impacting student performance.
- AY26-1 assessed two outcomes under the Department's cyclical schedule: SLO 3 (Construct and Evaluate Multiple Integrals) and SLO 5 (Use Technology to Visualize and Analyze).
- SLO 3 (Construct and Evaluate Multiple Integrals) met the standard with some weakness (Overall Evaluation C): of three direct TEE mechanisms, only the double integral (Problem 6b, 94.2%) met the 73% criterion, while surface area (Problem 3, 57.2%) and surface integral (Problem 8, 67.0%) fell below it.
- SLO 5 (Use Technology to Visualize and Analyze) met the standard with some weakness (Overall Evaluation C): the single direct measure—the TEE visualization item (Problem 6a)—was met by 71.0% of cadets, just below the 73% criterion. Evidence is limited to one mechanism; a scored technology-lab rubric is recommended to strengthen future SLO 5 evidence.
- Assessment modifications implemented in AY26-1 emphasized integral setup over full evaluation for high-difficulty problems and continued the balance of conceptual versus computational questions.
- Technology integration remains a challenge for SLO 5; students report that the Mathematica learning curve and the time-constrained environment limit its effectiveness as a learning tool.
- As of July 2023, 38 courses list MA205 as either a prerequisite or co-requisite across seven departments: Electrical Engineering and Computer Science (4), Civil and Mechanical Engineering (7), Mathematical Sciences (19), Physics and Nuclear Engineering (3), Chemistry and Life Science (3), Social Sciences (1), and Geography and Environmental Engineering (1).

MA205

AY25-2 Syllabus

Printed on: 5/22/2025

Textbook: Calculus - Ninth Edition, Early Transcendentals by James Stewart				
Lesson	***Date***	***Topic (Reading)***	***Homework (WebAssign)***	***Lesson Objectives***
BLOCK 1: VECTORS AND GEOMETRY OF SPACE				
Lesson 1	8/18/2025	Course Intro & 3D Coord Systems (12.1)	**12.1:** 7, 10, 37, 39	**1.1.** Understand how to model position and distance in 3D-space.
Lesson 2	8/19/2025	Geometry of Vectors (12.2)	**12.2:** 13, 21, 25	**1.2.** Understand the properties of vectors: addition, scalar multiplication, length, magnitude, standard basis vectors, and position vectors.
Lesson 3	8/20/2025	Dot Product (12.3)	**12.3:** 8, 9, 12, 19, 41, 51	**1.3.** Solve for and understand the definition and utility of the dot and cross products for practical applications between two vectors.
Lesson 4	8/21/2025	Cross Product (12.4)	**12.4:** 1, 39, 41	**1.4.** Solve for and understand the definition and utility of the dot and cross products for practical applications between two vectors.
Lesson 5	8/25/2025	Equations of Lines & Planes (12.5)	**12.5:** 5, 27, 31, 71 **12.3:** 14	**1.5.** Model and understand relationships between lines and planes in 3D-space. **1.6.** Determine the distance between different objects/points in 3D-space.
Lesson 6	8/26/2025	Vector Functions & Space Curves (13.1)	**13.1:** 1, 22, 49, 25, 27, 29, 30 **Quiz 1**	**1.7.** Model space curves in 3D using vector functions.
Lesson 7	8/28/2025	Derivatives & Integrals of Vector Functions (13.2)	**13.2:** 10, 21, 37, 43, 44, 49	**1.8.** Compute derivatives and integrals of vector functions. **1.9.** Compute the tangent line to a space curve and understand its significance.
Lesson 8	9/2/2025	Arc Length (13.3 - stop at Curvature p. 906	**13.3:** 4, 17	**1.10.** Determine distance along a 3D-space curve.
Lesson 9	9/3/2025	Motion in Space 1 (13.4 - stop at "Tangential and Normal" p. 919)	**13.4:** 3, 11 **Quiz 2**	**1.11.** Model the motion of an object in 3D-space.
Lesson 10	9/4/2025	Motion in Space 2 (13.4 - stop at "Tangential and Normal" p. 919)	**13.4:** 18, 23, 25	
Lesson 11	9/8/2025	**Mastery Day**		
Lesson 12	9/9/2025	**WPR 1**		
BLOCK 2: MULTIVARIABLE DIFFERENTIATION				
Lesson 13	9/11/2025	Multivariable Functions (Cartesian) (10.3 - stop at Example 7; 14.1)	**14.1:** 5, 21, 32, 34	**2.1.** Understand the definition of a function of two variables in cartesian and polar coordinates and how it can be represented algebraically, numerically, graphically, or verbally.
Lesson 14	9/15/2025	Partial Derivatives (14.3 - stop at Partial Differential Equations p. 968)	**14.3:** 20, 22, 47, 82	**2.2.** Given a function of several variables, calculate its partial derivatives. **2.3.** Understand the relationship and differences between average rate of change and instantaneous rate of change of a multivariable function using partial derivatives.
Lesson 15	9/16/2025	Tangent Planes (14.4 - stop at Differentials p. 979)	**14.4:** 3, 6, 25, 27	**2.4.** Given a function of two variables and a point, find the equation of the plane tangent to the surface at the given point. **2.5.** Approximate the value of a multivariable function using a linear equation.
Lesson 16	9/17/2025	Chain Rule (14.5 - stop at implicit differentiation p. 990)	**14.5:** 5, 7, 16, 44 **Quiz 3**	**2.6.** Given a multivariable function and parameterizations for the function, use the chain rule to solve a rates of change problem.
Lesson 17	9/18/2025	Gradient and Directional Derivative I		**2.7.** Understand what a directional derivative is in terms of a rate of change graphically, algebraically, and numerically. **2.8.** Given a function f(x,y) of two variables, find the directional derivative of f(x,y) at a given point in any direction. **2.9.** Given a function of two variables, find the gradient vector at a specified point. **2.13.** Employ technology to calculate, visualize, and understand the directional derivative and gradient of a multivariable function.
Lesson 18	9/22/2025	Gradient and Directional Derivative II	**14.6:** 1, 9, 13, 17, 30, 39	**2.10.** Understand that the gradient vector gives the direction of the greatest increase in functional value at a given point for a differentiable function. **2.11.** Understand that the magnitude of the gradient vector gives the maximum rate of change of a differentiable function at a given point. **2.12.** Understand geometrically how the gradient vector is related to level curves. **2.13.** Employ technology to calculate, visualize, and understand the directional derivative and gradient of a multivariable function.
Lesson 19	9/23/2025	Multivariable Optimization I (Min/Max) (14.7)	**14.7:** 3, 13*, 21*, 25* *Technology Problems	**2.14.** Employ technology to calculate, visualize, and understand optimization problems. **2.15.** Understand the definition of a critical point and be able to find the critical points of a function of two variables. **2.16.** Use the Second Derivatives Test to classify a critical point as a local maximum, local minimum, or saddle point
Lesson 20	9/24/2025	Multivariable Optimization II (Closed Interval Method) (14.7)	**14.7:** 38, 46, 47, 51, 54, 55 **Quiz 4**	**2.17.** Find the absolute maximum and minimum of a given continuous function f(x,y) on a closed and bounded set using the Closed Interval Method for functions of two variables. **2.18.** Model and solve unconstrained and constrained multivariable optimization problems.
Lesson 21	9/25/2025	Lagrange Multipliers I (Single-Constraint) (14.8)		**2.19.** Solve optimization problems for functions of several variables subject to a single constraint using the Method of Lagrange Multipliers.
Lesson 22	9/29/2025	Lagrange Multipliers II (Multi-Constraint) (14.8)	**14.8:** 1, 5, 9, 33, 49	**2.20.** Solve optimization problems for functions of several variables subject to two constraints using the Method of Lagrange Multipliers.
Lesson 23	10/1/2025	**Mastery Day**		
Lesson 24	10/2/2025	**WPR 2**		

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AY26-1

Lesson	Date	Topic (Reading)	Homework (WebAssign)	Lesson Objectives
BLOCK 3: MULTIVARIABLE INTEGRATION				
Lesson 25	10/6/2025	Double Integration over Rectangular Surfaces I (15.1)		**3.1.** Given a function f(x,y), use a double Riemann sum to estimate volume under a surface. **3.2.** Understand the concept of volume under a surface as the limit of the sums of volumes of rectangular columns.
Lesson 26	10/7/2025	Double Integration over Rectangular Surfaces II (15.1)	**15.1:** 1, 6, 7, 15, 19, 53, 42	**3.3.** Use level curves to estimate the volume under a surface. **3.4.** Estimate the average value of a function of two variables.
Lesson 27	10/8/2025	Double Integration over General Regions (15.2)	**15.2:** 5, 7, 8, 20, 39	**3.5.** For any region in the xy-plane, sketch the region, label the boundaries, and label the points of intersection. **3.6.** Compute the iterated integral of a multivariable function over a specified domain.
Lesson 28	10/9/2025	Multivariable Functions (Polar) (10.3 - stop at Example 7)	**10.3:** 12, 21, 24	**2.1.** Understand the definition of a function of two variables in cartesian and polar coordinates and how it can be represented algebraically, numerically, graphically, or verbally.
Lesson 29	10/14/2025	Double Integration in Polar Coordinates (15.3)	**15.3:** 6, 11, 30, 33, 45	**3.7.** Understand and use the polar coordinate system to solve iterated integrals.
Lesson 30	10/15/2025	Applications of Integrals (15.4)	**15.4:** 5, 11,13, 30, 33, 34	**3.8.**Use integration techniques in application settings of two independent variables.
Lesson 31	10/16/2025	Cylinders and Quadric Surfaces (12.6)	**12.6:** 23-30 **Quiz 5**	**1.6.** Model quadric surfaces in 3D-space. **1.12.** Employ technology to visualize quadric surfaces.
Lesson 32	10/20/2025	Triple Integrals I (15.6)		**3.9.** Evaluate a triple integral over a general region.
Lesson 33	10/21/2025	Triple Integrals II (15.6)	**15.6:** 4, 5, 9, 13, 18	**3.10.** Employ technology to compute triple integrals.
Lesson 34	10/23/2025	Triple Integrals in Cylindrical Coordinates I (15.7)		**3.11.** Understand when and how to apply the cylindrical coordinate system to solve iterated integrals.
Lesson 35	10/27/2025	Triple Integrals in Cylindrical Coordinates II (15.7)	**15.7:** 10, 15, 21, 32	
Lesson 36	10/28/2025	Triple Integrals in Spherical Coordinates I (15.8)	**Quiz 6**	**3.12.** Understand when and how to apply the spherical coordinate system to solve iterated integrals.
Lesson 37	10/30/2025	Triple Integrals in Spherical Coordinates II (15.8)	**15.8:** 21, 23, 28, 29	
Lesson 38	11/3/2025	**Mastery Day**		
Lesson 39	11/4/2025	**WPR 3**		
BLOCK 4: VECTOR CALCULUS				
Lesson 40	11/6/2025	Vector Fields (16.1)	**16.1:** 7*, 19, 11*, 27, 13, 15, 16, 31, 33 *Technology Problems	**4.1.** Understand vector fields and their different applications. **4.14.** Use technology to plot different types of vector fields.
Lesson 41	11/10/2025	Line Integrals I (16.2)		**4.2.** Understand the meaning and compute the line integral along a curve in space.
Lesson 42	11/12/2025	Line Integrals II (16.2)	**16.2:** 3, 7, 17, 22, 41	**4.3.** Understand different applications of line integrals. **4.15.** Employ technology to pass a vector defined function to a function. **4.16.** Employ technology to visualize a particle's path along a vector field.
Lesson 43	11/13/2025	Fundamental Theorem for Line Integrals (16.3 - stop at Conservation of Energy p. 1150)	**16.3:** 1, 3, 4, 7, 18, 23 **Quiz 7**	**4.4.** Understand and apply the Fundamental Theorem for line integrals.
Lesson 44	11/17/2025	Curl and Divergence (16.5 - stop at Vector Forms of Green's Theorem p. 1167)	**16.5:** 1, 3, 7*, 11, 18 *Technology Problem	**4.6.** Understand the "del" operator and use it to compute curl and divergence. **4.7.** Understand the physical representation of curl and divergence. **4.17.** Employ technology to calculate curl and divergence of a vector defined function.
Lesson 45	11/18/2025	Green's Theorem (16.4)	**16.4:** 3, 5, 7, 9, 17	**4.5.** Understand and apply Green's Theorem to general 2D regions.
Lesson 46	11/19/2025	Parametric Surfaces (16.6 - stop at Surface Area p. 1177)	**Quiz 8**	**4.8.** Understand how vector functions can be used to model surfaces in 3D-space.
Lesson 47	11/20/2025	Parametric Surface Area (15.5, 16.6 - start at Surface Area p. 1177)	**16.6:** 40, 39, 45 **15.5:** 4	**4.9.** Determine the area of a parametric surface.
Lesson 48	11/24/2025	**Mastery Day**		
Lesson 49	11/25/2025	**WPR 4**		
Lesson 50	12/2/2025	Surface Integrals (16.7)	**16.7:** 23, 24, 26	**4.10.** Understand the meaning and compute surface integrals of a scalar function over a given surface.
Lesson 51	12/3/2025	Stokes' Theorem (16.8)		**4.12.** Understand and apply Stokes' Theorem to general surfaces.
Lesson 52	12/4/2025	The Divergence Theorem II (16.9)	**16.9:** 1, 3, 5, 7	**4.13.** Understand and apply the Divergence Theorem to general solids.
Lesson 53	12/8/2025	**Problem Solving Lab**		
Lesson 54	12/9/2025	**Problem Solving Lab**		
Lesson 55	12/10/2025	**Project Briefs 10-12 December**		
Lesson 56	12/15/2025	**TEE Review**		

TAB B AY2025 COURSE ASSESSMENT

DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

MADN-MATH 5 June 2025

MEMORANDUM THRU

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COL Michael Scioletti, Professor and Head, Department of Mathematical Sciences, USMA, West Point, NY 10996

FOR Office of the Dean, AARS, USMA, West Point, NY 10996

SUBJECT: MA205 Calculus II Executive Summary (AY25)

1. **Purpose:** The purpose of this memorandum is to assess the MA205 curriculum.

2. **Assessment Summary:** Assessments for course student learning outcomes (SLOs) – see Enclosure 1 – and contributions to the assessment of the math sciences program SLOs. Enclosure 4 contains more detailed discussions.

Course SLOs	Strength of Attainment	Strength of Evidence	Overall Evaluation
SLO 6: Apply Multivariate Derivative	B	B	B
SLO 8: Apply Vector Operations and Model Motion of an Object	D	B	C
SLO 9: Demonstrate Calculus of Vector Fields/Line and Surface Integrals	A	A	A

Table 1: *MA205 SLO Evaluations. Key: A = exceeds the standard; B = meets the standard; C = meets the standard with some weakness; D = fails the standard (can be addressed in the short-term); F = catastrophically fails the standard (requires immediate remediation)*

Program SLOs	Strength of Attainment	Strength of Evidence	Overall Evaluation
SLO 1a	B/D/– $\rightarrow$ C	B/B/– $\rightarrow$ B	C
SLO 1b	B/–/A $\rightarrow$ B	B/–/A $\rightarrow$ B	B
SLO 1c	B/D/A $\rightarrow$ B	B/B/A $\rightarrow$ B	B
SLO 3a	B/–/A $\rightarrow$ B	B/–/A $\rightarrow$ B	B
SLO 3d	–/D/– $\rightarrow$ D	–/B/– $\rightarrow$ B	C
SLO 7	B/D/A $\rightarrow$ B	B/B/A $\rightarrow$ B	B

Table 2: *Contributions to the Math Sciences Program SLO Assessment. We aggregate the attainment and evidence evaluations from the course SLOs to obtain a measure for the program SLO attainment and evidence scores. These two scores then provide the overall evaluation. The letter keys are the same as Table 1.*

3. **Implementation Summary:** We implemented the new/adjusted mechanisms for course SLOs 1, 4, and 7 in accordance with Enclosure 3. These implementations are not grounded in historical assessment data. See Appendix A in Enclosure 4 for more information on the individual mechanisms.

a. SLO 1. WebAssign submission states, technology lab preparation assignments, and major applications report and video assignments.

(1) We have initial concerns regarding WebAssign submission states based on observations of Cadet behavior. We intended Cadets to self-pace and develop demonstrate good scholarly habits, however, we found many waited until the assignment due-date (typically a weekend) to begin work. They were unable to utilize additional instruction or class time to help resolve issues.

b. SLO 4. Follow-on course performance.

c. SLO 7. Single question on WPR 2 and five questions on the TEE.

4. **Plan Summary.** This year we planned the assessment of SLOs 2, 3, and 5 for next year. See Appendix A in Enclosure 4 for more information on the individual mechanisms.

a. SLO 2. Grade sequence of communication-focused assignments throughout the semester.

b. SLO 3. Technology lab in-class assignments.

c. SLO 5. Technology lab in-class assignments and major applications report and video assignments.

5. **Course Observations:** This section presents an overview of findings for AY25. Enclosure 4 contains more detailed discussions.

a. Course Changes and SLO Impacts.

(1) *75- to 55-minute Redesign.* The course redesign required various combinations of lesson mapping from the 40-lesson to the 56-lesson semester. This results in a different academic experience for the Cadets and a potential negative impact on SLOs. To mitigate this effect, we compared topic exposure hours to historical calendars and accepted reduced exposure for lessons in which Cadets demonstrated at-or-above acceptable performance as defined in Enclosure 4. This also permitted us to increase exposure to lessons associated with historical underperformance.

(2) *Formative Assessments for Final Three Topics.* Providing a formative assessment for each of the final three topics may detract from developing understanding for Cadets with a traditional lesson, thus negatively impacting SLO 9. However, a well-designed formative assessment supplements a Cadet's knowledge by facilitating a reasonable cognitive challenge and providing immediate feedback to certify understanding.

(3) *Initial Schedule Compression.* Compression of the first two blocks potentially negatively affects SLOs 6 and 8 due to Cadets not having as much processing time for the material outside of class. The compression may also preclude development and implementation of good scholarly habits (SLO 1); conversely, Cadets who do not drown in the course material possess average to above-average scholarly habits.

b. Course Strengths: the course's performance data, coupled with targeted future improvements to assessment timing and schedule structure, reflect a strong, adaptive learning environment that continues to uphold academic rigor and cadet success despite structural changes.

(1) MA205 demonstrated strong student performance in AY25, with Cadets achieving success in eight out of twelve total assessment mechanisms across the course. The shortfall in performance was isolated to four mechanisms, all on the TEE, suggesting that the length of time between instruction and final assessment—particularly for SLOs 6 and 8—may have impacted retention. Notably, SLO 6 showed stronger attainment than SLO 8, likely due to more balanced and immediate reinforcement through earlier WPRs, underscoring the value of timely assessment.

(2) Compared to previous academic years, MA205 maintained consistent performance, with only three of twenty-four inter-year question comparisons showing statistically significant lower results in AY25. These differences occurred in early blocks affected by a new 55-minute class format and compressed schedule, but later blocks showed no such decline, affirming the course's overall instructional effectiveness under the revised modality. The resilience of SLO 9, with no adverse performance signals, further supports this conclusion.

c. Course Improvements.

(1) Material presented earlier in the semester and not explicitly assessed again until the TEE challenges strength of evidence, particularly with the level of difficulty of the latter content. We should determine how to reduce the callback timespan for this earlier content. Some options are to include this content in focused questions on later assessments or to include the content as part of a larger problem to be developed and solved in formative/summative assessments.

(2) The course did not have a unifying theory on how to deploy and engage with technological visualizations to support learning the most complicated concepts in three dimensions. We are working to develop a more comprehensive approach to equip our instructors with the tools necessary to teach good 3D visualization processes. This aligns with the planning of SLO 3.

6. The point of contact for this memorandum is MAJ James Sherrell, MA205 Course Director, at james.sherrell@westpoint.edu or james.q.sherrell.mil@army.mil.

Encls
1. SLOs
2. Mapping SLOs to APGs
3. SLO Assessment Cycle
4. CDAR

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Enclosure 1: Student Learning Outcomes

At the end of the course, cadets will be able to...

SLO 1: Demonstrate Good Scholarly Habits. Students demonstrate good scholarly habits for progressive intellectual development.

SLO 2: Communicate Effectively. Students improve the quality of both their written and verbal communication of mathematics.

SLO 3: Apply Technology. Students are able to leverage technology when solving complex mathematical problems.

SLO 4: Develop Problem Solving and Reasoning Skills. Students develop the problem-solving and reasoning skills necessary for continued study in mathematics, the physical sciences, the social sciences, and engineering.

SLO 5: Develop Interdisciplinary Perspective. Students develop an interdisciplinary perspective that supports knowledge transfer across disciplinary boundaries to solve complex problems.

SLO 6: Apply Multivariate Derivative. Students apply the basic concepts of the multivariate derivative to solve multivariable unconstrained and constrained optimization problems.

SLO 7: Apply Multivariate Integral. Students apply multivariate integration to solve accumulation problems.

SLO 8: Apply Vector Operations and Model Motion of an Object. Students apply the basic properties and operations of vectors, quadric surfaces and space curves in three-dimensions, and model the motion of an object in three-dimensional space.

SLO 9: Demonstrate Calculus of Vector Fields/Line and Surface Integrals. Students demonstrate the calculus of vector fields, line and surface integral techniques, and the connections between these new types of integrals and the single, double, and triple integrals that are given by the higher dimensional versions of the Fundamental Theorem of Calculus.

Our SLOs are under review and will change to match those of MA204 to align the two standard track multivariable calculus courses.

Enclosure 2: Mapping SLOs to APGs

Academic Program Goals / M255 Assessment Techniques				
APG	1.) Communication	2.) Creative Thinking and Creativity	3.) Lifelong Learning	5.) STEM
WGCD	1.3.) Effectively convey meaningful information to diverse audience using appropriate forms and media.	2.4.) Reason both quanitatively and qualitatively.	3.1.) Demonstrate the willingness and ability to learn independently.	5.1.) Apply mathematics, science, and computing to model devices systems, processes, or behaviors
Core Math Goal	Communication	Technology Skills and Problem Solving	Habits of Mind	Interdisciplinary Problem Solving
MA205 CLO	2.) Communicate Effectively	3.) Apply Technology; 4.) Develop Problem Solving and Reasoning Skills 6.) Apply Multivariate Derivative; 7.) Apply Multivariate Integral; 9.) Demonstrate Calculus of Vector Fields/Line and Surface Integrals	1.) Demonstrate Good Scholarly Habits	5.) Develop Interdisciplinary Perspective 8.) Apply Vector Operations and Model Motion of an Object
Assessment	3x Projects (oral presentations)	4x WPRs and 1x TEE with leson objective mapping tied to CLOs; 3x Projects with Mathematica Requirements to assess students ability to leverage technology.	Nightly webassign homework; 8x quizes designed to test class preparation	3x Projects (interdeciplinary projects focused on applying calculus tecniquies to complex scenarios)

Figure 1: *Assessment Crosswalk*

Enclosure 3: SLO Assessment Cycle

WGCD	SLO	AY25	AY26	AY27
3.1	1	Implement	Plan	Assess
1.3	2	Plan	Assess	Implement
2.4, 5.1	3	Plan	Assess	Implement
3.1, 5.1	4	Implement	Plan	Assess
5.1	5	Plan	Assess	Implement
2.4, 4.4*	6	Assess	Implement	Plan
2.4	7	Implement	Plan	Assess
2.4	8	Assess	Implement	Plan
2.4	9	Assess	Implement	Plan

Figure 2: *Three-year cyclic assessment schedule for SLOs. This plan will be updated as the SLOs are modified.*

Enclosure 4: Course Assessment Plan and Overall Student Performance

Course Summary. There are a total of **56** required attendances consisting of 49 regular lessons, 3 technology labs, and 4 exams.

Course Evaluation Plan. See Table 3 for the distribution of course points.

Graded Event	Points
WPR 1	75
WPR 2	100
WPR 3	100
WPR 4	100
WebAssign	100
Labs	90
Application Paper/Video	20
Quizzes	75
TEE	300
Instructor Points	40
Total	1000

Table 3: *Point Distribution*

Grading. Table 4 summarizes student performance in MA205 AY25-1.

Letter Grade	Numerical Grade	% Students	Description
A+	97-100	6.3	Beyond Expectations
A	93-96.99	14.1	Dominates the material
A-	90-92.99	14.4	Mastery
B+	87-89.99	11.1	Excellent performance
B	83-86.99	18.0	Good understanding
B-	80-82.99	11.4	Proficient; aptitude for the subject
C+	77-79.99	13.5	Can build on foundations
C	73-76.99	6.0	Passing; Proficient
C-	70-72.99	3.9	Short-range understanding
D	65-69.99	1.2	Marginal performance/some learning
F	< 65	0	Failed to demonstrate understanding

Table 4: *Guidelines for Assigning Grades*

Enclosure 4: Course Design and Analysis Report.

Continued on the next page.

MA205 AY25 Course Design and Analysis Report

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1 Introduction

The Department of Mathematical Sciences at the United States Military Academy (USMA) teaches multi-variable and vector calculus (MA205) course to all cadets that major in science, technology, engineering, and mathematics (STEM). As discussed in Section 1.2, MA205 more closely aligns with a third iteration of calculus (e.g., Calculus III) at a traditional 4-year institution. Per USMA's academic catalog, the course seeks to *"provides a foundation for the continued study of mathematics and for the subsequent study of the physical sciences, social sciences, and engineering."* (Office of the Dean, 2023).

We present our annual design and analysis report over the next five sections. First, we present the historical context of MA205, focusing primarily on its evolution over the last decade and plans for the continued changes to the course; then conduct a review of literature and comparable institutions' calculus curricula. The second section summarizes the course's design and discusses the deliberate changes this AY. In section three, we identify the means to assess the course. Finally, sections four and five present the results of our assessments, analysis, and concluding thoughts.

1.1 Background

Calculus courses permeate U.S. higher-education. The D/F/W rates – ratios of students obtaining either D or F letter grades or withdrawing from the course – associated with calculus are near 30% (Schumacher and Siegel, 2015). West Point introduced MA205, in the 1950s, as a course for engineers. It has since been adopted by all STEM majors as a pre-requisite. It sits in an odd space on the spectrum of Engineer to Mathematics. Mathematics is about discovery, while engineering is about applications. This dichotomy is open for interpretation, but the way in which Calculus is delivered to students varies from theoretical to computational. The "chicken or the egg?" question is often compared to Calculus in terms of what do students need to know first: the theory or how to do the calculations? This makes teaching it very difficult. Instructors within the same department may hold opposing perspectives.

Over the previous seven years, MA205 undertook several changes, initiated by both program leadership and institutional demand. They include removal from the core curriculum, redesigning MA205 from

55-minute to 75-minute instruction periods, changing computational software, and major curriculum overhaul.

1.1.1 Core Mathematics Restructuring

The MA205 AY17-1 executive summary (DeGregory, 2017) contains a single sentence regarding its leadership's responsibility for evaluating what graduates can do (i.e., the USMA Academic Program Goals): *"No longer required for MA205 with removal of MA205 from core curriculum"*.

Horton (2015) references recent changes to USMA's core curriculum which, at the time, would result in many Cadets not taking MA205. Subsequently, MA104 and MA205 exchanged course material in order for Cadets to meet core goals of both D/MATH and USMA academics. In addition to altering the specific topics within MA205, Horton also noted MA205's removal from USMA's core curriculum would decrease course credit hours from 4.5 to 4 as well as, naturally, reducing future Cadet annual enrollment by approximately 60–70-percent.

1.1.2 Course Time Change

In AY18, MA205 consisted of fifty-six 55-minute lessons. This changed in AY19-1 when MA205 transitioned to forty 75-minute lessons (Evans, 2019). The opportunity to teach in a 75-minute instruction period was new to USMA. Colloquially, the math department needed to test some courses using the new 75-minute class setup. MA205 was a good candidate to test this new setup for reasons stated in the previous year's executive summary (Kuiper, 2018). The reasoning for this change focused on the gains of

> *"... consolidating our lessons into more dense and efficient blocks of instruction, with less start-up time per lesson, allowing for students to prepare more thoroughly for classes and assessments, and enabling instructors to prepare and deliver higher quality lessons."*

The "higher quality lessons" refers to allowing instructors to "go deeper in[to] individual topics during extended lesson periods (Kuiper, 2018)." At the time, there was a slight increase in Cadet performance following the transition from 55-minute classes to 75-minute classes. The change between AY18-1 and AY19-1 was not statistically significant, however, and Cadet performance on the TEE remained nearly constant (within one standard deviation). The AY21-1, AY21-2, and AY22-1 (all 75-minute classes) were uncharacteristically low-performing semesters which coincided with the COVID-19 remote and transition-from-remote instructional environments. Because most semesters' TEE performance remains within one standard deviation of the mean (with exception of these three semesters), class length seems not to play a major role in student performance. The course returned to 55-minute classes in AY24-2.

1.1.3 Course Curriculum Redesign

The AY22-2 end-of-course brief also provides insight into the fourth major change to MA205: the curriculum change. MA205's curriculum for AY23 would shift to *"... traditional Calculus III content"* (Trujillo et al., 2023), replacing instruction on both sequences/series and differential equations with geometry of space and vector calculus.

The origins of the redesign stem from a briefing by the MA205 and engineering mathematics (MA364) leadership to the USMA curriculum committee towards the end of AY22 (Trujillo et al., 2022). From the

MA205 perspective, this change was necessary to expand more fully on the subject of calculus while also going more in depth in certain lessons. AY23 was the first full year implementing the new curriculum and this will continue into AY25.

MA205's identity has changed significantly over the years, however there has been a constant of supporting the core goals of D/MATH and USMA's academic program goals (APGs). The Core Math Book (Lindquist, 2021) elaborates on both sets of goals.

As of July 2023, 38 courses list MA205 as either a pre-requisite or co-requisite (Office of the Dean, 2023). Table 1 aggregates these courses into their associated departments.

Department	Number of Courses
Electrical Engineering and Computer Science	4
Geography and Environmental Engineering	1
Mathematical Sciences	19
Civil and Mechanical Engineering	7
Physics and Nuclear Engineering	3
Social Sciences	1
Chemistry and Life Science	3

Table 1: *USMA academic departments and the number of courses which require MA205 as either a pre-requisite or co-requisite as of July 2023.*

While tempting to align MA205's content to the needs of Table 1's individual courses, *"MA205's material should not be determined based on the demands of courses which list it as a prerequisite"* (Evans, 2019).

AY25-1 was the fifth iteration of the new curriculum change including vector calculus. The sequencing of courses between MA205 and MA364 does not seem to negatively affect Cadets as they progress in their majors. Most Cadets in our course take MA205 in the fall of their Yearling year. Nearly all of our Cadets are taking one or two courses in their major, usually an introductory course not listing a mathematics course as a pre-requisite. Therefore, receiving instruction on sequences/series and differential equations later in their major is not a negative effect.

Some majors (e.g. Economics and Engineering Management) may never use vector calculus in their majors and may not go on to take MA364. Some major tracks require MA205 while other majors offer it as an elective or complementary support course. However, those majors do not necessarily need the former MA205 material, so the change of material is irrelevant to them.

1.2 Literature Review

The Mathematical Association of America (MAA) offers several reports and guides (MAA, 2015) related to curriculum design, examples of practices from various colleges and universities, as well as efforts to uniformly move higher-education calculus to modernized standards.

Saxe and Braddy (2015) discuss an impetus to change higher education in math due to various factors including student preparedness, technology advances, and quantitative skills demanded by other disciplines. The sections discussing course structure and workforce preparation encourage math instructors to move away from traditional lectures and use more hands-on instructive periods. Additionally, they suggest demonstrating techniques with real-world applications and reinforcing concepts using technology for visualization and experimentation. The MAA Press's 2022 report on calculus sequence case studies

(Johnson et al., 2022) presents a subset of results from data collection efforts beginning in 2010. The takeaways from different data sets are presented in approximately 90 case studies across five domains: rethinking student support programs; alternative course pathways; variations in approaches to the content of calculus; professional development and course coordination; and program assessment and the use of local data. Schumacher and Siegel (2015) emphasize that instructors should put more effort into active learning due to availability of extensive online resources. In fact, Abell et al. (2018) conclude the classroom practices section with *"... the learning of mathematical ideas is much more profound, ..., when students are active participants in the learning process, ..."*

The MAA's current stance is clear: instructors should adopt active learning strategies. The design of curriculums should feed the stakeholder's purpose. As a service course, Calculus often has many stakeholders, but mostly STEM departments.

1.2.1 Comparative Analysis

We compare West Point's MA205 description and syllabus to that of the U.S. Naval Academy (USNA) and the U.S. Air Force Academy (USAFA). Additionally, we perform comparisons with three liberal arts colleges based on U.S. News rankings (U.S. News & World Report, 2023) – Williams College in Amherst, MA (highest rank), Pomona College in Claremont, CA (lowest acceptance rate), and University of North Carolina (UNC)–Asheville in Asheville, NC (top-10 highest enrollment) – excluding the alphabetical and tuition-focused ranking schemes. We include acceptance rate under the assumption that low acceptance rates imply higher-class education. We also select high enrollment because all three of the U.S. service academies have the highest enrollment rates.

USNA provides Midshipmen "Calculus 3 with Vector Fields" (SM221) and "Calculus 3 with Optimization" (SM223) (Office of the Provost, 2023). Both courses combined cover the same topics as MA205, however Midshipmen only take one. The courses include the phrase "calculus 3" because USNA's core curriculum presents calculus content spanning three separate semesters. All USNA Midshipmen must take either SM221 or SM223.

USAFA offers its Cadets "Calculus 3" (Math 243/253) which matches MA205 up to Green's Theorem (United States Air Force Academy, 2022). Similar to USNA, this course is third in a sequence of calculus courses taught at the academy. Math 243/253 does not progress beyond Green's Theorem

Of the three liberal arts colleges we considered, both Pomona and UNC-Asheville are similar to USNA in that they cover all or nearly all the breadth of our MA205 material across two courses (Pomona College, 2022; University of North Carolina - Asheville, 2023). Pomona College culminates with Stokes' Theorem while UNC-Asheville ends with coverage of the divergence theorem. Of note, Pomona College's vector calculus course, MATH067, requires students to complete linear algebra prior. The third college, Williams College, offers one similar course (MATH 151) which ends with Stokes' Theorem (Williams College, 2023).

1.2.2 Instructional Design and Student Experience

In their "Variations in approaches to the content of calculus" section, Johnson et al. (2022) offer four vignettes on four areas of pedagogical emphasis. The first vignette begins with a course structured in a lecture-lab style. The school's "calculus 2" program appears to follow a four-day-per-week theoretical emphasis with a single lab day focused on applying lecture material to problems. The fourth vignette, following a description of how the school flipped the traditional calculus 1 sequence, emphasized their program's use of graphical tools.

The MAA also published work which purports to establish *"... who takes calculus 1, ... "* and identifies *" ... institutional practices that contribute to the retention of STEM students"* (Bressoud et al., 2015). Although the subject does not align with ours -– calculus 1 versus MA205 — the publication offers some insight into the distribution of theory, to numerical, and to applications based on survey findings. In a survey of faculty teaching calculus 1, 10-percent of instructors focused on proofs and justifications, 20-percent focused on problems on graphical interpretations, 20-percent wanted students to learn through repetition via word problems, and 50-percent chose to focus on techniques to implement computations.

Other resources concerning the distribution of student focus comes from the University of Twente in the Netherlands (Craig et al., 2022) and Carroll College in Helena, MT, USA (Sullivan and Melvin, 2016). In the former, the authors discuss their adaptation to online resources by designing lectures and weekly quizzes towards theory and four-hours-per-week of hands-on learning exercising the mechanics of solutions and using different 3D graphing software to explore vector calculus topics. The latter authors, whose multivariable course primarily feeds higher-level engineering courses, place heavy emphasis on mathematical modeling and computation, using software to accelerate/enhance student learning.

2 Course Design

A Cadet completing MA205 should demonstrate proficiency across nine student learning outcomes (SLOs). This occurs across four content sections (blocks), each demanding varying levels of effort (i.e., time commitment).

2.1 Student Learning Outcomes

MA205's nine SLOs are:

1. **Demonstrate Good Scholarly Habits.** Students demonstrate good scholarly habits for progressive intellectual development.

2. **Communicate Effectively.** Students improve the quality of both their written and verbal communication of mathematics.

3. **Apply Technology.** Students are able to leverage technology when solving complex mathematical problems.

4. **Develop Problem Solving and Reasoning Skills.** Students develop the problem solving and reasoning skills necessary for continued study in mathematics, the physical sciences, the social sciences, and engineering.

5. **Develop Interdisciplinary Perspective.** Students develop an interdisciplinary perspective that supports knowledge transfer across disciplinary boundaries to solve complex problems.

6. **Apply Multivariate Derivative.** Students apply the basic concepts of the multivariate derivative to solve multivariable unconstrained and constrained optimization problems.

7. **Apply Multivariate Integral.** Students apply multivariate integration to solve accumulation problems.

8. **Apply Vector Operations and Model Motion of an Object.** Students apply the basic properties and operations of vectors, quadric surfaces and space curves in three-dimensions, and model the motion of an object in three-dimensional space.

9. **Demonstrate Calculus of Vector Fields/Line and Surface Integrals.** Students demonstrate the calculus of vector fields, line and surface integral techniques, and the connections between these new types of integrals and the single, double, and triple integrals that are given by the higher dimensional versions of the Fundamental Theorem of Calculus.

2.2 Semester Layout

The course is divided into four blocks of instruction, each containing various topical lessons and course-wide events which support attaining the SLOs. We provide the Cadets with the following descriptions of each block:

1. **Vectors and Geometry of Space.** This block introduces vectors and three-dimensional coordinate systems, ultimately constructing the foundation for the study of multi-variable functions. You will become familiar with vectors and use them to express lines and planes in space. You will represent functions using analytic, numerical, verbal, and graphical methods, with emphasis on graphical methods. You will be able to model and analyze three-dimensional problems incorporating position, velocity, and acceleration.

2. **Multivariable Differentiation.** This block seeks to develop in the student, a solid understanding of rates of change in multi-dimensions. They will first focus on the physical concept of a partial derivative and how to compute them. Then, they will be introduced to the gradient and will solve optimization problems, with and without constraints.

3. **Multivariable Integration.** This block will develop the students' understanding of continuous accumulation and will further develop their comfort of graphical expression of multivariable functions. They will apply this understanding to compute volumes, surface areas, masses, and centers of mass of general regions. Students will also be able to apply some MA206 knowledge (if they have taken it) when they compute joint probability functions. Students will express regions in rectangular, polar, cylindrical, and spherical coordinates.

4. **Vector Calculus.** This block focuses on the study of the calculus of vector fields and serves as a foundation for future mathematics and engineering courses. Students will explore vector fields and be able to apply this knowledge toward solving engineering problems involving different types of fields. Students will extend their understanding of arc length and surface area into the study of line and surface integrals. The block culminates by developing the connection between the Fundamental Theorem of Calculus, double integrals, and triple integrals and the Fundamental Theorem for Line Integrals, Green's Theorem, and Stokes' Theorem.

We align the SLOs with the course structure to aid further course development.

		Student Learning Outcomes								
		1	2	3	4	5	6	7	8	9
Block	1	X	X	X	X	X	-	-	X	-
	2	X	X	X	X	X	X	-	-	-
	3	X	X	X	X	X	-	X	-	-
	4	X	-	-	X	X	-	-	-	X

Figure 1: *Deciding on the alignment of SLOs to the course layout facilitates further course development.*

Due to the density and relative challenge of the block four material, we exclude some SLO mechanisms (see Appendix A) from the design. This allows both students and faculty to focus on the content aligned with SLO 9.

2.3 Cadet Requirements and Support

MA205 is a four-credit-hour course with 56 lessons bringing a total Cadet commitment of approximately ***167 hours*** of combined in- and out-of-class work. Table 2 provides a breakdown of the time commitment. For a four-credit-hour course, the Academy requires 160 total hours of Cadet commitment.

Event Timing Calculations	**Total Hours**
(49 standard lessons)× (1 contact hour)	49 contact hours
(3 technology labs)×(1 contact hour)	3 contact hours
(2 WPRs)×(1 contact hour)+(2 WPRs)×(1.25 contact hours)	4.5 contact hours
(1 TEE)×(3.5 contact hours)	3.5 contact hours
(37 standard lessons)×(1 contact hour)×(2 prep hours)	74 prep hours
(4 WPRs)×(2.5 prep hours)	10 prep hours
(1 TEE)×(6 prep hours)	6 prep hours
(3 technology labs)×(2 prep hours)	6 prep hours
(3 technology labs)×(2 reflections)×(1 prep hour)	6 prep hours
(1 applications paper & video)×(5 prep hours)	5 prep hours
	167 Hours

Table 2: *The total time commitment of Cadets to MA205, including in and outside of class.*

To facilitate preparation, MA205 makes the syllabus and calendar available to all Cadets on Canvas. Canvas's integrated assignment due-date feature also helps Cadets allocate their time and effort. Canvas displays the due-dates and dates-of-occurrence for all graded course events in a single, time-ordered list as well as within their individual calendars.

Outside of MA205, there are numerous free resources for Cadets to exploit. Free, course-related content may be found on YouTube, as well as Khan Academy (Khan, 2023) and MIT's OpenCourseWare platform (Auroux, 2006, 2010). USMA's Center for Enhanced Performance (CEP), in addition to tutoring services, has, cumulatively, 16 historical quizzes, WRITs, and WPRs available on their website. Only some of the CEP's historical material maintains relevance due to curriculum changes discussed in Section 1.1.3.

2.4 Course Alterations

We implemented several changes to MA205 in AY25. The biggest change was the return of the course to 56 lessons with 55-minute in-class times from 40 lessons with 75-minute class times. We describe the changes in detail. Course design alterations must not detract from the SLOs, so we also discuss how each change may affect SLO assessments and the controls we emplaced.

2.4.1 75- to 55- minute Redesign

The course redesign required various combinations of lesson mapping from the 40-lesson semester to the 56-lesson semester: one-to-one, split, and compression. *One-to-one mappings* yield one 55-minute in-class meeting for a given 75-minute in-class meeting. *Split mappings* yield more than one 55-minute in-class meeting for a given 75-minute in-class meeting. *Compression mappings*, for any given sequence of 75-minute in-class meetings, yield a lesser quantity of 55-minute in-class meetings.

The resultant lesson mappings provide a different academic experience for the Cadets which, potentially resulting in a performance drop associated with SLOs 6–9. To mitigate this effect, we compared topic exposure hours to historical calendars and accepted reduced exposure for lessons in which Cadets demonstrated at-or-above acceptable performance as defined in 3. This consideration also permitted us to increase exposure to lessons associated with historical underperformance.

2.4.2 Formative Assessments for Final Three Topics

we encouraged all instructors to provide some type of formative assessment for the final three course topics: surface integrals, Stokes's Theorem, and the Divergence Theorem. We gave instructors 30 additional instructor points for this purpose and provided all with approved documents for such purposes, allowing them to also create their own.

Providing a formative assessment for each of the final three topics may detract from developing understanding for Cadets with a traditional lesson, thus negatively impacting SLO 9. However, a well-designed formative assessment supplements a Cadet's knowledge by facilitating a reasonable cognitive challenge and providing immediate feedback to certify understanding.

2.4.3 Initial Schedule Compression

A minor change of utilizing Dean's Hour periods (75 minutes) to administer exams required a major compression of the course schedule up front. To accommodate the available Dean's Hours, we had to complete the first two blocks within the first six weeks of the semester. As the course progressed, we found reason to abandon the Dean's Hour exams for blocks three and four through more focused exam development.

Compression of the first two blocks potentially negatively affected SLOs 6 and 8 due to Cadets not having as much processing time for the material outside of class. The compression also potentially precluded development and implementation of good scholarly habits (SLO 1); conversely, Cadets who do not drown in the course material possess average to above-average scholarly habits.

2.4.4 Lesson Relocation

We reorganized the lesson sequence by introducing polar coordinates and quadric surfaces in block three instead of block one. The course instruction since AY22 did not fully develop these topics, instead having them serve in a recall capacity for block three. While reaching back to previous material is an effective pedagogical strategy, shifting these lessons into the block where we primarily leveraged them allowed us to stay focused on the second most challenging topic for students.

Shifting the quadric surfaces has a direct affect on SLO 8's containment within block one, but does not overall prevent students from engaging with the subject. We can argue either way for the effect on SLO 4 – reachback or immediately accessible tools for problem solving – so we don't believe the net effect on this SLO is non-zero.

3 Assessment and Evaluation Plans

We discuss our plans to assess and evaluate MA205 during AY25. This relies on deliberate data gathering efforts tied to three of our nine SLOs. For each SLO, we detail the assessment plan and discuss how we use this data to evaluate the course design.

The Department reinvigorated cyclical course assessment plans this AY. We carefully considered what mechanisms would yield evidence in support of assessing our SLOs in accordance with DPOM 5-07 (Yankovich, 2019). Appendix A details all SLO assessment mechanisms along with the reasoning for their selection.

3.1 SLO 6. Apply Multivariate Derivative

The goal of this SLO is that Cadets apply the basic concepts of the multivariate derivative to solve multivariable unconstrained and constrained optimization problems. To judge the attainment of this SLO, we use three exam questions as assessment mechanisms: WPR2 Questions 2 and 3 and TEE Question 10.

We focus on the two specific questions from the WPR rather than the entire assessment because the other questions do not directly demonstrate SLO achievement. Focusing on differentiation fundamentals , we included them on the assessment to provide insight into underlying issues Cadets may be facing for future course improvement. This reasoning also applied to the WPR questions for SLOs 8 and 9.

We drafted the WPR questions during the summer prior to AY25-1 by first analyzing how a question would permit assessment of a SLO. We then determined which lesson objectives (LOs) aligned with this assessment plan. For WPR2 Question 2, we utilized LOs 2.15 and 2.16:

- **LO 2.15:** Understand the definition of a critical point and be able to find the critical points of a function of two variables.
- **LO 2.16:** Use the Second Derivatives Test to classify a critical point as a local maximum, local minimum, or saddle point

For WPR2 Question 3, we utilized LO 2.19:

- **LO 2.19:** Solve optimization problems for functions of several variables subject to a single constraint using the Method of Lagrange Multipliers.

TEE Question 10 did not substantively change from previous semesters; the review process clarified wording and presentation so the question more clearly aligned as support for SLO 6 assessment. This question also used LO 2.19.

Refer to Appendix B for the actual questions.

3.2 SLO 8. Apply Vector Operations and Model Motion of an Object

The goal for this SLO is that Cadets apply the basic properties and operations of vectors, quadric surfaces, and space curves in three-dimensions and model the motion of an object in three-dimensional space. To judge the attainment of this SLO, we use three exam questions as assessment mechanisms: WPR1 Question 2 and TEE Questions 4 and 7.

Following the same process for WPR1 Question 2 as we did for WPR2 questions with SLO6, we decided to incorporate LOs 1.8 and 1.11:

- **LO 1.8:** Compute derivatives and integrals of vector functions.
- **LO 1.11:** Model the motion of an object in 3D-space.

We altered the fourth question on the TEE from previous semesters, replacing a question on the multi-variable chain rule with one on space curves, planes, and their interactions in 3D space. The previous question did not directly tie into a SLO assessment and the first block of material was underrepresented on this final assessment. We designed the new TEE Question 4 by deciding LOs 1.4, 1.7, and 1.9 could appropriately be used to craft a question aligned with SLO 8.

- **LO 1.4:** Model and understand relationships between lines and planes in 3D-space.
- **LO 1.7:** Model space curves in 3D using vector functions.
- **LO 1.9:** Compute the tangent line to a space curve and understand its significance.

The substance of TEE Question 7 did not change from previous instances, but we did adjust the question. The intent of the question had previously been to leverage LOs 1.8 and 1.11, but we experienced multiple instances of Cadets using formulas memorized from their physics classes to answer the questions. Thus, although many achieved correct answers, they did not demonstrate either the functionality of the LOs or achievement of the SLO. The phrasing of the new sub-questions, as well as the information provided, removed these issues. This year's TEE Question 7 focused on LOs 1.8 and 1.11 (both previously listed) and LO 1.10:

- **LO 1.10:** Determine distance along a 3D-space curve.

Refer to Appendix B for the actual questions.

3.3 SLO 9. Demonstrate Calculus of Vector Fields/Line and Surface Integrals

The goal of this SLO is that Cadets demonstrate the calculus of vector fields, line and surface integral techniques, and the connections between these new types of integrals and the single, double, and triple integrals that are given by the higher dimensional versions of the Fundamental Theorem of Calculus.

To judge the attainment of this SLO, we use six exam questions as assessment mechanisms: WPR4 Questions 2 and 4 and TEE Questions 1, 2, 3, and 8.

We replicated the WPR question process from SLO6 for WPR4 questions 2 and 4. WPR4 Question 2 relied on LOs 4.2 and 4.6:

- **LO 4.2:** Understand the meaning and compute the line integral along a curve in space.
- **LO 4.6:** Understand the "del" operator and use it to compute curl and divergence.

WPR4 Question 4 focused on LO 4.5:

- **LO 4.5:** Understand and apply Green's Theorem to general 2D regions.

TEE Questions 1, 2, and 3 did not change from AY24-2, though we changed TEE Question 2 from AY24-1 to match with the department's MA204 and MA255 course's TEE for cross-course comparison. The substance of TEE Question 2 did not change. These TEE questions were the first summative assessment of the associated topics so as to not overwhelm Cadets on the block four's WPR. TEE Questions 1, 2, and 3 focused on LOs 4.11, 4.12, and 4.12, respectively.

- **LO 4.11:** Use the surface integral to determine the flux across a solid region.
- **LO 4.12:** Understand and apply Stokes' Theorem to general surfaces.
- **LO 4.13:** Understand and apply the Divergence Theorem to general solids.

We altered TEE Question 8 from a scalar field line integral to a vector field line integral. We implemented this change to give Cadets a "second-chance" to solve a line integral to what they experienced with WPR4 Question 2. TEE Question 8 also leveraged LO 4.2 in its design to support SLO 9.

Refer to Appendix B for the actual questions.

4 Results

Grades communicate something to both Cadets and future instructors. Throughout the course, MA205 utilized the following holistic grading standard to tie grades to qualitative performance feedback:

Letter Grade	Numerical Grade (%)	Description
A+	97-100	Beyond Expectations
A	93-96.99	Dominates the material
A-	90-92.99	Mastery
B+	87-89.99	Excellent performance
B	83-86.99	Good understanding
B-	80-82.99	Proficient; aptitude for the subject
C+	77-79.99	Can build on foundations
C	73-76.99	Passing; proficient
C-	70-72.99	Short-range understanding
D	65-69.99	Marginal performance\some learning
F	< 65	Failed to demonstrate understanding

Table 3: *MA205 Holistic Grading Scheme*

The rubrics of all course graded events, except WebAssign, used Table 3 for point allocation.

In consultation with the Curriculum and Assessment committee (Success Meeting, February 14), we define success in MA205 for SLOs 6, 8, and 9 in Table 4.

70% of Cadets achieve a *C- or greater* on each WPR question.
73% of Cadets achieve a *C or greater* on each TEE question.

Table 4: *The AY25 MA205 SLO Success Criteria.*

Table 3 aids the interpretation of the success criteria. We tie both the percentage of Cadets achieving a target and the target itself to our holistic grading standards. We aim for Cadets to depart MA205 with an individual "passing" assessment; we use the same passing assessment for the course. The first success criteria is quantitatively lower because we expect performance to improve on the blocks with the TEE - having had time to reflect and shore up knowledge.

We present the data gathered from our assessment mechanisms - with slight elaboration - before moving to a more detailed analysis which includes with our course assessment.

4.1 Data

We obtained our data from the final gradebook following the TEE.

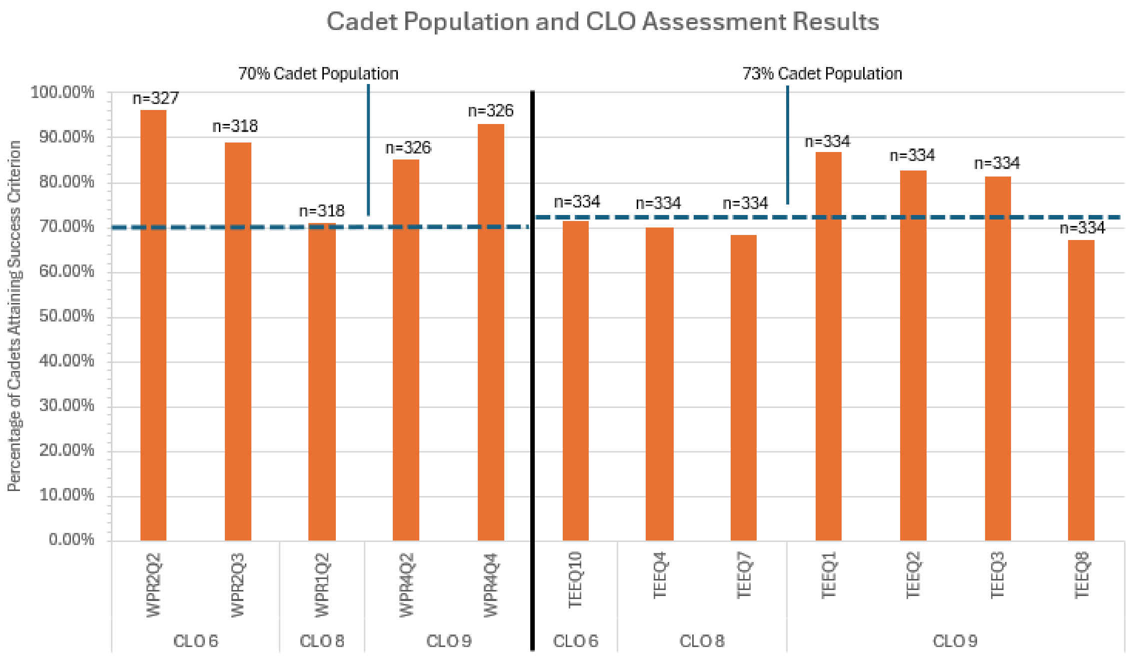

Figure 2: *The percentage of Cadets meeting SLO success criteria.*

We present more detailed information for each SLO and their assessment mechanism outcomes.

4.1.1 SLO 6: Apply Multivariate Derivative

Table 5 presents data associated with SLO 6 from Figure 2 coupled with the success criteria from Table 4.

Mechanism	Goal	Results
WPR 2 Question 2	70% achieve a C-	96.23%
WPR 2 Question 3	70% achieve a C-	88.99%
TEE Question 10	73% achieve a C	71.56%

Table 5: *A summary of the SLO6 success criteria and results.*

The MA205 population achieved two of the three success criteria. TEE Question 10 asked Cadets to implement the method of Lagrange multipliers to determine the point which minimized the given objective function while satisfying the given constraint.

4.1.2 SLO 8: Apply Vector Operations and Model Motion of an Object

Table 6 presents data associated with SLO 8 from Figure 2 coupled with the success criteria from Table 4.

Mechanism	Goal	Results
WPR 1 Question 2	70% achieve a C-	70.95%
TEE Question 4	73% achieve a C	70.06%
TEE Question 7	73% achieve a C	68.26%

Table 6: *A summary of the SLO8 success criteria and results.*

The MA205 population achieved one of the three success criteria. TEE Question 4 contained two, independent sub-questions. The first sub-question asked Cadets to develop the vector equation of a line using information related to a provided space curve. The second sub-question asked Cadets to determine the point a line intersected a plane - we provided equations for both.

4.1.3 SLO 9: Demonstrate Calculus of Vector Fields/Line and Surface Integrals

Table 7 presents data associated with SLO 9 from Figure 2 coupled with the success criteria from Table 4.

Mechanism	Goal	Results
WPR 4 Question 2	70% achieve a C-	85.28%
WPR 4 Question 4	70% achieve a C-	93.01%
TEE Question 1	73% achieve a C	86.83%
TEE Question 2	73% achieve a C	82.93%
TEE Question 3	73% achieve a C	81.44%
TEE Question 8	73% achieve a C	67.37%

Table 7: *A summary of the SLO9 success criteria and results.*

The MA205 population achieved five of the six success criteria. TEE Question 8 required Cadets to solve a line integral in a vector field. We provided the integral bounds, vector field, and parametric equation form of the space curve.

4.2 Discussion

When considering whether the course succeeded this year, we must consider the strength of our evidence and the strength of attainment. While the academy does not have an authoritative source document unifying these concepts for the departments, our communication with the Curriculum and Assessment committee (Evidence Meeting, June 2024) enabled our interpretations.

Strength of evidence relates to the assessment mechanism(s) used in SLO assessment. We consider the qualitative/quantitative nature of the mechanism(s) and the density and distribution of SLO assessment mechanisms. For example if we only use an end of course feedback question to assess a CLO then we could say the strength of evidence is probably pretty weak. Another example is, if we planned on using a question from WPR3 and we come to realize that there was a flaw in the way we asked the question, we could categorize that strength of evidence as being weak. Ultimately, this is for us to evaluate how well we are doing at capturing the required information to make an informed assessment.

Strength of attainment relates to the actual performance or information provided by the student measured against the preset target. If we decide a positive mechanism assessment is more than 90% of Cadets achieving a C or better and that happens, then we have a high strength of attainment. This necessitates defining success ahead of time - as we did with Table 4.

Enrolled Cadets on average succeeded in eight of the twelve assessment mechanisms in AY25. Two of the unsuccessful results occurred for mechanisms aligned with SLO 8 on the TEE. The remaining two also occurred on the TEE, one each for SLOs 6 and 9. Since all failures occurred on TEE-related mechanisms, we conclude the weight of the material presence and length of time since exposure to material play a factor, particularly for the questions aligned with SLOs 6 and 8. Following the "time since exposure" line of thinking, this also may be the underlying cause for achieving SLO 6 but not SLO 8. Two of the three assessment mechanisms for SLO 6 occur in the WPR given during the associated material, but only one of the three mechanisms for SLO 8 are on its relevant WPR. This means SLO 8 is more prone to experience the negative results of "time since exposure," biasing our conclusions. Future course iterations should reduce this bias by evenly distributing Cadet reach-back across all SLO assessment mechanisms and increasing the pool of mechanisms for each SLO.

This analysis leads to our final assessment of the course presented in Table 8.

Course Learning Outcome	**Strength of Attainment**	**Strength of Evidence**	**Overall Evaluation**
SLO 6: Apply Multivariate Derivative	B	B	B
SLO 8: Apply Vector Operations and Model Motion of an Object	D	B	C
SLO 9: Demonstrate Calculus of Vector Fields/Line and Surface Integrals	A	A	A

Table 8: *SLO Evaluations. Key: A = exceeds the standard; B = meets the standard; C = meets the standard with some weakness; D = fails the standard (can be addressed in the short-term); F = catastrophically fails the standard (requires immediate remediation)*

Based on the synthesis of each overall evaluation, we conclude MA205 met the standard in AY25.

5 Conclusion

This report presented historical background to cement the current state of the course while also providing future course directors a more aggregated source of information. Following a review of comparable academic institutions and the subject of calculus, we discussed the course design for AY25 and how we would assess ourselves. We finally presented our data and analysis, concluding MA205 succeeded in attaining its assessed SLOs this year.

In our conclusion, we shift to a broader perspective of the course. We compare this year's instantiation of MA205 to historical semesters to gain some understanding of how broader course design changes impacted assessments. Finally, we provide considerations for the future of the course.

5.1 Final Design Reflections

We compare this semester's results with those of the previous two semesters. For each summative assessment from previous AYs, we identified similar questions. Table 9 provides the mapping to historical questions.

	AY25-1	AY24-1	AY23-1
SLO6	WPR 2 Question 2	WPR 2 Question 5	WPR 2 Question 5
	WPR 2 Question 3	WPR 2 Question 2	WPR 2 Question 2
	TEE Question 10	TEE Question 10	TEE Question 10
SLO8	WPR 1 Question 2	WPR 1 Question 2	WPR 1 Question 2
	TEE Question 4	Incomparable	Incomparable
	TEE Question 7	TEE Question 7	TEE Question 7
SLO9	WPR 4 Question 2	WPR 4 Question 1	WPR 4 Question 1
	WPR 4 Question 4	WPR 4 Question 3	WPR 4 Question 3
	TEE Question 1	TEE Question 1	TEE Question 1
	TEE Question 2	TEE Question 2	TEE Question 2
	TEE Question 3	TEE Question 3	TEE Question 3
	TEE Question 8	Incomparable	Incomparable

Table 9: *Questions from historical summative assessments that are similar to our current SLO assessment mechanisms.*

We assume the difference in mean scores between comparable questions is normally distributed. This enables us to use 24 inter-year comparisons of the independent samples using non-paired, two-sample t-test. Since we modified the course from a 75- to a 55-minute modality, we want to ensure the Cadets are not negatively impacted. Therefore, our null hypothesis is the AY25 mean is at least as large as the previous years; we use a 0.05 significance level. In Table 10, we highlight the comparisons which indicate violations of our null hypothesis, of which there are only three of the twenty-four total inter-year comparisons.

	Assessment Mechanism	Historical Comparable	2-sample t-test p-value
SLO6	AY25-1 WPR2 Q2	AY24-1 WPR2 Q5	2.2e-16
SLO6	AY25-1 WPR2 Q3	AY23-1 WPR2 Q2	0.02126
SLO8	AY25-1 WPR1 Q2	AY23-1 WPR1 Q2	3.411e-11

Table 10: *Comparisons between the AY25-1 assessment mechanisms and historical, comparable mechanisms which evident a violation of our null hypothesis and indicate AY25-1 Cadets did not, on average, perform as well.*

There are many potential explanations for this drop in performance. From a course design perspective, we defer to the 55-minute in-class change and time-compression of the first two blocks.

We weakly conclude the 55-minute in-class time change impacted SLO performance because we did not see the same effect in latter blocks and each assessment mechanism in Table 10 only compares negatively with one of the two comparable historical examples.

Since we do not see SLO9 in the table, the initial schedule compression is more likely a cause for the results in Table 10. Each of blocks one and two, with their respective summative assessments, spanned two-and-a-half weeks. This conclusion is likely since these particular assessment mechanisms resided entirely within the compressed schedule. Future iterations of the course will mitigate this possibility by addressing the root cause of the up-front schedule compression: exams during Dean's hour periods. We demonstrated in-class, summative assessments work well which allows us to open the schedule of the first two blocks by adding more course X-hours.

5.2 Program SLO Impact

The department's Math Sciences program hosts MA205 as a service course in addition to several other courses. (Scioletti, 2024) For AY25, the program assessed its own SLOs:

1. Demonstrate competence in modeling physical, information and social phenomena by
   a. Identifying and articulating assumptions, metrics and constraints
   b. Applying appropriate solution techniques
   c. Interpreting results within the appropriate context
3. Achieve mathematical proficiency in breadth and depth
   a. Understand and apply theorems and algorithms
   d. Understand and apply graphical methods
7. Understand the role of mathematical sciences (in our world) by analyzing applied problems through disciplinary, multidisciplinary and interdisciplinary approaches

Table 11 details MA205's assessed SLOs nest within the program's SLOs.

		MA205 SLOs		
		6	8	9
Math Sciences SLOs	1a	X	X	
	1b	X		X
	1c	X	X	X
	3a	X		X
	3d		X	
	7	X	X	X

Table 11: *Identifying the nested relationship between the SLOs of MA205 and the Math Science program.*

Combining the SLO crosswalk and the MA205 SLO evaluations in Table 8, we conclude our contribution to the program SLOs as identified in Table 12.

Program SLO	Strength of Attainment	Strength of Evidence	Overall Evaluation
SLO 1a	B/D/– $\rightarrow$ C	B/B/– $\rightarrow$ B	C
SLO 1b	B/–/A $\rightarrow$ B	B/–/A $\rightarrow$ B	B
SLO 1c	B/D/A $\rightarrow$ B	B/B/A $\rightarrow$ B	B
SLO 3a	B/–/A $\rightarrow$ B	B/–/A $\rightarrow$ B	B
SLO 3d	–/D/– $\rightarrow$ D	–/B/– $\rightarrow$ B	C
SLO 7	B/D/A $\rightarrow$ B	B/B/A $\rightarrow$ B	B

Table 12: *Program SLO Assessment Contributions. We aggregate the attainment and evidence evaluations from the course SLOs to obtain a measure for the program SLO attainment and evidence scores. These two scores then provide the overall evaluation. The letter keys are the same as Table 8.*

Our assessed contribution to the program SLOs does not account for any weighting to account for the number of contributing course assessment factors.

5.3 Future Considerations

We have an opportunity to think critically about the course's design - particularly how we assess the SLOs. The subjectivity on how we defined success with Table 4 and the assessment mechanisms in Appendix A gives us flexibility in future iterations, but we should be leery of redefining these. Changing the success metrics does not alter root causes of under-performance.

As a final thought, we turn towards the SLOs themselves. If we set aside the ambiguity and poor wording which permeates the SLOs in section 2.1, we can still ask outselves if these are truly what we want Cadets to take from the course. While admirable, the purpose for MA205 since its removal from the core curriculum is to broaden technical math skills aligned with multi-variable calculus. From the description in Office of the Dean (2023):

> *"'This course provides a foundation for the continued study of mathematics and for the subsequent study of the physical sciences, social sciences, and engineering. ..."*

The 2022 curriculum change request (Trujillo, 2022) reaffirms the SLOs and provides an overarching purpose for the higher-level ambitions of SLOs 1-5. However, within the same document, the author notes two of the three stakeholders – follow-on courses, the Accreditation Board for Engineering and Technology, and the generic mathematical discipline – do not necessarily need these loftier SLOs. The academy seems to agree with this interpretation having removed MA205 from the core curriculum. Care should be given to analyzing the SLOs and determining which are appropriate for future iterations of the course given its place within the academic curriculum. Clarified and removed SLOs provide opportunities to more deliberately invest time and energy into the remaining outcomes, producing better experiences for enrolled students.

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A Assessment Mechanisms

In this appendix we describe the reasoning behind the mechanisms selected to assess the nine SLOs. This appendix serves as a basis off which future course directors should pivot for the purposes of improving the course.

1. **Demonstrate Good Scholarly Habits**
   - WebAssign submission states: we altered the due dates of WebAssign homework to be due prior to the first lesson of the following week. This gave Cadets freedom to choose when they completed the assignments. In theory, Cadets who attempt/complete these during the regular week have opportunities to engage in discussions within class, unlike those who postpone until the weekend. Since the WebAssign assignments are fairly lengthy, Cadets attempting to cram over the weekend will be less likely to complete them all, which offers an indicator of scholarly habits.
   - Technology Labs 1/2/3 Prep (assignments): these assignments opened at the beginning of the associated blocks of material (e.g., Tech Lab 1 Prep occurred in block 1). By making these available and the due date known, we left Cadets the freedom of when to complete and submit - the due date specifically used language of “due no later than ...” Cadets who failed to submit the assignment or achieved relatively poor scores exhibit poor scholarly habits.
   - Major Applications Paper & Video (assignments): we make these assignments available at the beginning of the course and due near the end. The logic is similar to the WebAssign and Technology Lab Prep assignments in that failure to submit indicates poor habits. This assignment differs, however, in that we direct the Cadets to perform research leveraging assets existing within the departments of their declared major. Cadets with citations tied to their associated departments, whether it be people, course information, or course textbooks, demonstrate good scholarly habits.
2. **Communicate Effectively**
   - Grade sequence of the communication-focused assignments: grading rubrics unite these five assignments by deliberately focusing on communication aspects and not technical, mathematical details. Through several iterations of formative feedback (technology lab reflections), we expect Cadets to obtain their highest scores on the Major Applications Paper & Video assignments. A positive trend in grades, on average, indicates the Cadets’ improvement of their ability to communicate technical material.
3. **Apply Technology**
   - Technology Labs 1/2/3 (assignments): these three low-stakes assignments present problems for Cadets to solve in Mathematica notebooks which they submit through the Canvas learning management system. These problems are solvable by hand, so Cadets know what the answers should be and must figure out how to use the Mathematica language to obtain the same answers.
4. **Develop Problem Solving and Reasoning Skills**

- Performance in follow-on courses listing MA205 as a prerequisite: MA205 students continue their academic journey to various courses across disciplines which use not necessarily overlapping subsets of the course material. We can measure student performance in follow-on courses by working with leadership of those courses to identify specific assessments or activities which explicitly utilize MA205 material. We focus on specific assessments and activities in order to isolate MA205-related skills.

5. **Develop Interdisciplinary Perspective**
    - Technology Labs 1/2/3 (assignments): the technology labs were all designed to focus on different disciplines which make use of MA205 content. These assignments provide evidence for the SLO by presenting problems through the lens of a unique discipline, such as military indirect fire trajectories. This requires students to identify how to leverage the course content to address these problems.
    - Major Applications Paper & Video (assignments): these coupled assignments force students to engage with their individual declared majors and develop an understanding of how the course content fits into their academic disciplines. Through research, writing, and speaking, students transcend the course and build a holistic picture of the structural support MA205 provides.

6. **Apply Multivariate Derivative**
    - WPR2 Questions 2 and 3. Block two provides three types of optimization methods: the second-derivatives test and closed-interval method extension for two-variable functions and Lagrange multipliers for multi-variable/constraint systems. These two questions address all three methods.
    - TEE Question 10. Previous TEEs include this question, permitting a standardized means of comparing AYs. The question rubric directly relates to the SLO description.

7. **Apply Multivariate Integral**
    - WPR3 Question 2. This is one of two questions on the assessment in which Cadets must both set up and evaluate an iterated integral. We use this question because it occurs earlier in the assessment when a Cadet is, in theory, not fatigued by problem solving, therein providing a more accurate snapshot of their knowledge.
    - TEE Questions 1, 2, 3, 8, and 9. These five problems span blocks 3 and 4. The rubric for each weights the integral set-up and solution, favoring the set-up. Setting up an integral is an implied task not directly addressed within this SLO. Thus, allowing for credit following set-up errors, we use the solution portion of these problems' rubrics to assess this SLO.

8. **Apply Vector Operations and Model Motion of an Object**
    - WPR1 Question 2. Historically there have been issues with Cadets attempting to recall memorized formulas from their physics classes to solve problems related to three-dimensional acceleration, velocity, and position problems. This question precludes this by positing two scenarios: one with a vertical acceleration with no standard gravitational effects and the other

with all equations provided. This results in Cadets demonstrating their understanding of how the language of calculus aids in solving these problems.

- TEE Questions 4 and 7. Previous TEEs include question seven, permitting a standardized means of comparing AYs. The question rubric directly relates to the SLO description. In AY25-1, we changed question four from a multi-variable chain rule problem (block two material - SLO 6) to one aligned with this SLO. We made this change because the TEE only included one question associated with this SLO and the previous question was not very well designed for assessing SLO 6.

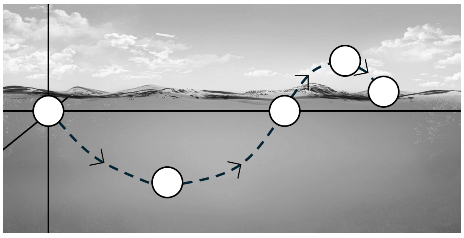
9. **Demonstrate Calculus of Vector Fields/Line and Surface Integrals**

   - WPR4 Questions 2 and 4. Questions two and four transcend straight-forward calculations and provide Cadets opportunities to demonstrate the applicability of various vector calculus theorems. Question two establishes that Cadets *cannot* leverage the fundamental theorem for line integrals and must solve the line integral using one of two appropriate techniques. Question four requires Cadets to demonstrating an understanding of how Green's Theorem relates the region and line integrals.
   - TEE Questions 1, 2, 3, and 8. Previous TEEs include questions one, two, and three, permitting a standardized means of comparing AYs. The questions' rubrics directly relate to the SLO description. In AY25-1, we changed question eight from a scalar field line integral to a vector field line integral because a majority of the material emphasizes vector fields and we did not want to catch the Cadets off guard.

B Questions

WPR 1 Question 2

A ball is pushed underwater. Assume the z-axis is orthogonal to the water's surface. This scenario uses *metric* units.



Underwater Information

- The ball has constant acceleration only towards the surface at 4 m/s^2
- $\mathbf{v}(0) = \langle 1,\ 10,\ -1 \rangle$

a. Determine the ball's acceleration, velocity, and position vectors while underwater.

b. When the ball pops out of the water at $t = \frac{1}{2}$, it has the acceleration, velocity, and position vectors

$$\mathbf{a}(t) = \langle 0,\, 0, -g \rangle, \qquad \mathbf{v}(t) = \left\langle 1,\ 10, -gt + 1 + \frac{g}{2} \right\rangle, \qquad \mathbf{r}(t) = \left\langle t,\ 10t,\ \frac{-g}{2}t^2 + \left(1 + \frac{g}{2}\right)t - \frac{g+4}{8} \right\rangle$$

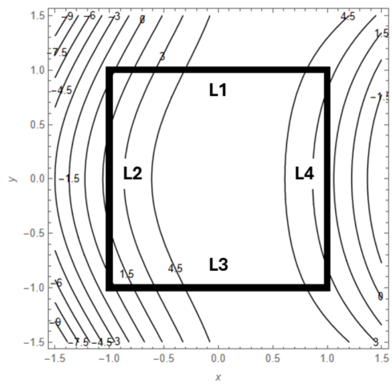
where $g \approx 9.81$ m/s^2 is the gravitational constant. At what time does the ball **return** to the water's surface? *hint: this is purely a vertical consideration*

WPR 2 Question 2

A worker's productivity is modeled by the function $f(x,y) = -4x^2 - \dfrac{y^2}{2} + 2xy^2 + 6$ where x represents a wage multiplier and y is a work hours multiplier.

a. Find and classify the critical points of the unconstrained function.

b. Consider the domain $-1 \le x \le 1,\ -1 \le y \le 1$ (*see image on bottom left*). There are eight boundary points on the domain boundaries L1, L2, L3, and L4 which are candidates for the global minimum or maximum. Find **THREE** of these boundary points across L3 and L4; do not use the intersection of L2 and L3.



WPR 2 Question 3

Consider the objective function $f(x,y) = 5x^2y + 2y^3$ subject to the constraint $y^2 = 25 - x^2$.

a. Use Lagrange Multipliers to set up the system of equations which optimizes $f(x,y)$. **DO NOT SOLVE.**

b. Is the point $P(5,0)$ a candidate for a maximum value? Show your work or explain how you know.

WPR 4 Question 2

Consider the vector field $\mathbf{F} = \langle xy,\ y^2,\ yz \rangle$

a. Perform the entire curl calculation to demonstrate $\mathbf{F}$ is not conservative.

b. Evaluate the line integral using $\mathbf{F}$ above where the curve C is given by $\mathbf{r}(t) = \langle 2t,\ 4,\ 3t \rangle$ on the interval $0 \le t \le 5$.

WPR 4 Question 4

The next two questions both refer to the path in Figure (2) and the non-conservative vector field

$$\mathbf{F}(x,y) = \langle x^2 + y,\ 1 - x \rangle$$

a. Use Green's Theorem to calculate the line integral by instead evaluating a double integral.

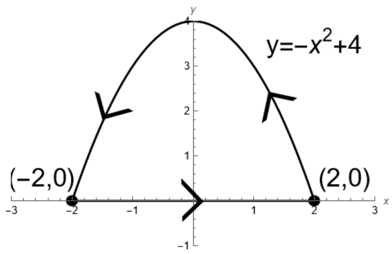


Figure 3: *A single traversal along a SLOsed path starting and ending at the point* $(2,\ 0)$.

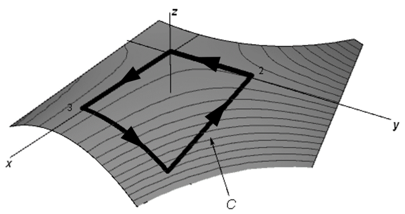
b. Fully set up the line integral. **DO NOT SOLVE.**

TEE Question 1

Find the flux of the field, $\mathbf{F}(x,y,z) = x\mathbf{i} + z\mathbf{j} + y\mathbf{k}$ across the surface, S, which is a cone with parametric equations $x = u,\ y = u\sin v,$ and $z = u\cos v$ where $0 \le u \le 3$ and $0 \le v \le \dfrac{3\pi}{2}$.

TEE Question 2

Consider the vector field $\mathbf{F}(x,y,z) = x^2\mathbf{i} + z^2\mathbf{j} + 2y\mathbf{k}$ and the curve, C, which is the boundary of the part of the surface $z = 1 - xy^2$ over the rectangular region $0 \le x \le 3$, $0 \le y \le 2$, with positive (counter-SLOckwise) orientation when viewed from above. (See the plot below.) Use Stokes' Theorem to evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$.



TEE Question 3

Use the Divergence Theorem to calculate the flux of $\mathbf{F}$ across S where $(x,y,z) = (x^3 + y^3)\mathbf{i} + (y^3 + z^3)\mathbf{j} + (x^3 + z^3)\mathbf{k}$, and S is a sphere centered at the origin of radius 5. *Recall the identity* $x^2 + y^2 + z^2 = \rho^2$ *in the spherical coordinate system.*

TEE Question 4

a. Find a vector equation or a set of parametric equations for the tangent line to the curve $\mathbf{r}(t) = \langle t,\ t^2,\ t^3 \rangle$ at time $t = 2$.

b. At what point (x, y, z) does the line $\mathbf{r}(t) = \langle 2+t,\ 4+4t,\ 8+12t \rangle$ intersect the plane $x + 2y + 3z = 169$?

TEE Question 7

a. A projectile in flight experiences acceleration $\mathbf{a}(t) = \langle 0,\ 0, -32 \rangle$ with $\mathbf{v}(0) = \langle 5,\ 4,\ 1 \rangle$. Determine the speed at $t = 1/32$. Assume distance is measured in feet and time in seconds.

b. Starting at ground-level, a projectile travels along the path $\mathbf{r}(t) = \langle 2t,\ 4t,\ -16t^2 + 64t \rangle$ for $t \geq 0$. Set up the integral which calculates how far it has traveled between the time $t = 0$ and when it returns to the ground (e.g. the z component is 0). **DO NOT SOLVE.**

TEE Question 8

Evaluate the line integral along the curve C in the vector field $\mathbf{F}$

$$C: \quad \begin{aligned} x &= t + 1 \\ y &= 2t \\ z &= t + 2 \\ 0 &\leq t \leq 2 \end{aligned} \qquad\qquad \mathbf{F} = \langle yz,\ x + z,\ y^2 \rangle$$

TEE Question 10

Use the method of Lagrange Multipliers to find the shortest squared-distance between the point $(-1, 2)$ and the line $x + y = 5$ using the following information:

Objective Function: $(x+1)^2 + (y-2)^2$

Constraint: $x + y = 5$

MA205

AY25-1 Syllabus

Printed on: 11/13/2024

Textbook: Calculus - Ninth Edition, Early Transcendentals by James Stewart

Lesson	Date	Topic (Reading)	Homework (WebAssign)		Lesson Objectives
BLOCK 1: VECTORS AND GEOMETRY OF SPACE					
Lesson 1	8/19/2024	Course Intro & 3D Coord Systems (12.1)	**12.1:** 3, 7, 10, 37	**WA Due: Aug 25**	**1.1.** Understand how to model position and distance in 3D-space.
Lesson 2	8/20/2024	Geometry of Vectors (12.2)	**12.2:** 13, 21, 25 **Tech Lab 1 Prep Assigned**		**1.2.** Understand the properties of vectors: addition, scalar multiplication, length, magnitude, standard basis vectors, and position vectors.
Lesson 3	8/21/2024	Dot Product (12.3)	**12.3:** 8, 9, 12, 19, 41, 51		**1.3.** Solve for and understand the definition and utility of the dot and cross products for practical applications between two vectors.
Lesson 4	8/22/2024	Cross Product (12.4)	**12.4:** 1, 39, 41		**1.3.** Solve for and understand the definition and utility of the dot and cross products for practical applications between two vectors.
Lesson 5	8/23/2024	Equations of Lines & Planes (12.5)	**12.5:** 5, 27, 31, 71 **12.3:** 14 **Quiz 1**		**1.4.** Model and understand relationships between lines and planes in 3D-space. **1.5.** Determine the distance between different objects/points in 3D-space.
Lesson 6	8/26/2024	Vector Functions & Space Curves (13.1)	**13.1:** 1, 22, 49, 25, 27, 29, 30 **Applications Memo & Video Assigned**	**WA Due: Sep 4**	**1.7.** Model space curves in 3D using vector functions.
Lesson 7	8/27/2024	Derivatives & Integrals of Vector Functions (13.2)	**13.2:** 10, 21, 37, 43, 44, 49		**1.8.** Compute derivatives and integrals of vector functions. **1.9.** Compute the tangent line to a space curve and understand its significance.
Lesson 8	8/28/2024	Arc Length (13.3 - stop at Curvature p. 906	**13.3:** 4, 17 **Quiz 2**		**1.10.** Determine distance along a 3D-space curve.
Lesson 9	8/29/2024	Motion in Space 1 (13.4 - stop at "Tangential and Normal" p. 919)	**13.4:** 3, 11		**1.11.** Model the motion of an object in 3D-space.
Lesson 10	8/30/2024	Motion in Space 2 (13.4 - stop at "Tangential and Normal" p. 919)	**13.4:** 18, 23, 25		
Lesson 11	9/3/2024	**Technology Lab 1**			
Lesson 12	9/5/2024	**WPR 1**			
BLOCK 2: MULTIVARIABLE DIFFERENTIATION					
Lesson 13	9/9/2024	Multivariable Functions (Cartesian) (14.1)	**14.1:** 5, 21, 32, 34 **Tech Lab 2 Prep Assigned**	**WA Due: Sep 15**	**2.1.** Understand the definition of a function of two variables in cartesian coordinates and how it can be represented algebraically, numerically, graphically, or verbally.
Lesson 14	9/10/2024	Partial Derivatives (14.3 - stop at Partial Differential Equations p. 968)	**14.3:** 20, 22, 47, 82		**2.2.** Given a function of several variables, calculate its partial derivatives. **2.3.** Understand the relationship and differences between average rate of change and instantaneous rate of change of a multivariable function using partial derivatives.
Lesson 15	9/11/2024	Tangent Planes (14.4 - stop at Differentials p. 979)	**14.4:** 3, 6, 25, 27 **Quiz 3**		**2.4.** Given a function of two variables and a point, find the equation of the plane tangent to the surface at the given point. **2.5.** Approximate the value of a multivariable function using a linear equation.
Lesson 16	9/12/2024	Chain Rule (14.5 - stop at implicit differentiation p. 990)	**14.5:** 5, 7, 16, 44		**2.6.** Given a multivariable function and parameterizations for the function, use the chain rule to solve a rates of change problem.
Lesson 17	9/13/2024	Gradient and Directional Derivative I		**WA Due: Sep 24**	**2.7.** Understand what a directional derivative is in terms of a rate of change graphically, algebraically, and numerically. **2.8.** Given a function f(x,y) of two variables, find the directional derivative of f(x,y) at a given point in any direction. **2.9.** Given a function of two variables, find the gradient vector at a specified point. **2.13.** Employ technology to calculate, visualize, and understand the directional derivative and gradient of a multivariable function.
Lesson 18	9/16/2024	Gradient and Directional Derivative II	**14.6:** 1, 9, 13, 17, 30, 39		**2.10.** Understand that the gradient vector gives the direction of the greatest increase in functional value at a given point for a differentiable function. **2.11.** Understand that the magnitude of the gradient vector gives the maximum rate of change of a differentiable function at a given point. **2.12.** Understand geometrically how the gradient vector is related to level curves. **2.13.** Employ technology to calculate, visualize, and understand the directional derivative and gradient of a multivariable function.
Lesson 19	9/17/2024	Multivariable Optimization I (Min/Max) (14.7)	**14.7:** 3, 13*, 21*, 25* *Technology Problems **Quiz 4**		**2.14.** Employ technology to calculate, visualize, and understand optimization problems. **2.15.** Understand the definition of a critical point and be able to find the critical points of a function of two variables. **2.16.** Use the Second Derivatives Test to classify a critical point as a local maximum, local minimum, or saddle point
Lesson 20	9/18/2024	Multivariable Optimization II (Closed Interval Method) (14.7)	**14.7:** 38, 46, 47, 51, 54, 55		**2.17.** Find the absolute maximum and minimum of a given continuous function f(x,y) on a closed and bounded set using the Closed Interval Method for functions of two variables. **2.18.** Model and solve unconstrained and constrained multivariable optimization problems.
Lesson 21	9/19/2024	Lagrange Multipliers I (Single-Constraint) (14.8)			**2.19.** Solve optimization problems for functions of several variables subject to a single constraint using the Method of Lagrange Multipliers.
Lesson 22	9/20/2024	Lagrange Multipliers II (Multi-Constraint) (14.8)	**14.8:** 1, 5, 9, 33, 49		**2.20.** Solve optimization problems for functions of several variables subject to two constraints using the Method of Lagrange Multipliers.
Lesson 23	9/23/2024	**Technology Lab 2**			
Lesson 24	9/25/2024	**WPR 2**			

Printed on: 11/13/2024

AY25-1 Syllabus

Lesson	Date	Topic (Reading)	Homework (WebAssign)		Lesson Objectives
BLOCK 3: MULTIVARIABLE INTEGRATION					
Lesson 25	9/27/2024	Double Integration over Rectangular Surfaces I (15.1)		WA Due: Oct 6	**3.1.** Given a function f(x,y), use a double Riemann sum to estimate volume under a surface. **3.2.** Understand the concept of volume under a surface as the limit of the sums of volumes of rectangular columns.
Lesson 26	9/30/2024	Double Integration over Rectangular Surfaces II (15.1)	**15.1:** 1, 6, 7, 15, 19, 53, 42 **Tech Lab 3 Prep Assigned**		**3.3.** Use level curves to estimate the volume under a surface. **3.4.** Estimate the average value of a function of two variables.
Lesson 27	10/1/2024	Double Integration over General Regions (15.2)	**15.2:** 5, 7, 8, 20, 39		**3.5.** For any region in the xy-plane, sketch the region, label the boundaries, and label the points of intersection. **3.6.** Compute the iterated integral of a multivariable function over a specified domain.
Lesson 28	10/3/2024	Multivariable Functions (Polar) (10.3 - stop at Example 7)	**10.3:** 12, 21, 24		**2.1.** Understand the definition of a function of two variables in polar coordinates and how it can be represented algebraically, numerically, graphically, or verbally.
Lesson 29	10/7/2024	Double Integration in Polar Coordinates (15.3)	**15.3:** 6, 11, 30, 33, 45	WA Due: Oct 15	**3.7.** Understand and use the polar coordinate system to solve iterated integrals.
Lesson 30	10/8/2024	Center of Mass and Probability (15.4 - skip Moment of Inertia pp. 1072-1074)	**15.4:** 1, 3, 15, 29, 31, 32		**3.8.** Find the mass, moments, and center of mass of any lamina whose area is a specified region in the xy-plane and whose density is a given function ρ(x,y). **3.9.** Employ technology to find the mass, moments, and center of mass of any lamina whose area is a specified region in the xy plane and whose density is a given function ρ(x,y). **3.10.** Understand that a joint density function is a probability density function of multiple continuous random variables. **3.11.** Given a joint density function of a pair of random variables X and Y determine the probability that f(x,y) lies in a region D. **3.12.** Employ technology to create joint density functions and calculate probabilities. **3.13.** Verify if multivariable functions are valid joint density functions.
Lesson 31	10/10/2024	Cylinders and Quadric Surfaces (12.6)	**12.6:** 23-30 **Quiz 5**		**1.6.** Model quadric surfaces in 3D-space. **1.12.** Employ technology to visualize quadric surfaces.
Lesson 32	10/16/2024	Triple Integrals I (15.6)		WA Due: Oct 20	**3.14.** Evaluate a triple integral over a general region.
Lesson 33	10/17/2024	Triple Integrals II (15.6)	**15.6:** 4, 5, 9, 13, 18		**3.15.** Employ technology to compute triple integrals.
Lesson 34	10/21/2024	Triple Integrals in Cylindrical Coordinates I (15.7)	**Quiz 6**	WA Due: Oct 29	**3.16.** Understand when and how to apply the cylindrical coordinate system to solve iterated integrals.
Lesson 35	10/22/2024	Triple Integrals in Cylindrical Coordinates II (15.7)	**15.7:** 10, 15, 21, 32		
Lesson 36	10/24/2024	Triple Integrals in Spherical Coordinates I (15.8)			**3.17.** Understand when and how to apply the spherical coordinate system to solve iterated integrals.
Lesson 37	10/25/2024	Triple Integrals in Spherical Coordinates II (15.8)	**15.8:** 21, 23, 28, 29		
Lesson 38	10/28/2024	**Technology Lab 3**			
Lesson 39	10/30/2024	**WPR 3**			
BLOCK 4: VECTOR CALCULUS					
Lesson 40	11/1/2024	Vector Fields (16.1)	**16.1:** 7*, 19, 11*, 27, 13, 15, 16, 31, 33 *Technology Problems	WA Due: Nov 3	**4.1.** Understand vector fields and their different applications. **4.14.** Use technology to plot different types of vector fields.
Lesson 41	11/4/2024	Line Integrals I (16.2)		WA Due: Nov 11	**4.2.** Understand the meaning and compute the line integral along a curve in space.
Lesson 42	11/5/2024	Line Integrals II (16.2)	**16.2:** 3, 7, 17, 22, 41		**4.3.** Understand different applications of line integrals. **4.15.** Employ technology to pass a vector defined function to a function. **4.16.** Employ technology to visualize a particle's path along a vector field.
Lesson 43	11/7/2024	Fundamental Theorem for Line Integrals (16.3 - stop at Conservation of Energy p. 1150)	**16.3:** 1, 3, 4, 7, 18, 23		**4.4.** Understand and apply the Fundamental Theorem for line integrals.
Lesson 44	11/12/2024	Curl and Divergence (16.5 - stop at Vector Forms of Green's Theorem p. 1167)	**16.5:** 1, 3, 7*, 11, 18 *Technology Problem **Quiz 7**	WA Due: Nov 17	**4.6.** Understand the "del" operator and use it to compute curl and divergence. **4.7.** Understand the physical representation of curl and divergence. **4.17.** Employ technology to calculate curl and divergence of a vector defined function.
Lesson 45	11/14/2024	Green's Theorem (16.4)	**16.4:** 3, 5, 7, 9, 17		**4.5.** Understand and apply Green's Theorem to general 2D regions.
Lesson 46	11/18/2024	Parametric Surfaces I (16.6 - stop at Surface Area p. 1177)	**Quiz 8**	WA Due: Nov 21	**4.8.** Understand how vector functions can be used to model surfaces in 3D-space.
Lesson 47	11/19/2024	Parametric Surfaces II (16.6 - stop at Surface Area p. 1177)	**16.6:** 19, 20, 33, 35		
Lesson 48	11/20/2024	Parametric Surface Area (15.5, 16.6 - start at Surface Area p. 1177)	**16.6:** 40, 39, 45 **15.5:** 4		**4.9.** Determine the area of a parametric surface.
Lesson 49	11/22/2024	**WPR 4**			
Lesson 50	12/3/2024	Surface Integrals I (16.7)		WA Due: Dec 8	**4.10.** Understand the meaning and compute surface integrals of a scalar function over a given surface.
Lesson 51	12/4/2024	Surface Integrals II (16.7)	**16.7:** 23, 24, 26		**4.11.** Use the surface integral to determine the flux across a solid region.
Lesson 52	12/6/2024	Stokes' Theorem I (16.8)		WA Due: Dec 13	**4.12.** Understand and apply Stokes' Theorem to general surfaces.
Lesson 53	12/9/2024	Stokes' Theorem II (16.8)	**16.8:** 9, 10, 19, 21		
Lesson 54	12/11/2024	The Divergence Theorem I (16.9)			**4.13.** Understand and apply the Divergence Theorem to general solids.
Lesson 55	12/12/2024	The Divergence Theorem II (16.9)	**16.9:** 1, 3, 5, 7		
Lesson 56	12/13/2024	TEE Review			

UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB C AY2024 COURSE ASSESSMENT

DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

MADN-MATH 23 May 2024

MEMORANDUM THRU

COL Joseph Lindquist, Director, Service Program, Department of Mathematical Sciences, USMA, West Point, NY 10996

COL Michael Scioletti, Professor and Head, Department of Mathematical Sciences, USMA, West Point, NY 10996

FOR Office of the Dean, AARS, United States Military Academy, West Point, NY 10996

SUBJECT: MA205 Multivariable and Vector Calculus Executive Summary (AY24)

1. **Purpose:** The purpose of this memorandum is to assess the MA205 curriculum.

2. **Assessment Summary:**

MA205 Course Learning Outcomes (CLOs)	Strength of Attainment	Strangth of Evidence	Overall Evaluation
CLO 1: Demonstrate Good Scholarly Habits	C	B	B-
CLO 6: Apply Multivatie Integral CLO 7: Apply Multivariate Integral	C	B	B-

Figure 1: CLO Evaluations. Key: A = exceeds the standard; B = meets the standard; C = meets the standard with some weakness; D = fails the standard (can be addressed in the short-term); F = catastrophically fails the standard (requires immediate remediation)

3. **Assessment Discussion:** This section presents an overview of assessment findings for AY23. A detailed discussion of direct and indirect indicators can be found in the Course Design and Analysis Report located in Enclosure 4.

a. Status of significant changes: In AY23-2, we increased the number of projects from two, smaller projects, to four, one per block. In AY23-1, we assigned a project during block two that assessed a student's ability to apply and execute material from both blocks one and two. Likewise, we assigned a project during block three that covered the rest of block two and most of block three. Each project consisted of in-class work requirements as well as out-of-class group work. In AY23-2, we changed to four projects near the end of each block with the intent to reinforce the block's material prior to the in-class graded event. An unintended consequence was the creation of an internal "Thayer Week" consisting of two major graded assignments in one week, as a result of four projects. In 24-1, we changed to three projects, each approximately one week after the end of the first three blocks, and the fourth project day became an additional day for block four material, which is helpful, given the challenging nature of the vector calculus concepts. The placement of each project was a deliberate attempt to address the reinforcement idea from the previous semester but without the "Thayer Week" drawback. We assessed the block four exam as the graded event that, historically, students had performed worst on, so we assumed that the additional in-class-day for that exam would lead to better student performance. What we found, however, was there was not a significant change in student performance on the assessment. The average on WPR 4 for AY23-1 (no review day, two projects) was a 73.52%. The average on WPR 4 for AY23-2 (no review day, four projects) was a 74.34%. The average on WPR 4 for AY24-1 (review day, three projects) was a 73.89%. This is not conclusive evidence given the difference in student population, aptitude, and perceived exam difficulty. We will continue to consider modifications and trade-offs for the project component of the course in future semesters.

b. Strengths in the course: The course has a good structure with daily homework problems, suggested board problems, two quizzes, (sometimes a project), and a WPR reinforcing the learning objectives for each block. Students get exposed to the material multiple times throughout the course, as opposed to the structure of a midterm/final exam or even just purely a final exam. This construct provides continual engagement in the feedback-loop between instructors and students. The course naturally builds on itself, exposing the students to the material multiple times and generating more intuition within the fourth block, Vector Calculus (the hardest and most applicable block). Our student population consists of STEM majors, most of whom have yet to take a course in their major, and therefore it provides the foundation for their future in advanced STEM material. Another strength of our course is that we incorporate problems, examples, and projects from multiple disciplines to give the students exposure to the material they are likely to see later in their major and to help strengthen buy-in from the students as to why they are learning the material. If they can see that it is applicable to something in which they are interested we believe they are more likely to engage with and learn the material.

c. Areas for improvement in the course: The WebAssign platform provides the students with an out-of-class engagement forcing function and prepares students to "Demonstrate Good Scholarly Habits". However, we identified that it falls short of achieving a construct-centered approach presented by Messick [1]. The construct consists of three simple questions; (i) What KSAs are the targets of instruction (and assessment)? (ii) What learner actions/behaviors will reveal these KSAs?, (iii) What tasks will elicit these specific actions or behaviors? We seek to improve the methods for achieving each of these three areas. Essentially, how are we instructing students to demonstrate good scholarly habits?, what student actions demonstrate proficiency?, and what are we doing to elicit/evaluate those actions? While our students performed at the "meets the standard" level, we believe our teaching and assessment plan can improve in this area. Students struggle to progress through Kolb's four-stage learning model, given the amount of material covered in MA205. The first block, consisting of vectors and geometry of space (all single variable), should ease students into the complex nature of the material. However, the course then moves quickly through two taxing blocks, consisting of partial derivatives and iterated integrals. We then pivot back to vectors during block four which includes a culmination of the prior three blocks. In its current setup the course spends approximately six weeks on the entirety of vector calculus when most colleges/universities dedicate half of a semester or an entire semester to this challenging material. A method to address the quantity of information we ask our students to engage with is to split MA205 and MA364 into a three-course sequence, consisting of (i) Multivariable Calculus, (ii) Vector Calculus, and (iii) Differential Equations. We recognize that this would generate a significant amount of turmoil to the cadets' 8TAP, but it might give the students a better chance to retain the material to be able to use it later in their major. We intend to continue to assess other ways to ensure graduates of this course are capable of handling downstream material in their respective majors.

d. Changes to be implemented in AY24: Starting in AY24-2, the course will change from a 40-lesson, 75-minute/lesson structure to a 56-lesson, 55-minute/lesson structure. This is believed to increase student engagement by increasing the number of days in which the class meets. Our assumption is that students will prepare the same amount for a math lesson regardless of how long they are in the classroom. Thus, having more lessons should increase overall preparation for the course. We plan to test this hypothesis through surveys to the students. During AY24-1, we collected survey responses on student preparation for 75-minute lessons. During AY24-2, we plan to collect survey responses on student preparation for 55-minute lessons. The surveys are scheduled for similar lessons between the two structures to try and compare preparation for similar material.

e. Proposed changes to the CLOs: We plan to combine CLOs 6 & 7 and CLOs 8 & 9. These new CLOs will read: "Apply Multivariate Differentiation and Integration" and "Demonstrate the Calculus and Applications of Vectors". This change consolidates similar objectives under one CLO. We plan to adjust CLO 5 from "Develop Interdisciplinary Perspective" to "**Reinforce an** Interdisciplinary Perspective". This change addresses building upon interdisciplinary perspectives gained from MA103 and MA104. The full list of current CLOs can be found in Enclosure 1. The proposed changes are administrative in nature, are not substantive, and are unlikely to warrant a full curriculum change committee review.

4. The point of contact for this memorandum is MAJ Alex Withenbury, MA205 Course Director, at 845-938-8137 or alexander.withenbury@westpoint.edu.

Encl

JAMES C. GRYMES
LTC, AG/FA49
Associate Program Director

Enclosure 1: Full Set of Course Learning Objectives (CLOs)

1. At the end of the course, cadets will be able to...

CLO 1: **Demonstrate Good Scholarly Habits.** Students demonstrate good scholarly habits for progressive intellectual development.

CLO 2: **Communicate Effectively.** Students improve the quality of both their written and verbal communication of mathematics.

CLO 3: **Apply Technology.** Students are able to leverage technology when solving complex mathematical problems.

CLO 4: **Develop Problem Solving and Reasoning Skills.** Students develop the problem-solving and reasoning skills necessary for continued study in mathematics, the physical sciences, the social sciences, and engineering.

CLO 5: **Develop Interdisciplinary Perspective.** Students develop an interdisciplinary perspective that supports knowledge transfer across disciplinary boundaries to solve complex problems.

CLO 6: **Apply Multivariate Derivative.** Students apply the basic concepts of the multivariate derivative to solve multivariable unconstrained and constrained optimization problems.

CLO 7: **Apply Multivariate Integral.** Students apply multivariate integration to solve accumulation problems.

CLO 8: **Apply Vector Operations and Model Motion of an Object.** Students apply the basic properties and operations of vectors, quadric surfaces and space curves in three-dimensions, and model the motion of an object in three-dimensional space.

CLO 9: **Demonstrate Calculus of Vector Fields/Line and Surface Integrals.** Students demonstrate the calculus of vector fields, line and surface integral techniques, and the connections between these new types of integrals and the single, double, and triple integrals that are given by the higher dimensional versions of the Fundamental Theorem of Calculus.

2. Anticipated changes to CLOs: We plan to combine CLOs 6 & 7 and CLOs 8 & 9. These new CLOs will read: “Apply Multivariate Differentiation and Integration” and “Demonstrate the Calculus and Applications of Vectors”. This change consolidates similar objectives under one CLO. We plan to adjust CLO 5 from “Develop Interdisciplinary Perspective” to “**Reinforce an** Interdisciplinary Perspective”. This change addresses building upon interdisciplinary perspectives gained from MA103 and MA104.

Enclosure 2: Mapping of Academic Program Goals to CLOs

Academic Program Goals / M255 Assessment Techniques				
APG	1.) Communication	2.) Creative Thinking and Creativity	3.) Lifelong Learning	5.) STEM
WGCD	1.3.) Effectively convey meaningful information to diverse audience using appropriate forms and media.	2.4.) Reason both quanitatively and qualitatively.	3.1.) Demonstrate the willingness and ability to learn independently.	5.1.) Apply mathematics, science, and computing to model devices systems, processes, or behaviors
Core Math Goal	Communication	Technology Skills and Problem Solving	Habits of Mind	Interdisciplinary Problem Solving
MA205 CLO	2.) Communicate Effectively	3.) Apply Technology; 4.) Develop Problem Solving and Reasoning Skills 6.) Apply Multivariate Derivative; 7.) Apply Multivariate Integral; 9.) Demonstrate Calculus of Vector Fields/Line and Surface Integrals	1.) Demonstrate Good Scholarly Habits	5.) Develop Interdisciplinary Perspective 8.) Apply Vector Operations and Model Motion of an Object
Assessment	3x Projects (oral presentations)	4x WPRs and 1x TEE with leson objective mapping tied to CLOs; 3x Projects with Mathematica Requirements to assess students ability to leverage technology.	Nightly webassign homework; 8x quizes designed to test class preparation	3x Projects (interdeciplinary projects focused on applying calculus tecniquies to complex scenarios)

Figure 2: Assessment Crosswalk

Enclosure 3: Cyclical schedule for assessment of CLOs and Student Outcomes

WGCD	CLO	AY24	AY25	AY26
3.1	1	Assess	Implement	Plan
1.1, 1.2, 1.5	2	Plan	Assess	Implement
5.1, 7.1	3	Implement	Plan	Assess
2.1, 2.2	4	Implement	Plan	Assess
7.1	5	Implement	Plan	Assess
2.2, 2.3, 5.1	6	Assess	Implement	Plan
5.1	7	Plan	Assess	Implement

Figure 3: Three-year cyclic assessment schedule for CLOs

Enclosure 4: Course Assessment Plan and Overall Student Performance

Course Summary. There are a total of **40** required attendances consisting of 33 regular lessons, 3 project lessons, and 4 exams.

Course Evaluation Plan. See Table 1 for the distribution of course points.

Graded Event	**Points**
WPR 1	75
WPR 2	100
WPR 3	100
WPR 4	100
WebAssign 1	20
WebAssign 2	21
WebAssign 3	24
WebAssign 4	35
Project 1	30
Project 2	35
Project 3	35
Quizzes	75
TEE	300
Instructor Points	50
Total	1000

Table 1: Point Distribution

Grading. A summary of student performance in MA205 AY24-1 is found in Table 2.

Letter Grade	**Numerical Grade**	**% Students**	**Description**
A+	97-100	1.2	Beyond Expectations
A	93-96.99	5.5	Dominates the material
A-	90-92.99	9.7	Mastery
B+	87-89.99	10.6	Excellent performance
B	83-86.99	14.2	Good understanding
B-	80-82.99	13.6	Proficient; aptitude for the subject
C+	77-79.99	11.2	Can build on foundations
C	73-76.99	15.8	Passing; Proficient
C-	70-72.99	6.7	Short-range understanding
D	65-69.99	8.8	Marginal performance/some learning
F	< 65	2.7	Failed to demonstrate understanding

Table 2: Guidelines for Assigning Grades

Enclosure 4: Detailed Course Design and Analysis Report.
Continued on the next page.

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1 Introduction and Literature Review

The Department of Mathematical Sciences at the United States Military Academy (USMA) teaches multi-variable and vector calculus (MA205) course to all cadets that major in science, technology, engineering, and mathematics (STEM). We define multi-variable calculus in the context of MA205 as the following blocks of instruction: (i) vectors and geometry of space, (ii) multivariable differentiation, (iii) multivariable integration, and (iv) vector calculus. From USMA's academic catalog the course seeks to *"provides a foundation for the continued study of mathematics and for the subsequent study of the physical sciences, social sciences, and engineering."* (Office of the Dean, 2023).

We present our analysis over five sections. First, the remainder of this section addresses an in depth review of literature and comparable institutions' calculus curricula. Second, we present the historical context of MA205, focusing primarily on its evolution over the last decade and plans for the continued changes to the course. We then analyze MA205's curriculum and discuss how technology plays a role in the course.

1.1 Calculus

Calculus is taught in every institution across the country that has STEM disciplines. The D/F/W rates associated with calculus are near 40% (do we have a reference). It often serves as a filter course. Calculus was primarily introduced as a course for engineers in the 1950's, but has since promulgated to all STEM major pre-requisites. It sits in an odd space on the spectrum of Engineer to Mathematics. Mathematics is about discovery, while engineering is about applications. This dichotomy is open for interpretation, but the way in which Calculus is delivered to students varies from theoretical to computational. The what comes first the chicken or the egg example is often compared to Calculus in terms of what do students need to know first the theory or how to do the math. This makes teaching it very difficult. Most professors even within a mathematics department hold dissimilar viewpoints. In an age where nearly all high school students seek secondary education and fundamental skills vary greatly between them, calculus is at a crossroads. Some schools have considered not making it mandatory for certain STEM majors, while others have opted for a more accessible approach often using words like “business, or biological" calculus.

1.2 Pedagogical Review

The Mathematical Association of America (MAA) offers several reports and guides (MAA, 2015) related to curriculum design, examples of practices from various colleges and universities, as well as efforts to uniformly move higher-education calculus to modernized standards.

Saxe and Braddy (2015) discuss an impetus to change higher education in math due to various factors including student preparedness, technology advances, and quantitative skills demanded by other disciplines. The sections discussing course structure and workforce preparation encourage math instructors to move away from traditional lectures and use more hands-on instructive periods. Additionally, they suggest demonstrating techniques with real-world applications and reinforcing concepts using technology for visualization and experimentation. The MAA Press’s 2022 report on calculus sequence case studies (Johnson et al., 2022) presents a subset of results from data collection efforts beginning in 2010. The takeaways from different data sets are presented in approximately 90 case studies across five domains: rethinking student support programs; alternative course pathways; variations in approaches to the content of calculus; professional development and course coordination; and program assessment and the use of local data. Schumacher and Siegel (2015) emphasize that instructors should put more effort into active learning due to availability of extensive online resources. In fact, Abell et al. (2018) conclude the classroom practices section with *“. . . the learning of mathematical ideas is much more profound, . . . , when students are active participants in the learning process, ...”*

The MAA’s current stance is clear: instructors should adopt active learning strategies. The design of curriculums should feed the stakeholder’s purpose. As a service course, Calculus often has many stakeholders, but mostly STEM departments.

1.3 Comparative Analysis

We compare USMA’s MA205 description and syllabus to that of the U.S. Naval Academy (USNA) and the U.S. Air Force Academy (USAFA). Additionally, we perform comparisons with three liberal arts colleges based on U.S. News rankings (U.S. News & World Report, 2023) – Williams College in Amherst, MA (highest rank), Pomona College in Claremont, CA (lowest acceptance rate), and University of North Carolina (UNC)–Asheville in Asheville, NC (top-10 highest enrollment) – excluding the alphabetical and tuition-focused ranking schemes. We include acceptance rate under the assumption that low acceptance rates imply higher-class education. We also select high enrollment because all three of the U.S. service academies have the highest enrollment rates.

USNA provides Midshipmen “Calculus 3 with Vector Fields” (SM221) and “Calculus 3 with Optimization” (SM223) (Office of the Provost, 2023). Both courses combined cover the same topics as MA205, however Midshipmen only take one. The courses include the phrase “calculus 3” because USNA’s core curriculum presents calculus content spanning three separate semesters. All USNA Midshipmen must take either SM221 or SM223.

USAFA offers its Cadets “Calculus 3” (Math 243/253) which matches MA205 up to Green’s Theorem (United States Air Force Academy, 2022). Similar to USNA, this course is third in a sequence of calculus courses taught at the academy. Math 243/253 does not progress beyond Green’s Theorem

Of the three liberal arts colleges we considered, both Pomona and UNC-Asheville are similar to USNA in that they cover all or nearly all the breadth of our MA205 material across two courses (Pomona College, 2022; University of North Carolina - Asheville, 2023). Pomona College culminates with Stokes' Theorem while UNC-Asheville ends with coverage of the divergence theorem. Of note, Pomona College's vector calculus course, MATH067, requires students to complete linear algebra prior. The third college, Williams College, offers one similar course (MATH 151) which ends with Stokes' Theorem (Williams College, 2023).

1.4 Instructional Design and Student Experience

In their "Variations in approaches to the content of calculus" section, Johnson et al. (2022) offer four vignettes on four areas of pedagogical emphasis. The first vignette begins with a course structured in a lecture-lab style. The school's "calculus 2" program appears to follow a four-day-per-week theoretical emphasis with a single lab day focused on applying lecture material to problems. The fourth vignette, following a description of how the school flipped the traditional calculus 1 sequence, emphasized their program's use of graphical tools.

The MAA also published work which purports to establish *"... who takes calculus 1, ... "* and identifies *" ... institutional practices that contribute to the retention of STEM students"* (Bressoud et al., 2015). Although the subject does not align with ours -– calculus 1 versus MA205 — the publication offers some insight into the distribution of theory, to numerical, and to applications based on survey findings. In a survey of faculty teaching calculus 1, 10-percent of instructors focused on proofs and justifications, 20-percent focused on problems on graphical interpretations, 20-percent wanted students to learn through repetition via word problems, and 50-percent chose to focus on techniques to implement computations.

Other resources concerning the distribution of student focus comes from the University of Twente in the Netherlands (Craig et al., 2022) and Carroll College in Helena, MT, USA (Sullivan and Melvin, 2016). In the former, the authors discuss their adaptation to online resources by designing lectures and weekly quizzes towards theory and four-hours-per-week of hands-on learning exercising the mechanics of solutions and using different 3D graphing software to explore vector calculus topics. The latter authors, whose multivariable course primarily feeds higher-level engineering courses, place heavy emphasis on mathematical modeling and computation, using software to accelerate/enhance student learning.

2 Background

Over the previous six years, MA205 undertook several changes, initiated by both program leadership and institutional demand. They are: removing MA205 from the core curriculum; redesigning MA205 from a 55-minute to 75-minute instruction period; changing graphical software; and introducing spatial geometry and vector calculus into MA205's curriculum.

2.1 Core Curriculum Redesign

The MA205 AY17-1 executive summary (DeGregory, 2017) contains a single sentence regarding its leadership's responsibility for evaluating what graduates can do (i.e., the USMA Academic Program Goals): *"No longer required for MA205 with removal of MA205 from core curriculum"*.

Horton (2015) references recent changes to USMA's core curriculum which, at the time, would result in many students not taking MA205. Subsequently, MA104 and MA205 exchanged course material in order for students to meet core goals of both D/MATH and USMA academics. In addition to altering the specific topics within MA205, Horton also noted MA205's removal from USMA's core curriculum would decrease course credit hours from 4.5 to 4 as well as, naturally, reducing future student annual enrollment by approximately 60–70-percent.

2.2 Course Time Change

In AY18, MA205 consisted of fifty-six 55-minute lessons. This changed in AY19-1 when MA205 transitioned to forty 75-minute lessons (Evans, 2019). The opportunity to teach in a 75-minute instruction period was new to USMA. Colloquially, the math department needed to test some courses using the new 75-minute class setup. MA205 was a good candidate to test this new setup for reasons stated in the previous year's executive summary (Kuiper, 2018). The reasoning for this change focused on the gains of

> *"... consolidating our lessons into more dense and efficient blocks of instruction, with less start-up time per lesson, allowing for students to prepare more thoroughly for classes and assessments, and enabling instructors to prepare and deliver higher quality lessons."*

The "higher quality lessons" refers to allowing instructors to "go deeper in[to] individual topics during extended lesson periods." At the time, there was a slight increase in student performance following the transition from 55-minute classes to 75-minute classes. The change between AY18-1 and AY19-1 was not statistically significant, however, and student performance on the TEE has remained nearly constant (within one standard deviation) each semester regardless of class length. AY17-01 (55-minute classes) and AY23-1 (75-minute classes) were uncharacteristically high-performing semesters while AY21-1, AY21-2, and AY22-1 (all 75-minute classes) were all uncharacteristically low-performing semesters. Because most semesters' TEE performance remains within one standard deviation of the mean (with exception of the 5 semesters just listed), it does not show a statistically significant difference based on the change of class length. However, in an effort to test this theory, the course will return to 55-minute classes in AY24-2.

2.3 Math Curriculum Redesign

The AY22-02 end-of-course brief also provides insight into the fourth major change to MA205: the curriculum change. MA205's curriculum for AY23 would shift to *"... traditional Calculus III content"* (Trujillo et al., 2023a), replacing instruction on both sequences/series and differential equations with geometry of space and vector calculus.

The origins of the redesign stem from a briefing by the MA205 and engineering mathematics (MA364) leadership to the USMA curriculum committee towards the end of AY22 (Trujillo et al., 2022). From the MA205 perspective, this change was necessary to expand more fully on the subject of calculus while also going more in depth in certain lessons. AY23 was the first full year implementing the new curriculum and this will continue into AY24.

MA205's identity has changed significantly over the years, however there has been a constant of supporting the core goals of D/MATH and USMA's academic program goals (APGs). The Core Math Book (Lindquist, 2021) elaborates on both sets of goals. With these in mind, updated MA205's course learning outcomes (CLOs), per Cammack et al. (2023), are: Demonstrate good scholarly habits; Communicate effectively; Apply technology; Develop problem solving and reasoning skills; Develop interdisciplinary perspective; Apply multivariate derivative; Apply multivariate integral; Apply vector operations and model motion of an object; and Demonstrate calculus of vector fields/line and surface integrals.

Currently, the course is divided into four blocks of instruction. Block one establishes a basis of three-dimensional coordinate systems and the mathematical objects used to explore this space – vectors. After setting vectors as the fundamental means with which we explore the nature of various manifolds and motion in 3D space, we move to the second block.

Block two follows the traditional calculus practice of introducing differentiation before integration. We allow for multivariate components of vectors and implement partial differentiation. We end the second block with three lessons of applications in the optimization field, including the Method of Lagrange Multipliers.

Block three introduces multivariable integration. A significant challenge for students in this block is the introduction to the concepts of changes of basis to different coordinate systems (i.e. cylindrical and spherical coordinate systems). Applications of this block include two-dimensional center of mass and probability of a joint density function.

With the formal language of vectors and multivariable differentiation/integration established, block four introduces vector calculus. Beginning with vector fields and line integrals, we move through curl and divergence and surface integrals to the culmination with Stokes' and Gauss' Theorems.

As of July 2023, 38 courses list MA205 as either a pre-requisite or co-requisite (Office of the Dean, 2023). Table 1 aggregates these courses into their associated departments.

Department	Number of Courses
Electrical Engineering and Computer Science	4
Geography and Environmental Engineering	1
Mathematical Sciences	19
Civil and Mechanical Engineering	7
Physics and Nuclear Engineering	3
Social Sciences	1
Chemistry and Life Science	3

Table 1: *USMA academic departments and the number of courses which require MA205 as either a pre-requisite or co-requisite.*

While tempting to align MA205's content to the needs of Table 1's individual courses, *"MA205's material should not be determined based on the demands of courses which list it as a prerequisite"* (Evans, 2019).

AY24-1 will be the third iteration of the new curriculum change including vector calculus. The sequencing of courses between MA205 and MA364 does not seem to negatively affect students as they progress in their majors. Most students in our course take MA205 in the fall of their Yearling year. Almost all of our students are taking one or two courses in their major, usually an introductory course not listing a mathematics course as a pre-requisite. Therefore, receiving instruction on sequences/series and differential equations later in their major is not a negative effect. During the curriculum change, it was identified that some students taking EE301 Fundamentals of Electrical Engineering would need the method of integrating factors, which was added to the end of MA104 starting last semester.

Vector calculus is, by far, the hardest block of material we teach in our course. On average, students do worse on blocks one and four compared to the course average and averages for blocks two and three. Some majors, i.e. Economics and Engineering Management, may never use vector calculus in their majors and may not go on to take MA364 (some major "tracks" require it and other majors offer it as an elective or Complementary Support Course). However, those majors do not necessarily need the "old" MA205 material (sequences/series and differential equations), so the change of material is somewhat irrelevant to them. Historically, students did similarly poorly on those blocks of material anyway (sequences/series and differential equations), so it seems irrelevant as to what that material is beyond multivariable differentiation and integration.

2.4 Supporting APGs

We focus on two specific APGs and how MA205's AY24-1 plan supports them. *APG 1: Graduates communicate effectively with all audiences*. We will distribute three group projects in AY24-01. They resolve to a final product of an oral presentation with supporting visual aides. These projects will aid Cadets' development of oral communication skills to both technical and non-technical target audiences.

APG 7: Graduates integrate and apply knowledge and methodological approaches gained through in-depth study of an academic discipline. The fourth block of instruction has been the most challenging for students since vector calculus was incorporated. The logical flow from block one to block three leads MA205 Cadets to leverage all they know to tackle problems associated with Stokes' Theorem and the divergence theorem. This practice is a localized version of APG 7's vision of graduates using what they already know to handle new challenges.

2.5 Cadet Requirements and Support

MA205 is a four-credit-hour course with 40 lessons bringing a total student commitment of approximately 162.5 hours as a combination of in- and out-of-class work. We expect students to work outside of the course in accordance with the following breakdown:

$$(33\ standard\ lessons) \times (1.25\ hours\ per\ lesson) \times (2\ hours\ per\ contact\ hour)\ =\ 82.5\ hours$$

$$(3\ projects) \times (approximately\ 4.67\ hours\ outside\ class\ per\ project)\ =\ 17.5\ hours$$

$$(4\ WPRs) \times (2.5\ hours\ of\ preparation\ per\ WPR)\ =\ 12.5\ hours$$

Adding these results, we expect Cadets to devote 112.5 hours outside of class to MA205 preparation. Additionally, incorporating the time for 40 lessons (1.25 hours each), total student commitment is approximately 162.5 hours. For a four-credit-hour course, the Academy requires 160 total hours of student commitment.

To facilitate preparation, MA205 makes the syllabus and calendar available to all students on Canvas, as well as Mathematica-based technology guides to facilitate the projects. Outside of MA205, there are numerous free resources for Cadets to exploit. Free, course-related content may be found on YouTube, as well as Khan Academy (Khan, 2023) and MIT's OpenCourseWare platform (Auroux, 2006, 2010).

USMA's Center for Enhanced Performance (CEP), in addition to tutoring services, has, cumulatively, 16 historical quizzes, WRITs, and WPRs available on their website. Of course, only some of the CEP's historical material maintains relevance due to curriculum changes discussed in Section 2.3.

In AY24-2, the course will pilot a 55-minute version instead of the traditional 75-minute classes to see if it improves student performance. In preparation for this change, the previous MA205 leadership team looked at the literature on changing the amount of time in the classroom each class period. Pedagogical literature is mixed on the subject. There is no definitive answer whether longer classes are more effective or more frequent classes are. The literature *does* support that two or three meetings per week is better than one meeting. However, other than short weeks (i.e. Thanksgiving or the week prior to TEEs), there are no weeks with only one class meeting regardless of class length. In 2019, the Department of Civil and Mechanical Engineering (CME) did a study on the effect of changing from thirty 75-minute classes to forty 55-minute classes for four of their mechanical engineering courses. There was no statistically significant change in the amount of prep time per class or performance of the average student between the two types of class design. There were, however, marginal differences in cumulative prep time across the semester (Rigney et al., 2019).

To assess how MA205 students respond to this type of change, we should, at a minimum, attempt to collect the same information as CME, using time-survey data. For their study, they had ten years of time-survey data to compare to, where we can, at most, collect that data in AY24-1 to compare to subsequent semesters. We could also analyze the impact on instructors having to teach more frequently. We would not necessarily focus on their workload but on their ability to provide additional instruction or adequately prepare for each lesson if they are teaching every day as opposed to every other day. As we would with any change, we would also look at student performance, although it is not always a good measure of the efficacy of such a change. Finally, through our mid-course and final surveys, we could ask questions of the students about how they feel their preparation/performance differs in a 55-minute class vs. a 75-minute class.

3 Experiment or System Design

In AY24, MA205 will continue the practice of utilizing Mathematica for computation and visualization. We also modify the course assessment methods by reducing the number of projects. This reduction from four to three projects stands in addition to numerous other means of student assessment which largely have not changed since AY23. Also, we discuss the option for MA205 to return to a 55-minute course.

3.1 Technology

The course executive summary for AY19-02 (Evans, 2019) sets forth, among various reflections, an intent to *"... enhance the MATLAB Tech Guide to establish a solid foundation for cadets to rely on for solving problem[s] ...."* Within this same report, LTC Evans discusses MA205's partial adoption of MATLAB (MathWorks, 2023) due to demand by the departments of civil and mechanical engineering (CME) and electrical engineering and computer science (EECS), whereas students from the previous two years only utilized Mathematica (Wolfram, 2023).

Over the next two years, our pilot program exposed students to both languages where they could choose either for coursework. After studying student performance on course-wide graded events and feedback about both softwares, the AY22-02 end-of-course brief (Trujillo et al., 2023a) recommended future MA205 courses return to solely Mathematica. In AY23, MA205 solely utilized Mathematica to fill the gap of rapid computation and visualization.

MATLAB was the computation platform used in MA205 at the request of a few departments that use it in their majors. For two semesters, under the previous leadership, the decision to keep MATLAB or move to Mathematica, which is familiar to cadets from MA103 and MA104, was considered. MATLAB served approximately 21% of the students, to include mechanical, chemical, and electrical engineering students (hard STEM); leaving 79% of our students, to include systems engineering, business management, and economics (soft STEM) students who will not use MATLAB again after the completion of MA205. This roughly 80/20 split was not the reason the course changed to Mathematica. The decision to change came from an assessment of the two platforms run concurrently between different sections in AY22-02. The assessment involved instructors using MATLAB or Mathematica based on their section composition (hard or soft STEM). Soft STEM sections used Mathematica and hard STEM sections used MATLAB. Across the course we had instructors that used Mathematica or MATLAB exclusively and the Program Director had one section of each. We assessed the use of technology on the same problem sets and project. After the project and problem sets, we surveyed the students on how comfortable they were using MATLAB and how comfortable they were using Mathematica. Of the students who responded to the survey, 27.4% were somewhat or very comfortable with multivariable differentiation and integration using MATLAB. Of the students who responded to the survey, 61.3% were somewhat or very comfortable with multivariable differentiation and integration using Mathematica. Students found Mathematica easier to use due to its use of symbolic pallets, and hence did not feel a barrier to learning another language to overcome while they were trying to take a partial derivative or integrate a triple integral, for example. From a survey with our instructors and instructors from other departments, many felt it was more important for students to understand multivariable calculus concepts than just know a computer program. Upon further exploration, Mathematica has a wider range of visualization options, allowing students to see more interactive and intuitive exploration with multivariable concepts. Of note, for the sections that used MATLAB the technology guide had examples in both Mathematica and MATLAB to help bridge the gap between Mathematica, which is used in previous math classes, and MATLAB which was new to most students. A major improvement in AY23-01 was our technology guides for Mathematica, especially for the visualization with triple integrals. Students were able to change a variable parameter while holding other variables constant and could see how changing their limits of integration changed the region of integration. In AY23, MA205 solely used Mathematica for its graphical visualizations, calculations, and projects (Cammack et al., 2023).

Through AY24, we plan to continue to use Mathematica for visualization in class and for the students to evaluate more complicated expressions. Mathematica provides a better user experience for our population and requires less onboarding than switching to a new language given their familiarization with it from MA205.

3.2 Assessments

Currently, the assessment and feedback plan for each block during AY24-01 for the students is one WPR, one project, two quizzes, and daily homework per block. The two quizzes happen midway through each block on identified difficult topics. Each quiz has one question similar to what students would see on a WPR. The idea is to give students an idea of the type of question and how well they understand the material before they go into the WPR. Each day, we have homework due (through WebAssign) at midnight the day of instruction. We encourage students to, at a minimum, attempt the homework prior to class to be able to ask relevant questions in class. Block four, because it goes through lesson 40, does not have a project to try and prevent too much work right before the TEE.

In AY23-01, the course had two projects: one in block two that covered block one and some of block two and the second project in block four that covered, mostly, block three material. We felt that since projects can help reinforce learning and extend concepts beyond what is taught in the classroom, we changed the course for AY23-02 to have four projects, one project per block and manage the time required per project to be the same amount of work across the semester (four small projects as opposed to two large projects). However, this created a sort of internal "Thayer week" for our students since we had a project due the lesson prior to the WPR. Instead of helping to reinforce the learning and help the students study for the WPR, it caused more stress and work for the students for just our class (regardless of what was happening across their other classes).

In AY24-01, we will have three projects, somewhat disconnected from the WPRs. Rather than having the projects due the lesson prior to the WPR, we have the projects due approximately one week after the WPR but on the material from the previous block. The idea behind this is to avoid the "Thayer week" while still reinforcing and extending material through the use of a project. Through our mid-course and final surveys, we will assess how students feel that change of project execution helped with their work load and learning of the material. This should help inform future semesters of MA205.

Homework questions can be answered in class at the instructor's discretion. We encourage our students to ask questions on the homework to drive the in-class material, rather than force them to go to AI. Quizzes are graded in class following execution. Right after the quiz is complete, we show the rubric on the screen and have the students write themselves feedback as to where they went wrong on the problem. This saves instructors some grading time and also allows the students to immediately see what they did well/poorly and write themselves feedback based on what they need to know rather than what instructors might say. WPRs are graded together, similar to core courses, and returned usually within two lessons of the exam. Finally, projects are graded on-the-spot with verbal feedback given following the cadets' presentation, and written feedback (including the grade) is given as soon as possible by the instructor. We have all groups present on the same day and short feedback is given to the group following their presentation with more in-depth feedback given later.

3.3 Time Change

In AY24-2 MA205 will transition from a 40-lesson 75-minute course to a 56-lesson 55-minute course. We completed a rough draft of this schedule at the end of AY23-02 which compares almost exactly to the advanced multivariable calculus course (MA255).

The time change results in two main benefits for students. First, students will engage with the material more often and for more duration. The current modality often results in engagement breaks of up to six days which is detrimental to both retention and recall. Second, the 55-minute modality provides course leadership with more lesson flexibility. With increased lessons at shorter times there exists more degrees of freedom to explore with different learning experiences and assessments. Students consistently struggle with the vector calculus block since implementing the new curriculum. In preparing the draft calendar, we spread certain topics across multiple days to give students more direct exposure to challenging material. AY23-2 leadership generally agreed that 75-minute courses facilitated student learning by allowing time for students to struggle with material in class and then work through one or two examples in a non-rushed manner (Trujillo et al., 2023b).

4 Assessment and Evaluation Plans

We discuss our plans to assess and evaluate MA205 during AY24. This relies on deliberate data gathering efforts tied to the three main efforts discussed in Section 3: technology, assessments, and time change. For each effort, we detail the data plan and discuss how we use this data to evaluate our designs.

4.1 Technology

We will continue to assess how well technology as a whole and specifically Mathematica helps cadets visualize multivariable calculus concepts and their understanding of the material. We have data from previous semesters related to this topic and will continue to ask of the cadets how comfortable they feel executing certain tasks in Mathematica through the mid-course and final surveys.

4.2 Assessments

Evaluation of our assessments at the course level will be focused on how we execute projects. Previous course leadership has assessed the efficacy of quizzing throughout the semester and the correlation between homework grades in WebAssign and performance on WPRs and the TEE. Since the largest change we have implemented in the course since the most recent restructure has been on the number of projects, that will be our focus of this AY. To assess our projects, we will include questions to the cadets in the mid-course and final surveys related to how the projects improved their understanding of the material. We will also get feedback from instructors in the course regarding the execution of projects and subsequent grading (i.e. oral vs. written and is grading projects on-the-spot the most effective way to provide feedback).

4.3 Time Change

To support the final decision of whether MA205 will return to a 55-minute instruction period in AY24-02, we rely on comparative measurements of our historical data as well as best practices from literature.

The many changes of MA205 over the years makes a simple before-and-after comparison of student performance difficult. Ideally, the AY19-1 change to 75-minutes and its implementation over the previous four years yields the data we need to support a decision. Unfortunately, the change in MA205 curriculum and a move to remote learning during the coronavirus pandemic during AY20/21 render this difficult. This being said, it is still possible to make use of data from this time frame and the upcoming semester.

Discussed in their general considerations, Trujillo et al. (2023b) allude to two measurements to resolve this course design question: conducting longitudinal studies of student performance in MA205 follow-on courses and student preparation. We may thus review follow-on course grades since the 75-minute course implementation and compare it to the upcoming 55-minute results; but this requires MA205 to be a 55-minute course for at least through AY25. We can also implement the apparently successful student peer-grading on self-preparation which was abandoned when the coronavirus pandemic shifted MA205 to remote learning.

5 Results

5.1 Technology

Cadets continue to struggle with the use of Mathematica to aid in their understanding of multivariable and vector calculus. Unfortunately, we do not have time during lessons to go through much use of Mathematica to help the students visualize the material or help solve complicated problems. Instead, we provide students with "technical guides" per block with working code to execute "every" command related to that block's material (sometimes needing to be supplemented with additional commands). We then expect the students to utilize Mathematica on projects as a way to deepen their understanding of the material. Repeatedly, to include AY24-1, students respond that they struggle with using Mathematica and that one of the most frustrating, and time-consuming, parts of project work is trying to use Mathematica. Rather than Mathematica being a tool to help the students better understand the material, visualize objects that they may otherwise not be able to, or execute tasks that would be tedious by hand (as most computer solftware should), it instead stands as a barrier to completing projects and projects become yet another boring task that must be completed.

We still believe that technology is important in the understanding of calculus. We still believe that Mathematica is the right tool to deepen that understanding. In the time-constrained environment that we have for MA205, where we are trying to teach differentiation, integration, and vector calculus in one semester, the use of technology becomes almost an afterthought. If the goal is to teach this material so that our students are better prepared for follow-on courses in their STEM major, we determine that it is more important for the students to understand the concepts rather than understand how to code the concepts into Mathematica. As a result, most of our work is done by-hand, and the use of technology is relegated to projects, again often as a frustration instead of a tool. We feel this is the best way to present the material for our student population and the only way we can incorporate technology in the course as it is currently structured.

5.2 Assessments

Our focus on assessments for this AY was on projects. Our students, on the aggregate, think that projects are overall helpful for their understanding of the material. Some of the feedback from students centered around the complexity of the projects. To instructors/course leadership, projects are a good way to deepen students' understanding by taking the concepts "one step" further than what the students have seen in homework problems, board problems, or even WPR problems. Projects are designed to take longer than, for example, a problem set, with the hope that through increased complexity, it develops a deeper understanding of the material. From the student perspective, however, some view projects as unnecessarily difficult. Rather than a slight extension from what they have worked before (two dimensions to three dimensions, for example), the projects seem like they are completely new information or significantly more difficult.

Although transparent to the students, we feel the change in timing for projects was an improvement. It forced a small amount of reach back to the material by moving approximately one week after the WPR.

And again, it removed some of the self-imposed "Thayer week" to the students. Some students, however, instead of realizing that they didn't understand a concept during the WPR and taking the time to learn it better to apply it on the project, instead just did poorly on that concept twice because they felt it beyond their comprehension.

Because our course occurs before most of our students have gotten far into their majors, we could use projects as an opportunity to expose our students to material related to their major. Having projects with a theme of civil engineering or economics, for example, could increase buy-in from students in those majors and give them a taste of what to expect from future courses. Often times, especially with STEM majors, students may not know exactly what to expect from topics in their major, and our class can be an opportunity to expose to them that information and how the math that we are learning is relevant to what they will do. Historically, we have sought increased participation from instructors in the course by having instructors outside of the course leadership participate in project creation. In AY24-1, we had the intent to incorporate multi-discipline projects, but did not place as much emphasis on diversifying the subjects as we could have. In future semesters, we could have a deliberate plan on subjects/disciplines for each project and then brainstorm specific ideas within that as the semester progresses.

5.3 Time change

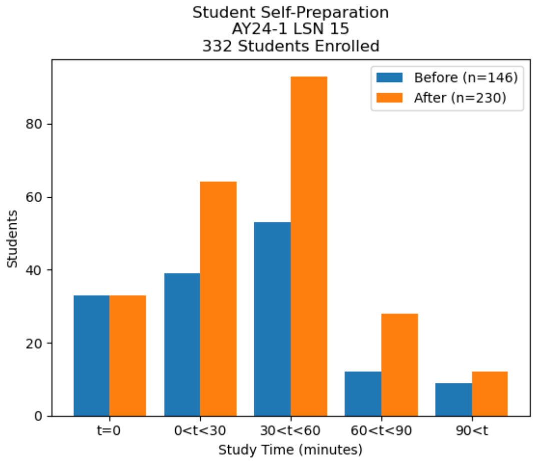
In AY24-1, we conducted six student surveys focused on student study habits both before and after lessons. We chose three lessons which retained a single day of class exposure for students, meaning the lesson was presented in one day in a 75- and would also be a single day in the 55-minute format. The method of collection was Microsoft forms and consisted of two questions. For the pre-class survey, these were

1. "Approximately how much time did you spend preparing for this lesson before class?"
2. "What is your MA205 section?"

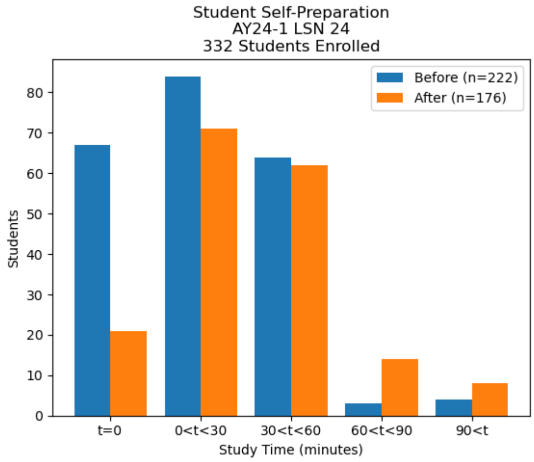
For the time-oriented question, students could respond by bubble selection of the following binned times

- "I did not prepare ahead of time."
- "Less than 30 minutes."
- "Between 30 and 60 minutes."
- "Between 60 and 90 minutes."
- "More than 90 minutes."

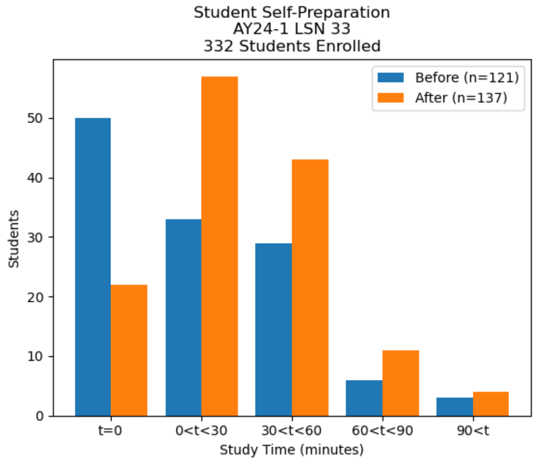
Similar questions followed for students in the follow-up survey presented after the lesson at the start of the following day. Surveys were administered by instructors utilizing QR codes and were also made available through the Canvas LMS as a course-wide announcement. Table 1 shows results from the AY24-1 surveys.



***(a)** AY24-1 LSN 15: MV Optimization 2 (CIM)*



***(b)** AY24-1 LSN 24: Triple Integrals*



***(c)** AY24-1 LSN 33: Green's Theorem*

***Figure 1:** Student self-professed study habits before (blue) and after (orange) lessons with a 1-to-1 coverage day mapping from AY24-1 to AY24-2.*

We keep the AY24-1 study habits as an expected baseline for the next on-cycle of MA205 in AY25-1. As of this writing, the AY24-2 self-study survey have not yet been administered. Interestingly, we do observe a trend in Fig. 1 of students spending less time studying before class indicated by the accumulation of blue columns towards the left side of the chart.

Because the material covered across MA205 did not fundamentally change over AYs 23 and 24, WPR performance is within the scope of course comparison. After administering the AY24-2 WPR1, the following summary statistic was included in the post-WPR report

AY	Mean Grade
23-2	64.03
24-1	74.21
24-2	69.95

***Table 2:** Mean WPR1 performance across previous three semesters.*

Except for two questions in the AY24-2 WPR1 which did not have a directly comparable question in previous exams, the results in Tab. 2 do not permit assessing the 55-minute format had a negative impact on material covered in the first block.

6 Conclusion

MA205 is a challenging course to cover so much calculus information in one semester. In a school that only offers two calculus courses, for some students who are already taking over 20 credit hours of courses each semester, we offer a deliberate plan to present the material to our students. Our focus for AY24-1 was to assess the quality of how we present projects to the students and AY24-2 was the best time structure to deliver the material (55-minute classes vs. 75-minute classes). After a relatively drastic restructure of the course for AY23-1, we felt our focus should be on refining the quality of instruction and methods for delivery rather than on content of the course.

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MA205: Multi-Variable and Vector Calculus
AY24-01 Syllabus

Printed on: 8/15/2023

Textbook: Calculus - Ninth Edition, Early Transcendentals by James Stewart

Lesson	*Date*	*Topic (Reading)*	*Homework (WebAssign)*	*Lesson Objectives*
BLOCK 1: VECTORS AND GEOMETRY OF SPACE				
Lesson 1	8/14/2024	3D Coord Systems & Geometry of Vectors (12.1 - 12.2)	**12.1:** 7, 10, 37, 39 **12.2:** 13, 21, 25	1.1. Understand how to model position and distance in 3D. 1.2. Understand the properties of vectors: addition, scalar multiplication, length, magnitude, standard basis vectors, and position vectors.
Lesson 2	8/16/2024	Dot and Cross Product (12.3 - 12.4)	**12.3:** 8, 9, 12, 19, 41, 51 **12.4:** 1, 39, 41	1.3. Solve for and understand the definition and utility of the dot and cross products for practical applications between two vectors.
Lesson 3	8/21/2024	Eqns. of Lines & Planes/Quadric Surfaces (12.5-12.6)	**12.5:** 5, 27, 31, 71 **12.6:** 23-30 **12.3:** 14 **Quiz 1**	1.4. Model and understand relationships between lines and planes in three-dimensional space. 1.5. Determine the distance between different objects/points in three-dimensional space. 1.6. Model quadric surfaces in three-dimensions. 1.12. Employ technology to visualize quadric surfaces.
Lesson 4	8/23/2024	Vector Functions & Space Curves (13.1)	**13.1:** 1, 22, 49, 25, 27, 29, 30	1.7. Model space curves in three-dimensions using vector functions.
Lesson 5	8/25/2024	Derivatives & Integrals of Vector Functions (13.2)	**13.2:** 10, 21, 37, 43, 44, 49 **Quiz 2**	1.8. Compute derivatives and integrals of vector functions. 1.9. Compute the tangent line to a space curve and understand its significance.
Lesson 6	8/28/2024	Arc Length & Motion in Space 1 (13.3 - stop at Curvature p. 906 13.4 - stop at "Tangential and Normal" p. 919)	**13.3:** 4, 17 **13.4:** 3, 11	1.10. Determine distance along a three-dimensional space curve. 1.11. Model the motion of an object in three-dimensional space.
Lesson 7	8/30/2024	Motion in Space 2 (13.4 - stop at "Tangential and Normal" p. 919)	**13.4:** 18, 23, 25	1.11. Model the motion of an object in three-dimensional space.
Lesson 8	9/5/2024	**WPR 1 (In-class)**		
BLOCK 2: MULTIVARIABLE DIFFERENTIATION				
Lesson 9	9/7/2024	Multivariable Functions (10.3 - stop at Example 7, 14.1)	**10.3:** 12, 21, 24 **14.1:** 5, 21, 32, 34	2.1. Understand the definition of a function of two variables in cartesian and polar coordinates and how it can be represented algebraically, numerically, graphically, or verbally.
Lesson 10	9/11/2024	Partial Derivatives/Chain Rule (14.3 - stop at Partial Differential Equations p. 968; 14.5 - stop at implicit differentiation p. 990)	**14.3:** 20, 22, 47, 82 **14.5:** 5, 7, 16, 44	2.2. Given a function of several variables, calculate its partial derivatives. 2.3. Understand the relationship and differences between average rate of change and instantaneous rate of change of a multivariable function using partial derivatives. 2.6. Given a multivariable function and parameterizations for the function, use the chain rule to solve a rates of change problem.
Lesson 11	9/13/2024	**Project 1**	**Project 1 (before-class work)**	
Lesson 12	9/15/2024	Tangent Planes (14.4 - stop at Differentials p. 979)	**14.4:** 3, 6, 25, 27 **Quiz 3**	2.4. Given a function of two variables and a point, find the equation of the plane tangent to the surface at the given point. 2.5. Approximate the value of a multivariable function using a linear equation.
Lesson 13	9/18/2024	Gradient and Directional Derivative (14.6)	**14.6:** 1, 9, 13, 17, 30, 39	2.7. Understand what a directional derivative is in terms of a rate of change graphically, algebraically, and numerically. 2.8. Given a function f(x,y) of two variables, find the directional derivative of f(x,y) at a given point in any direction. 2.9. Given a function of two variables, find the gradient vector at a specified point. 2.10. Understand that the gradient vector gives the direction of the greatest increase in functional value at a given point for a differentiable function. 2.11. Understand that the magnitude of the gradient vector gives the maximum rate of change of a differentiable function at a given point. 2.12. Understand geometrically how the gradient vector is related to level curves. 2.13. Employ technology to calculate, visualize, and understand the directional derivative and gradient of a multivariable function.
Lesson 14	9/20/2024	MV Optimization 1 (Min/Max) (14.7)	**14.7:** 3, 13*, 21*, 25* *Technology Problems **Quiz 4**	2.14. Employ technology to calculate, visualize, and understand optimization problems. 2.15. Understand the definition of a critical point and be able to find the critical points of a function of two variables. 2.16. Use the Second Derivatives Test to classify a critical point as a local maximum, local minimum, or saddle point
Lesson 15	9/25/2024	MV Optimization 2 (Closed Interval Method) (14.7)	**14.7:** 38, 46, 47, 51, 54, 55	2.17. Find the absolute maximum and minimum of a given continuous function f(x,y) on a closed and bounded set using the Closed Interval Method for functions of two variables. 2.18. Model and solve unconstrained and constrained multivariable optimization problems.
Lesson 16	9/27/2024	Lagrange Multipliers (14.8)	**14.8:** 1, 5, 9, 33, 49	2.19. Solve optimization problems for functions of several variables subject to a single constraint using the Method of Lagrange Multipliers. 2.20. Solve optimization problems for functions of several variables subject to two constraints using the Method of Lagrange Multipliers.
Lesson 17	9/29/2024	**WPR 2 (Dean's Hour)**		

Printed on: 8/15/2023

Lesson	*Date*	*Topic (Reading)*	*Homework (WebAssign)*	*Lesson Objectives*
		BLOCK 3: MULTIVARIABLE INTEGRATION		
Lesson 18	10/2/2024	Volume Estimation/Iterated Integrals (15.1)	**15.1:** 1, 6, 7, 15, 19, 53, 42	3.1. Given a function f(x,y), use a double Riemann sum to estimate volume under a surface. 3.2. Understand the concept of volume under a surface as the limit of the sums of volumes of rectangular columns. 3.3. Use level curves to estimate the volume under a surface. 3.4. Estimate the average value of a function of two variables.
Lesson 19	10/4/2024	General Regions (15.2)	**15.2:** 5, 7, 8, 20, 39	3.5. For any region in the xy-plane, sketch the region, label the boundaries, and label the points of intersection. 3.6. Compute the iterated integral of a multivariable function over a specified domain.
Lesson 20	10/10/2024	Polar Regions (15.3)	**15.3:** 6, 11, 30, 33, 45	3.7. Understand and use the polar coordinate system to solve iterated integrals.
Lesson 21	10/12/2024	**Project 2**	**Project 2 (before-class work)**	
Lesson 22	10/16/2024	Center of Mass (8.3 - skim, 15.4 - stop at Moment of Inertia p. 1072)	**15.4:** 1, 3, 15	3.8. Find the mass, moments, and center of mass of any lamina whose area is a specified region in the xy-plane and whose density is a given function ρ(x,y). 3.9. Employ technology to find the mass, moments, and center of mass of any lamina whose area is a specified region in the xy plane and whose density is a given function ρ(x,y).
Lesson 23	10/18/2024	Probability (15.4 - start at Probability p. 1074)	**15.4:** 29, 31, 32 **Quiz 5**	3.10. Understand that a joint density function is a probability density function of multiple continuous random variables. 3.11. Given a joint density function of a pair of random variables X and Y determine the probability that f(x,y) lies in a region D. 3.12. Employ technology to create joint density functions and calculate probabilities. 3.13. Verify if multivariable functions are valid joint density functions.
Lesson 24	10/20/2024	Triple Integrals (15.6)	**15.6:** 4, 5, 9, 13, 18	3.14. Evaluate a triple integral over a general region. 3.15. Employ technology to compute triple integrals.
Lesson 25	10/23/2024	Triple Integrals in Cylindrical Coordinates (15.7)	**15.7:** 10, 15, 21, 32 **Quiz 6**	3.16. Understand when and how to apply the cylindrical coordinate system to solve iterated integrals.
Lesson 26	10/25/2024	Triple Integrals in Spherical Coordinates (15.8)	**15.8:** 21, 23, 28, 29	3.17. Understand when and how to apply the spherical coordinate system to solve iterated integrals.
Lesson 27	10/30/2024	**WPR 3 (In-class)**		
		BLOCK 4: VECTOR CALCULUS		
Lesson 28	11/1/2024	Vector Fields (16.1)	**16.1:** 7*, 19, 11*, 27, 13, 15, 16, 31, 33 *Technology Problems	4.1. Understand vector fields and their different applications. 4.14. Use technology to plot different types of vector fields.
Lesson 29	11/3/2024	Parametric Equations, Line Integrals (16.2)	**16.2:** 3, 7, 17, 22, 41	4.2. Understand the meaning and compute the line integral along a curve in space. 4.3. Understand different applications of line integrals. 4.15. Employ technology to pass a vector defined function to a function. 4.16. Employ technology to visualize a particle's path along a vector field.
Lesson 30	11/6/2024	**Project 3**	**Project 3 (before-class work)**	
Lesson 31	11/8/2024	Fundamental Theorem for Line Integrals (16.3 - stop at Conservation of Energy p1150)	**16.3:** 1, 3, 4, 7, 18, 23	4.4. Understand and apply the Fundamental Theorem for line integrals.
Lesson 32	11/13/2024	Curl and Divergence (16.5 - stop at Vector Forms of Green's Theorem p1167)	**16.5:** 1, 3, 7*, 11, 18 *Technology Problem **Quiz 7**	4.6. Understand the "del" ∇ operator and use it to compute curl and divergence. 4.7. Understand the physical representation of curl and divergence. 4.17. Employ technology to calculate curl and divergence of a vector defined function.
Lesson 33	11/15/2024	Green's Theorem (16.4)	**16.4:** 3, 5, 9, 17	4.5. Understand and apply Green's Theorem to general two-dimensional regions.
Lesson 34	11/20/2024	Green's/Parametric Surfaces (16.6 - stop at Surface Area p1177)	**16.4:** 7 **16.6:** 19, 20, 33, 35 **Quiz 8**	4.8. Understand how vector functions can be used to model surfaces in three-dimensional space.
Lesson 35	11/28/2024	Surface Area (15.5, 16.6 - start at Surface Area p1177)	**16.6:** 40, 39, 45 **15.5:** 4	4.9. Determine the area of a parametric surface.
Lesson 36	11/30/2024	Exam review	Block 4 practice problems worksheet	
Lesson 37	12/4/2024	**WPR 4 (Dean's Hour)**		
Lesson 38	12/6/2024	Surface Integrals (16.7)	**16.7:** 23, 24, 26	4.10. Understand the meaning and compute surface integrals of a scalar function over a given surface. 4.11. Use the surface integral to determine the flux across a solid region.
Lesson 39	12/8/2024	Stokes' Theorem (16.8)	**16.8:** 9, 10, 19, 21	4.12. Understand and apply Stokes' Theorem to general surfaces.
Lesson 40	12/11/2024	The Divergence Theorem (16.9)	**16.9:** 1, 3, 5, 7	4.13. Understand and apply the Divergence Theorem to general solids.

TAB D AY2023 COURSE ASSESSMENT

DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
DEPARTMENT OF MATHEMATICAL SCIENCES
WEST POINT, NY 10996-1197

15 February 2023
MADN-MATH

MEMORANDUM THRU

COL Lindquist, Service Program Director, Department of Mathematical Sciences, USMA, West Point, NY 10996-1786
COL Hartley, Professor and Head, Department of Mathematical Sciences, USMA, West Point, NY 10996-1786

FOR Office of the Dean, AARS, United States Military Academy, West Point, NY 10996-1926

SUBJECT: Annual Executive Summary for MA205, AY23-01.

1. **Course Description.** MA205: Multivariable and Vector Calculus provides a foundation for the continued study of mathematics and for the subsequent study of the physical sciences, social sciences, and engineering. MA205 covers topics in vectors and geometry of space, vector functions, multivariable differential and integral calculus, and vector calculus. Throughout the course mathematical models motivate the study of topics such as optimization, accumulation, change in several variables, and other topics from the natural, social, and decision sciences. An understanding of course material is enhanced through the use of the computer algebra system.

2. **Curriculum change completed from AY22-2 to AY23-1.** In AY22-2, MA205 continued to teach multivariable differentiation and integration, sequences and series, and an introduction to ordinary differential equations (ODEs). To facilitate delivery of material that eliminated redundancy and increases depth between a two-course sequence, for STEM majors, topics in MA205 and MA364 (Engineering Mathematics) were realigned without loss of material through the curriculum change process in Spring of 2022. The resequencing of topics led to removing sequences and series and ODEs from MA205. These topics, already covered in MA364, will continue to be part of MA364. MA205 replaced these topics with vectors and geometry of space and vector calculus making it more of a traditional multivariable calculus course. This change causes students to see sequences and series and ODEs later but see vector calculus earlier. Rationale for the change is provided in Annex G of the curriculum change documents provided as part of this EXSUM submission. In summary, the net result of resequencing material between MA205 and MA364 was eight new lessons in vectors and geometry of Euclidean space added to MA205. All material previously taught in the two courses will still be covered albeit in a new sequence. By restructuring the material, MA205 follows the *Calculus: Early Transcendentals 9E* Stewart textbook exactly from Chapter 12-16. This provides a full treatment of multivariable calculus similar to that of

peer institutions such as Massachusetts Institute of Technology (two-course calculus sequence) and Cornell University (three-course calculus sequence). The decision to include sequences and series in MA364 and not MA205 is deliberate. By including this topic in MA364, we gain the pedagogical benefit of developing Taylor and Fourier Series in a course that then uses them to derive solution methods for partial differential equations. The resequencing of topics between MA205 and MA364 follows a natural progression, allowing cadets to make connections for how concepts are related and the basis for derived solution methods.

The goal to eliminate redundancy and increase depth for the STEM two-course sequence was motivated by the fact that MA205 is no longer a required core course for all USMA cadets. It was assessed, by looking at the course topics and lesson objectives between MA205 and MA364, that a two-course progression of mathematical topics for STEM majors could be streamlined and improved by following pedagogical research on the sequencing of topics, such as timing of sequences and series and ODEs in undergraduate mathematics, which were originally part of the MA205 curriculum, to better serve cadets when MA205 was a core course and determine the most effective sequence of topics that prepared our cadets for STEM progression. This offered the MA205 course leadership an opportunity to engage with downstream stakeholders from eight departments to discuss the challenges and benefits of topic realignment considering follow-on coursework.

MA205 and MA364 course leaders coordinated with all stakeholders from February to April 2022, presented the change request to the Curriculum Committee on 14 April 22, and received approval from the General Committee on 12 May 2022. One important note to mention in this EXSUM, which is further discussed in the curriculum change request (Annex G), is the addition of a lesson on 1st Order, Linar Differential Equations to MA104 so that cadets that choose Systems Engineering as their major will see a solution technique prior to taking EE301 – Fundamentals of Electrical Engineers (a course for non-Electrical Engineering majors) since these cadets are not required to take MA364. All other material needed for follow-on coursework, for all other majors, will be learned before it is required, and for all others that take MA205/MA364, they will receive the mathematical content required for STEM degrees. Thus, beginning in AY23-1, MA205 covers chapters 12-16 in Stewart 9E and MA364 covers sequences/series, ODEs, and PDEs.

3. **AY23-1 Course Learning Outcomes**. The new MA205 course learning outcomes are:

a. Demonstrate good scholarly habits.

b. Communicate effectively.

c. Apply technology.

d. Develop problem solving and reasoning skills.

e. Develop interdisciplinary perspective.

f. Apply multivariable derivatives.

g. Apply multivariate integrals.

h. Apply vector operations and model motion of an object.

i. Demonstrate calculus of vector fields/line and surface integrals.

More information on these course learning outcomes can be found in the MA205 Course Curriculum Change memorandum for MA205 in Enclosure 1.

4. **Student Population Change**: In AY22-2, civil engineering major cadets entered the MA205 population. Also, in AY23-1, the environmental engineering major cadets started to take MA205. The addition of these two majors as students in MA205 is because of the closure of MA366. Of special note, in AY22-2 and AY23-1, we had several students who stated they took an Advanced Placement calculus class in high school but did not feel comfortable validating MA104 as that was the year COVID-19 remote teaching protocols were put in place and they did not feel as if they learned much after their high school went remote.

5. **At-Risk Population AY22-2**: An analysis of the performance of students who took MA104 and MA205 in the previous three semesters (AY21-2 to AY22-2) revealed that students who received a C- or below in MA104 did not pass MA205 on the first time they took MA205. This trend held true across all semesters included in the analysis seen in Figure 1. Cadets below the 73% grade in MA104 received a grade below 65% in MA205. The cadets with a score above 65% in MA205 and below 73% in MA104 in Figure 1 are cadets who were taking MA205 for the second time. In light of this information, COL Hartley sent a letter to the Department Academic Counselors (DACs) of the students who received a C- or below in MA104 and who need to take MA205, highlighting the importance of understanding the material covered in MA104 and encouraging students to retake the course if necessary before taking MA205. One student took the suggestion and decided to retake MA104. He ended up with a B- (previously a D) and had a better understanding of the material. MA205 recommended a minimum of a C in MA104 to be able to understand the concepts in MA205. Also, MA205 repeaters typically changed their majors while taking MA205 the second time – however the cadets still needed to take MA205 to grade replace the F from the previous semester.

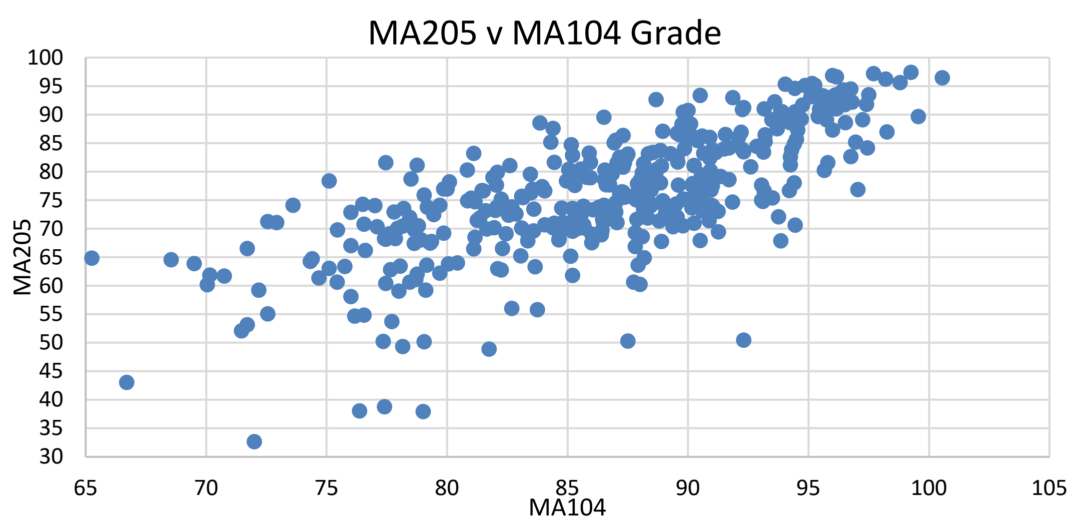

Figure 1. Graph of MA104 vs MA205 grades for AY21-2, AY22-1, and AY22-2.

6. **At-Risk Population AY23-1**: At the end of AY23-1, two cadets who received a C- or below in MA104 passed MA205. We hope this trend continues since MA104 changed their course to encompass a greater body of knowledge and less dependence on calculators and technology on graded events. In addition, as COVID remote protocols are removed, we anticipate a higher level of learning from in-class teaching. Students still need to demonstrate an in-depth level of understanding in MA104 before entering MA205 to continue to grow on their calculus body of knowledge and succeed.

7. **Computer Aided System: Mathematica**. MATLAB was the computation platform used in MA205 at the request of a couple of departments that use it in their majors. For two semesters, under the current leadership, the decision to keep MATLAB or move to Mathematica, which is familiar to cadets from MA103 and MA104, was considered. MATLAB served approximately 21% of the students, to include mechanical, chemical, and electrical engineering students (hard STEM); leaving 79% of our students, to include systems engineering, business management, and economics (soft STEM) students who will not use MATLAB again after the completion of MA205. This roughly 80/20 split was not the reason the course changed to Mathematica. The decision to change came from an assessment of the two platforms run concurrently between different sections in AY22-2. The assessment involved instructors using MATLAB or Mathematica based on their section composition (hard or soft STEM). Soft STEM sections used Mathematica and hard STEM sections used MATLAB. Across the course we had instructors that used Mathematica or MATLAB exclusively and the Program Director had a section of each. We assessed the use of technology on the same problem sets and project. After the project and problem sets, we surveyed the students on how comfortable they were using MATLAB and how comfortable they were using Mathematica. Of the students who responded to the survey, 27.4% were somewhat or very comfortable with multivariable differentiation and integration using MATLAB. Of the students who responded to the survey, 61.3% were somewhat or very comfortable with multivariable differentiation and integration using Mathematica. Students found Mathematica easier to use due to its use of symbolic pallets, and hence did not feel a barrier to learning another language to overcome while they were trying to take a partial derivative or integrate a triple integral. From a survey with our instructors and instructors from other departments, many felt it was more important for students to understand multivariable calculus concepts than just know a computer program. Upon further exploration, Mathematica has a wider range of visualization options, allowing students to see more interactive and intuitive exploration with multivariable concepts. Of note, for the sections that used MATLAB the technology guide had examples in both Mathematica and MATLAB to help bridge the gap between Mathematica, which is used in previous math classes, and MATLAB which was new to all students. A major improvement in AY23-1 was our technology guides for Mathematica, especially for the visualization with triple integrals. Students were able to change a variable parameter while holding other variables constant and could see how changing their limits of integration changed the region of integration. An example of this is seen in Figure 2. In AY23-1, MA205 solely used Mathematica for its graphical visualizations, calculations, and projects.

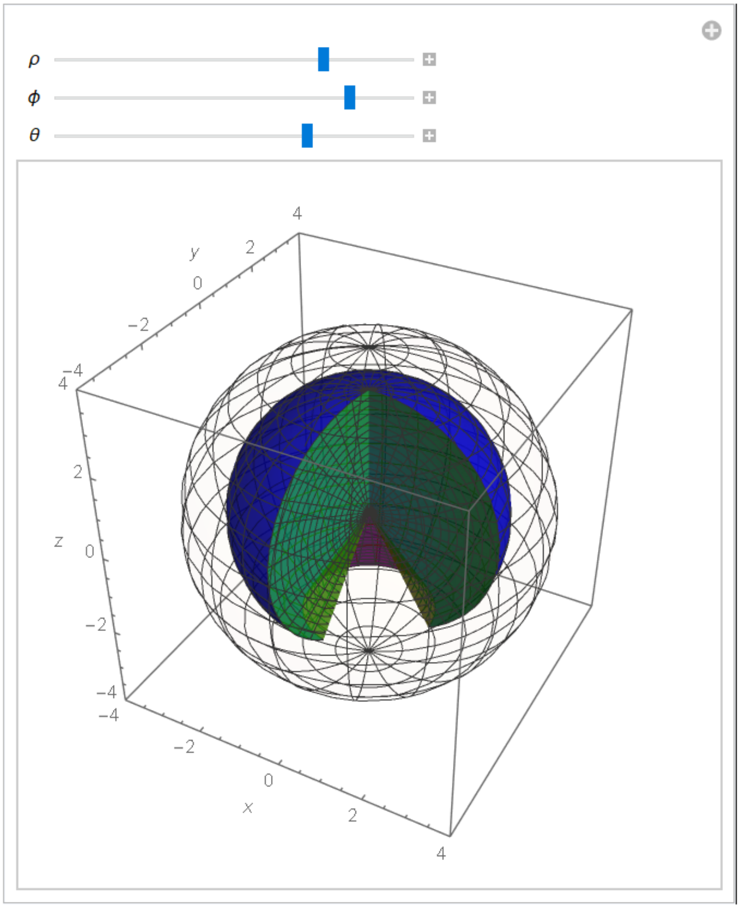

Figure 2. An interactive visualization of spherical coordinates in Mathematica.

8. Assessment Summary.

Academic Program Goal	Program (Course Goal)	Assessment Indicator	Assessment
1. Communication: Graduates communicate effectively with all audiences	1. Communicate Effectively	1. Project Presentations 2. Discussion of board answers / participation 3. Instructor writing assignments	**2**
2. Critical Thinking and Creativity: Graduates think critically and creatively	1. Apply Technology 2. Apply Multivariate Derivative 3. Apply Multivariate Integral 4. Apply Vector Operations and Model Motion of an Object 5. Demonstrate Calculus of Vector Fields/Line and Surface Integrals	Course Assessments a. In-class quizzes b. Written Partial Reviews (WPRs) c. Course Projects d. Technology Labs e. Term End Exam (TEE)	**2**

3. Lifelong Learning: Graduates demonstrate the capability and desire to pursue progressive and continued intellectual development	1. Demonstrate Good Scholarly Habits 2. Develop Problem Solving and Reasoning Skills	1. WebAssign Scores 2. Survey Responses a. Before/After Out of Class Work b. Resources c. Course Structure	**3**
5. Science, Technology, Engineering, and Mathematics (STEM): Graduates apply science technology, engineering, and mathematics concepts and processes to solve complex problems	1. Apply Technology 2. Develop Problem Solving and Reasoning Skills 3. Develop Interdisciplinary Perspective	1. Course Projects 2. Course Assessments	**2**

Scale: 1 = Outstanding (A+, A), 2 = Very Good (A-, B+), 3 = Good (B-, C+), 4 = Fair (C-, D), 5 = Poor (F)

9. **APG 1: Graduates communicate effectively with all audiences.**

i. **Summary:** MA205 supports APG 1 through MA205 Course Goal 2.

ii. **Assessment:** MA205 assesses "Communicate Effectively," primarily through the course project, board work, and some instructor specific writing assignments. Overall, our students' communication skills improved throughout the semester, focusing on both public speaking and written communication. Our instructors used group discussions and projects to assess the students' abilities and provided feedback to help them improve. From a survey of our instructors, most felt their students were initially quiet but became more engaged and willing to contribute as the semester progressed. Instructors observed improvements in the students' ability to explain complex work and convey information.

In AY22-2, the SpaceX themed course project included an in-person oral presentation and a written technical report. In order to give cadets a chance to improve their oral communication skills, in AY23-1, the course moved from one to two projects and focused on oral presentations for both of them. The move to two oral presentations was due to the amount of technical writing practice and signature writing events cadets are already mandated to receive in their core math curriculum, which culminates with a statistical technical report in MA206. In addition, with the new course material added to MA205 in AY23-1, there became an opportunity to include more modeling, solving, and analysis for two course projects. The two projects focused on optimal trajectory, projectile designs, and maximizing profits.

Instructors observed their students improved their technical communication skills between the first and second projects, with the second project presentations being more professional and competent. They attribute this improvement to the fact that the second project was easier for students to comprehend and the fact that students factored in feedback from the first project to improve their second project. Our instructors also observed that while the students were better at oral communication, they still struggled to organize and develop proper slides to convey their ideas. For the second project, we asked students to submit a video instead of an in-person presentation. The video submissions of the projects were better as students were not as nervous, likely rehearsed, or did not feel the same pressure as having the instructor sitting right there in the room. Cadet confidence throughout the course to verbally communicate with different types of audiences on a mildly to strong level rose from 47% to 62%.

10. **APG 2: Graduates think critically and creatively.**

i. **Summary:** MA205 supports APG 2 through MA205 Course Goals 3, 6, 7, 8, and 9.

ii. **Assessment: Evaluation of Cadet Learning Objectives.** The assessments throughout the semester are planned around the Course Learning Objectives. This deliberate effort resulted in the evaluation of 70 of the 73 Course Learning Objectives (CLOs) during AY22-2. CLOs were assessed on Rumbles (in-class quizzes), two WPRs, two WRITs, a TEE, a course project, and two technology labs. In AY23-1, the course assessed 57 out of 57 CLOs on eight Rumbles, four WPRs, two projects, and a TEE. The course also assessed 6 out of 10 technology focused CLOs during the two course projects. A list of CLOs from each semester are in Enclosures 4 and 5.

The following tables in Figure 3 include the four WPRs topics and the averages across different semesters. The first and fourth blocks, vectors and geometry of space and vector calculus, were new in AY23-1. They do not have any other comparable averages from previous semesters. We expect the averages to be higher next year, since a MA205 given assessment example will be available for cadets to see and use during their studies on the CEP website. During AY23-1, these two blocks were the lowest performing, most likely due to the lack of previous WPRs to prepare from. Given the complexity of the material and newness from previous semesters, the cadets who prepared well, did an exceptional job. Our highest performer during WPR 1 received a 99.67%. Our highest performer during WPR 4 received a 100%. To gain a better understanding of vector calculus, block 4 should be made longer for students to process the concepts. Currently there are 10 lessons devoted to vector calculus. If more time was given to Curl and Divergence, Green's, Stokes', and Divergence Theorem's, then students may have a better understanding of the similarities and differences between them. This time may be created by placing more of the single variable topics such as vectors and parametric equations in MA103 and MA104 – these align with Stewart's Chapter 12 and 13. The more students get exposed to in those classes, the more they will feel comfortable seeing vectors and parametric equations again in MA205. In AY23-1 blocks 2 and 3, multivariable differentiation and integration covered the same material from previous semesters. A table outlining the WPR grades are in Figure 2.

The recent return to in-person classes for college students led to a noticeable increase in grades from AY22-1 to AY22-2. This can be attributed to several factors, including the fact that many students had previously taken a single variable calculus course in high school during the height of the COVID remote teaching. They did not feel confident enough to take the MA104 validation exam and chose to take another chance to brush up on single variable calculus material again. They possibly had a stronger foundation before taking MA205. Additionally, the AY22-2 semester's students featured a new student population, a large number of civil engineering majors, who as a group performed higher than the class average. These reasons may have led to a higher overall class average for what is typically the MA205 off-cycle.

Vectors and Geometry of Space		
AY22-1	AY22-2	AY23-1
n/a	n/a	73.69

Multivariable Differentiation		
AY22-1	AY22-2	AY23-1
70.85	84.88	81.51
Multivariable Integration		
AY22-1	AY22-2	AY23-1
73.2	77.98	79.88
Vector Calculus		
AY22-1	AY22-2	AY23-1
n/a	n/a	73.81
TEE*		
AY22-1	AY22-2	AY23-1
68.5	78.57	81.79

Figure 3. WPR and TEE assessment scores across the different blocks from AY22-1 to AY23-1.
*TEE for AY23-1 is different from AY22-1 and AY22-2

From AY22-1 to AY22-2, MA205 kept the same TEE. Student performance during AY22-2 was better than AY22-1 for the same questions. The change in AY23-1 kept the classroom material focus on multivariable calculus concepts the entire semester. The last block, vector calculus, continued building on multivariable differentiation and integration. This differs from the previous semesters where students would also learn sequences and series and introduction to ordinary differential equations, which are primarily single variable concepts. Since the students were continuing to solve multivariable derivatives and integrals, the median scores on the triple integral and Lagrange multiplier questions on the TEE in AY23-1 students were higher than the median of AY22-1 and AY22-2 students. The AY23-1 students continually had an opportunity to practice and refine their multivariable differentiation and integration techniques during the vector calculus block of instruction.

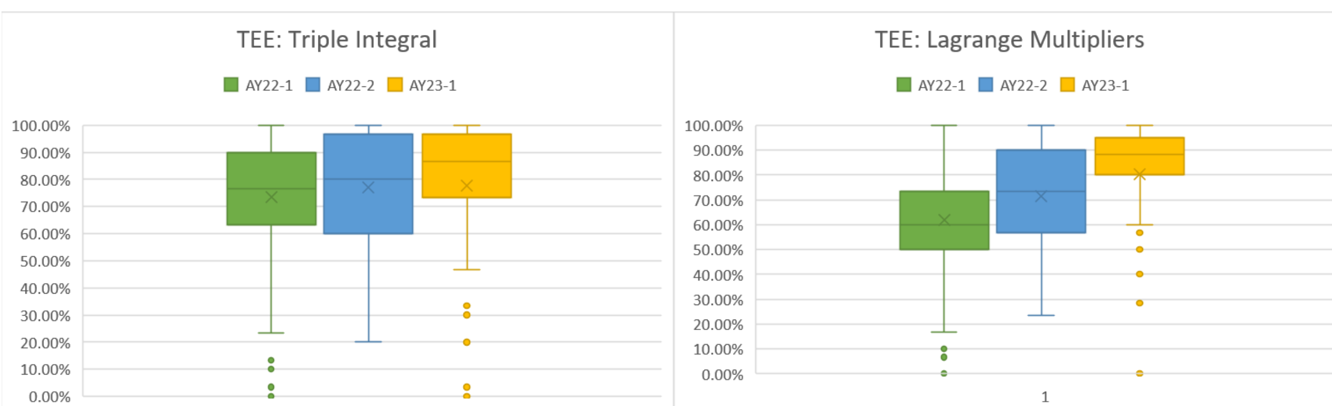

Figure 4. TEE Performance of Triple Integrals and Lagrange Multipliers
after the Curriculum Change

Small stakes in-class assessments, such as MA205s Rumbles, continue to be a valuable tool for improving student performance on WPRs and the TEE. As Figure 5 illustrates, Rumbles allow students to practice taking a timed assessment in class, which in turn helps them to better understand and retain the material covered in class. This is evident when looking at the progression of comprehension for students after completing homework, taking the Rumble, WPR, and finally the TEE. Additionally, when the data is broken down by grade range, it becomes clear that students who performed at a B, C, or D level benefited the most from taking these in-class quizzes. One of the most important part of Rumbles is the immediate grading after taking it – this provides students with immediate feedback. Students grade themselves while the instructor goes over the rubric and solution. Students who demonstrate intellectual development write themselves notes on a concept they missed or highlight a possible miscalculated step. The use of in-class assessments, like Rumbles, is a valuable tool for instructors to improve student performance and for students to experience test-like conditions on a smaller stakes assessment.

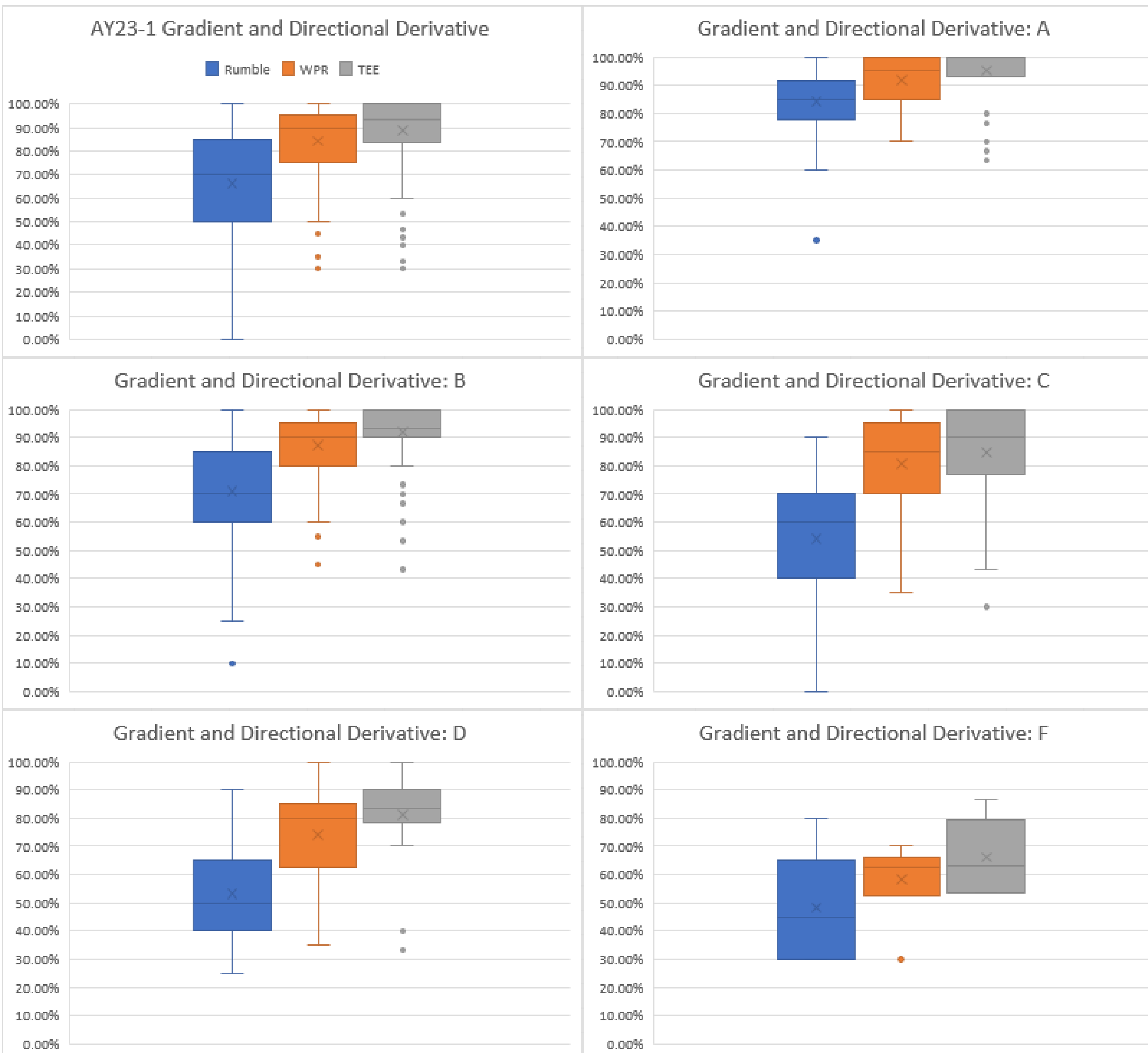

Figure 5. Rumble, WPR, and TEE performance for the Gradient and Directional Derivative CLOs.

11. **APG 3: Graduates demonstrate the capability and desire to pursue progressive and continued intellectual development.**

i. **Summary:** MA205 supports APG 3 through MA205 Course Goals 4, and 5.

ii. **Assessment:** Students who demonstrate good scholarly habits in this course are those who actively engage with the material both in and out of class. This is evident in the success of students with high WebAssign scores, as well as those who actively take notes and participate in class discussions. Additionally, it appears that these students are more likely to prepare for class by doing homework and reviewing solutions to past homework and rumbles. If a student did not understand the material and did not come to Additional Instruction with their instructor, they turned to Khan Academy, YouTube videos, Organic Chemistry Tutor, videos on WebAssign to help guide them through an example. The Read-Watch-Do videos on WebAssign helped approximately 72% students understand the steps to solve a problem. However, for some students, this eventually led to dependence on them and not reading the textbook. This is a crucial point as the material in the textbook is more in-depth and includes more examples which can help students to have a better understanding of the subject matter. Figure 6 depicts what resources students used in order to complete their homework.

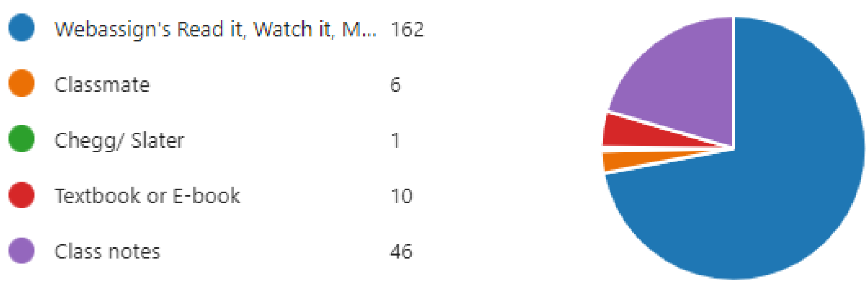

Figure 6. Responses to what tools cadets use most to complete WebAssign homework.

Instructor observations suggest that students who struggle with good scholarly habits include students who have lower performances in the first two blocks of the course. These students are often disengaged in class and are unable to answer basic questions about the material. Conceptual questions are key to understanding how to proceed in a question, and without understand the base concepts, it was difficult for students to dive deeper into the material. Furthermore, the dependence of Read-Watch-Do videos on WebAssign led students to not read their books outside of class, which likely contributed to their struggles with the material. Figure 7 shows how students performed on their homework vs their graded assessments. Without the help of the Read-Watch-Do videos during timed assessments, cadets who did not adequately prepare for the graded assessments, did not perform as well as they did on their homework. Most cadets complete their homework, but do not gain the understanding they need to truly succeed in the course. WebAssign made more Read-Watch-Do videos from the previous two semesters, and the dependency on them is growing.

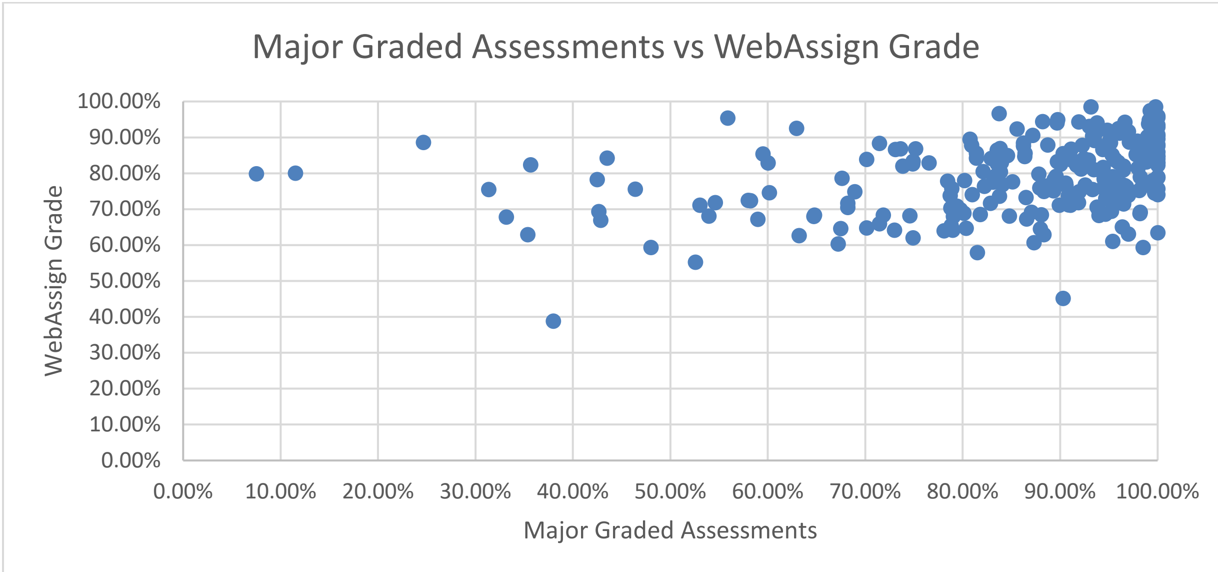

Figure 7. A comparison of WebAssign grades to the sum of the major graded assessments

The instructor's approach to teaching the course may also be contributing to the challenges that students are facing. The use of the Thayer Method, which emphasizes preparation outside of class, seems to be a challenge as students generally come to class without having fully read the section or attempted problems. Additionally, instructors observed students often wait until the last minute to complete assignments and seek assistance, which likely hinders their ability to fully engage with the material. Overall, it seems that many students in this course are struggling to establish good scholarly habits and may benefit from additional support and guidance in this area. For some students, this is their first "hard" math course that is not a review from high school. Students will benefit from some time at the beginning of the course set aside for "good study habits" or tips and instruction on "how to prepare for this course". While time consuming, if students see how to properly read the textbook or find outside resources, it will benefit the student in the long run.

On the topic of the structure of the course as it applies to good scholarly habits, during course design of the new MA205 format, we considered, and will continue to assess, if it is pedagogically beneficial to have a 40-lesson, 75-minute or ~55-lesson, 55-minute design. We ultimately decided for AY 23-1 that a 40-lesson, 75-minute design would provide longer in-class instruction for more complex lessons and provide opportunity for more board work and hands-on discussion with instructors. From instructor feedback, 75-minute lessons worked exactly as designed for lessons that were more difficult, especially in the vector calculus block. However, with a 75-minute, 40-lesson design we did not have room for labs or assessment preparation. Moreover, there is a concern that students have less contact with the material when coming to the classroom every other day. Given the amount of material to cover, a 40-lesson design means new material every class which forced instructors to carve out review time in their lesson plan. Instructors felt that with more time for practice problems on some lessons it meant that too much material was being crammed together in another lesson (feedback from an instructor that previously taught MA255, which has similar material and a 55-minute course design). As with any course, the challenge for instructors is getting through new material, conducting reviews of current material or fundamentals from previous classes (if needed), and providing ample time for practical application at the boards, which is consistently the #1 method cadets list as the most helpful use of their in-class time. To the aforementioned point that not all cadets come to class prepared, a certain amount of time is usually spent going over the lesson, which leads to a balancing act of how much discussion/lecture should occur and how much time cadets have at the boards. Other considerations for the 55-minute v. 75-minute discussion have been explored during faculty development discussions at the department level and continued at the course level during Program Director's Time. Some takeaways from these discussions follow.

The most frequent reasons for a 55-minute design are more repetition, more frequent "touches", and less days in-between classes which have been regarded in literature as beneficial for the learning/teaching of mathematics. On the other hand, the benefits of a 75-minute design include more responsibility on the cadet to learn the material, possibly more board time, and more time available to schedule additional instruction outside of class. While each of these reasons can be used to support the argument for why to conduct 55-minute or 75-minute class meetings, we are still assessing the right balance for our population of STEM students and the amount of material we cover in MA205. Admittedly, MA205 instructors prefer the 75-minute format as it allows more time for class preparation, more time to provide feedback on homework, projects, and quizzes, and more time in class to go over complex topics. At this time, we've selected the 75-minute format and are focused on the question, "Do cadets prepare more or less time outside of class from the usual 55-minute classes they've been used to?", to inform future decisions about the format that best supports APGs.

In AY 22-2 and AY 23-1, we conducted two surveys each semester to try to answer the question of how much effort cadets spend outside of class. It is important to note that the number of participants in each survey were 88 cadets, 43 cadets, 167 cadets, and 225 cadets for the four surveys conducted as outlined below. We specifically asked how much time cadets spend before class preparing for the lesson and how much time they spend after class completing their assigned homework on the same lesson. Under the current course design, we assign WebAssign homework for every lesson. Cadets have access to the homework before every lesson and have until midnight the day of the in-class lesson to complete the homework. Our rationale for this format is to give cadets the opportunity to attempt the WebAssign homework before class and then complete it afterwards, so that they can ask questions during class. Each instructor is given freedom on how they prompt questions in class and how they respond, but will not solve homework problems since they are for points. The results of self-reported numbers for time spent outside of class are in Figure 8.

AY 22-2 Block I Results

	Minutes	%		Minutes	%
Before Class	0-30	11.4%	After Class	0-30	29.5%
	31-60	42.0%		31-60	44.3%
	61-90	30.7%		61-90	13.6%
	91-120	12.5%		91-120	10.2%
	121+	3.4%		121+	2.3%

AY 22-2 Block II Results

	Minutes	%		Minutes	%
Before Class	0-30	23.3%	After Class	0-30	30.2%
	31-60	41.9%		31-60	41.9%
	61-90	23.3%		61-90	25.6%
	91-120	9.3%		91-120	2.3%
	121+	2.3%		121+	0.0%

AY 23-1 Block I & II Results

	Minutes	%		Minutes	%
Before Class	0-30	31.1%	After Class	0-30	24.6%
	31-60	49.1%		31-60	46.7%
	61-90	16.8%		61-90	22.2%
	91-120	3.0%		91-120	4.2%
	121+	0.0%		121+	2.4%

AY 23-1 Block III Results

	Minutes	%		Minutes	%
Before Class	0-30	46.2%	After Class	0-30	23.1%
	31-60	40.4%		31-60	48.9%
	61-90	12.9%		61-90	22.7%
	91-120	0.4%		91-120	4.4%
	121+	0.0%		121+	0.9%

Figure 8. Cadet's self-reported time spent outside of class (before and after each lesson)

In the surveys conducted after Blocks I and II (AY 22-2) and Blocks I/II and III (AY23-1), it was first important to know if cadets were putting in effort to prepare and what resources they were using. From Figure 8, we see that most cadets range from 0-90 minutes of study outside of class for each lesson. This is a wide range of values and expected given each cadet prioritizes differently, requires different amounts of time to study, and time will vary for each cadet based on their familiarity with the subject. To better understand the results, we used averages to compute estimates of total time to see if any trends emerged. A summary of our estimated values can be seen in Figure 9. For a 4.0 credit course with 75-minute in-person class time, we expect cadets to spend 150 minutes outside of class. The cadets' self-reported outside of class effort does not necessarily match what instructors reported during discussions throughout the term. As previously stated, instructors observed cadets wait until the last minute to complete work and did not necessarily come prepared for each lesson.

	Before Class (minutes)	After Class (minutes)	Total (minutes)
AY 22-2 Block I	61.2	48.3	**109.5**

AY 22-2 Block II	52.6	45	**97.6**
AY 23-1 Blocks I & II	72.3	48.8	**121.1**
AY 23-1 Block III	35.3	48.3	**83.6**

Figure 9. Estimates of before and after class time spent for AY 22-2 and 23-01

While the self-reported numbers of time spent outside of class seem appropriate, the next questions are, "How are cadets preparing?" or "How can cadets improve their preparation?". As previously mentioned, we are looking at the design of the course and the incorporation of "good study habits" demonstrations, but we do not feel that cadets have been performing at a level that is a concern under the current design. Through discussions with MA255 it is not clear if a 55-minute course design for MA205 is the "best" approach because the population of students is different. We acknowledge spreading the material over more lessons could make it more "digestible", but a course being digestible is predicated on "good scholarly habits", and from instructor feedback habits between MA255 and MA205 populations are not equal. Moreover, MA255 has more class time as a 4.5 credit course. Nonetheless, a more granular approach to collecting data is being considered to make a more informed decision about the course structure. This includes a deeper look at the use of WebAssign for homework and collecting daily time reports from each cadet. We will continue to consult pedagogical literature and continue to challenge/meet cadets at their entry-level to develop intellectual and scholarly habits.

12. **APG 5: Graduates apply science, technology, engineering, and mathematics concepts and processes to solve complex problems.**

i. **Summary:** MA205 supports APG 5 through MA205 Course Goals 3, 4, and 5.

ii. **Assessment:** The MA205 course projects served as a great way for students to apply their calculus-based body of knowledge to solve questions from varying disciplines. New this past calendar year, cadets were given a worksheet to guide them through the problem-solving process. The worksheet for each problem served as a set of sub-steps to the greater problem, helping students break down the problem into smaller, manageable portions. This allowed them to see the bigger picture and understand how the smaller parts fit together. The projects covered a wide range of topics, including optimal trajectory, gradient and directional derivatives for target placement, optimizing profits, center of mass, projectile design, triple integrals applying spherical and cylindrical coordinate systems, calculating probabilities using joint density functions, and optimization using Lagrange multipliers. Through these projects, students were able to apply the concepts they learned in class to real-world situations, gaining a deeper understanding of the material and developing valuable problem-solving skills.

The projects are designed to challenge the students and push them to think outside the box, while using Mathematica to help solve the complex problem. The worksheet served as a roadmap for the project, outlining the steps and concepts that are needed to solve the problem. It helps students to stay on track and not get overwhelmed by the complexity of the problem. The projects were also a great opportunity for students to work in teams, as they learned to collaborate and communicate effectively to solve the problem. Some cadets found the group projects invaluable as they were able to work in teams, even if it was not with people they would normally work with. The course projects were a great way for students to develop interdisciplinary perspective, work with cadets from outside their major, and develop problem solving skills.

12. **Significant planned or continuing-into-the-next-academic-year curricular change.** During AY 23-1, the course leadership met with representatives from several of the departments that our course serves. We extended the invite to all eight departments that that require MA205, but only representatives from Economics, Geography and Environmental Engineering, and Department of Systems Engineering attended. The purpose of the meeting included a briefing on the changes to MA205 and how those could potentially impact early courses in our served majors if their students did not have sequences/series and/or ODEs. We also discussed possible application problems from across the STEM disciplines to help increase interest in MA205 projects from the cadets. Our assumption is that if cadets can see how the material we teach in MA205 can be applied to their major, then they are more likely to master it and retain it. Finally, we talked about how these majors use technology throughout a cadet's tenure. In past semesters of MA205, the course used either MATLAB or Mathematica, or both; so, we wanted to gauge how the departments use technology to inform our decisions on what technology platform to use and incorporate commands and workflow the cadets see in downstream courses. In AY23-02 and beyond, we plan to continue the relationships with these departments to help plan projects and synchronize material between courses.

13. **Proposed changes to the Cadet Learning Outcomes and the justification for doing so.** In AY23-2, we removed one Learning Objective from AY23-1: "1.11 Understand how the normal, binormal, and curvature of a space curve determine its shape.". This lesson objective came from Lesson 06: "Arc Length and Curvature", which we changed to be "Arc Length and Motion in Space I" with the next lesson being "Motion in Space II". The course leadership felt that the material on curvature and unit tangent-normal-binormal frames was unnecessarily confusing for the students. The material itself is hard and then we never used the material again in the course, and only tangentially in the vector calculus block (i.e., there are specific ways of utilizing Green's Theorem using the unit tangent vector). The Stewart calculus textbook only introduces these topics in the one section and then doesn't utilize them again in the sections our course covers. We determined that removing this material would not significantly impact the students' learning and may even allow them to focus more on the other material that *will* come up again later in the course.

14. **Significant planned or implemented assessment process changes.** The only significant assessment process change implemented in AY 23-2 is that we have changed the projects from two, "large" projects, to four "small" projects. We expect the same overall amount of time spent throughout the semester on projects but spread among four projects instead of two. With this new plan, we have one project per block, to assess Learning Objectives in order to help improve the students' understanding of the material prior to each WPR. Currently, we have no, significant assessment process changes planned for AY 24-1.

15. **Updated current assessment methodology and schedule.** Course assessments are essential in evaluating student comprehension of class material and the effectiveness of instruction. Assessments provide a baseline measurement of student performance and allow instructors to make necessary adjustments to the course. The new course assessment methodology and schedule can be found in Table 5 of Enclosure 1. Since this was the first semester with the new course, we assessed each source of evidence of each outcome to get a baseline. Students were assessed on their comprehension of class

material via their homework and major graded events. The course also asked feedback from the students and instructors via Microsoft Forms. The collection of feedback from both students and instructors further enhances the course by providing valuable insights into the areas where improvements can be made. The course will now follow the schedule in Enclosure 1.

16. Point of contact is the undersigned at josephine.cammack@westpoint.edu, or phone 845-938-5615.

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5 Encls
1. MA205 Curriculum Change Memo
2. AY22-2 Calendar
3. AY23-1 Calendar
4. CLOs for AY22-2
5. CLOs for AY23-1

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AY 23-1		MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
Block 1	1	**15-Aug-22** **1-1** Lesson 1 3D Coord Systems & Geometry of Vectors	**16-Aug-22** **N/A** Study Day No Class	**17-Aug-22** **1-2** Lesson 2 Dot and Cross Product	**18-Aug-22** **2-1** No Class	**19-Aug-22** **1-3** Lesson 3 Equ of Lines & Planes/ Quadric Surfaces Rumble 1
	2	**22-Aug-22** **1-4** Lesson 4 Vector Functions and Space Curves	**23-Aug-22** **2-2** No Class	**24-Aug-22** **1-5** Lesson 5 Derivatives & Integrals of Vector Functions Rumble 2	**25-Aug-22** **2-3** No Class	**26-Aug-22** **1-6** Lesson 6 Arc Length and Curvature
	3	**29-Aug-22** **1-7** Lesson 7 Motion in Space: Velocity & Acceleration	**30-Aug-22** **2-4** No Class	**31-Aug-22** **N/A** Study Day WPR 1	**1-Sep-22** **1-8** Lesson 8 Class Drop	**2-Sep-22** **2-5** No Class
Block 2	4	**5-Sep-22** **N/A** Labor Day No Class	**6-Sep-22** **2-6** No Class	**7-Sep-22 Modified** **1-9** Lesson 9 Multivariable Functions	**8-Sep-22** **2-7** No Class	**9-Sep-22** **1-10** Lesson 10 Partial Derivatives/ Chain Rule
	5	**12-Sep-22** **1-11** Lesson 11 Tangent Planes Rumble 3	**13-Sep-22** **2-8** No Class	**14-Sep-22** **N/A** Study Day No Class	**15-Sep-22** **1-12** Lesson 12 Gradient and Directional Derivative	**16-Sep-22** **2-9** No Class
	6	**19-Sep-22** **1-13** Lesson 13 Optimization Project (In-class work)	**20-Sep-22** **2-10** No Class	**21-Sep-22** **1-14** Lesson 14 MV Optimization 1 (Min/Max) Rumble 4	**22-Sep-22** **2-11** No Class	**23-Sep-22** **1-15** Lesson 15 MV Optimization 2 (CIM)
	7	**26-Sep-22** **1-16** Lesson 16 Lagrange Multipliers	**27-Sep-22** **2-12** No Class	**28-Sep-22** **N/A** Study Day/ Honorable Living Day WPR 2	**29-Sep-22** **1-17** Lesson 17 Class Drop	**30-Sep-22** **2-13** No Class
	8	**3-Oct-22** **1-18** Lesson 18 Project 1 Presentations	**4-Oct-22** **2-14** No Class	**5-Oct-22** **1-19** Lesson 19 Project 1 Presentations	**6-Oct-22** **2-15** No Class	**7-Oct-22** **1-20** Lesson 20 Vol. Est., Iterated Integrals
Block 3	9	**10-Oct-22** **N/A** Columbus Day No Class	**11-Oct-22** **2-16** No Class	**12-Oct-22** **1-21** Lesson 21 General Regions	**13-Oct-22** **2-17** No Class	**14-Oct-22** **1-22** Lesson 22 Polar Regions
	10	**17-Oct-22** **1-23** Lesson 23 Application 1: Center of Mass	**18-Oct-22** **2-18** No Class	**19-Oct-22** **N/A** Study Day No Class	**20-Oct-22** **1-24** Lesson 24 Application 2: Probability Rumble 5	**21-Oct-22** **2-19** No Class
	11	**24-Oct-22** **1-25** Lesson 25 Interdiscplinary Project (In-class work)	**25-Oct-22** **2-20** No Class	**26-Oct-22** **1-26** Lesson 26 Triple Integrals	**27-Oct-22** **2-21** No Class	**28-Oct-22 Modified** **1-27** Lesson 27 Triple Integrals in Cylindrical Coord Rumble 6
	12	**31-Oct-22** **1-28** Lesson 28 Triple Integrals in Sphereical Coord	**1-Nov-22** **2-22** No Class	**2-Nov-22** **N/A** Study Day WPR 3	**3-Nov-22** **1-29** Lesson 29 Class Drop	**4-Nov-22** **2-23** No Class
Block 4	13	**7-Nov-22** **1-30** Lesson 30 Vector Fields	**8-Nov-22** **2-24** No Class	**9-Nov-22** **1-31** Lesson 31 Parametric Equations, Line Integrals	**10-Nov-22** **2-25** No Class **Project 2 Presentations Due**	**11-Nov-22** **N/A** Veteran's Day No Class
	14	**14-Nov-22** **1-32** Lesson 32 Fundamental Theorem for Line Integrals	**15-Nov-22** **2-26** No Class	**16-Nov-22** **N/A** Study Day No Class	**17-Nov-22** **1-33** Lesson 33 Curl and Divergence	**18-Nov-22** **2-27** No Class
	15	**21-Nov-22** **1-34** Lesson 34 Green's Theorem Rumble 7	**22-Nov-22** **2-28** No Class	**23-Nov-22** **N/A** No Class	**24-Nov-22** **N/A** Thanksgiving No Class	**25-Nov-22** **N/A** Training Holiday No Class
	16	**28-Nov-22** **1-35** Lesson 35 Green's/ Parametric Surfaces	**29-Nov-22** **2-29** No Class	**30-Nov-22** **1-36** Lesson 36 Surface Area Rumble 8	**1-Dec-22** **2-30** No Class	**2-Dec-22** **1-37** Lesson 37 Surface Integrals
	17	**5-Dec-22** **1-38** Lesson 38 WPR 4	**6-Dec-22** **N/A** Study Day No Class	**7-Dec-22** **1-39** Lesson 39 Stokes' Theorem	**8-Dec-22** **N/A** Study Day No Class	**9-Dec-22 Modified** **1-40** Lesson 40 The Divergence Theorem
TEE	18	**12-Dec-22** **N/A** TEE Week	**13-Dec-22** **N/A** TEE Week	**14-Dec-22** **N/A** TEE Week	**15-Dec-22** **N/A** TEE Week	**16-Dec-22** **N/A** TEE Week

KEY	No Class	Project	Course-wide Graded Event	Rumble	**Assignment or Modified Sched.**

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AY 23-1		MONDAY		TUESDAY		WEDNESDAY		THURSDAY		FRIDAY	
Block 1	1	15-Aug-22	1-1	16-Aug-22	N/A	17-Aug-22	1-2	18-Aug-22	2-1	19-Aug-22	1-3
		Lesson 1 3D Coord Systems & Geometry of Vectors		Study Day No Class PD Meeting: Rumbles, questions		Lesson 2 Dot and Cross Product		No Class		Lesson 3 Equ of Lines & Planes/ Quadric Surfaces Rumble 1	
	2	22-Aug-22	1-4	23-Aug-22	2-2	24-Aug-22	1-5	25-Aug-22	2-3	26-Aug-22	1-6
		Lesson 4 Vector Functions and Space Curves		No Class PD Meeting: WPR 1		Lesson 5 Derivatives & Integrals of Vector Functions Rumble 2		No Class		Lesson 6 Arc Length and Curvature	
	3	29-Aug-22	1-7	30-Aug-22	2-4	31-Aug-22	N/A	1-Sep-22	1-8	2-Sep-22	2-5
		Lesson 7 Motion in Space: Velocity & Acceleration		No Class		Study Day WPR 1		Lesson 8 Class Drop		No Class	
Block 2	4	5-Sep-22	N/A	6-Sep-22	2-6	7-Sep-22 Modified	1-9	8-Sep-22	2-7	9-Sep-22	1-10
		Labor Day No Class		No Class PD Meeting: Project 1		Lesson 9 Multivariable Functions		No Class		Lesson 10 Partial Derivatives/ Chain Rule	
	5	12-Sep-22	1-11	13-Sep-22	2-8	14-Sep-22	N/A	15-Sep-22	1-12	16-Sep-22	2-9
		Lesson 11 Tangent Planes Rumble 3		No Class PD Meeting		Study Day No Class		Lesson 12 Gradient and Directional Derivative		No Class	
	6	19-Sep-22	1-13	20-Sep-22	2-10	21-Sep-22	1-14	22-Sep-22	2-11	23-Sep-22	1-15
		Lesson 13 Optimization Project (In-class work)		No Class PD Meeting		Lesson 14 MV Optimization 1 (Min/Max) Rumble 4		No Class		Lesson 15 MV Optimization 2 (CIM)	
	7	26-Sep-22	1-16	27-Sep-22	2-12	28-Sep-22	N/A	29-Sep-22	1-17	30-Sep-22	2-13
		Lesson 16 Lagrange Multipliers		No Class		Study Day/ Honorable Living Day WPR 2		Lesson 17 Class Drop		No Class	
	8	3-Oct-22	1-18	4-Oct-22	2-14	5-Oct-22	1-19	6-Oct-22	2-15	7-Oct-22	1-20
		Lesson 18 Project 1 Presentations		No Class		Lesson 19 Project 1 Presentations		No Class		Lesson 20 Vol. Est., Iterated Integrals	
	9	10-Oct-22	N/A	11-Oct-22	2-16	12-Oct-22	1-21	13-Oct-22	2-17	14-Oct-22	1-22
		Columbus Day No Class		No Class PD Meeting		Lesson 21 General Regions		No Class		Lesson 22 Polar Regions	
	10	17-Oct-22	1-23	18-Oct-22	2-18	19-Oct-22	N/A	20-Oct-22	1-24	21-Oct-22	2-19
		Lesson 23 Application 1: Center of Mass		No Class PD Meeting		Study Day No Class		Lesson 24 Application 2: Probability Rumble 5		No Class	
	11	24-Oct-22	1-25	25-Oct-22	2-20	26-Oct-22	1-26	27-Oct-22	2-21	28-Oct-22 Modified	1-27
		Lesson 25 Interdiscplinary Project (In-class work)		No Class PD Meeting		Lesson 26 Triple Integrals		No Class		Lesson 27 Triple Integrals in Cylindrical Coord Rumble 6	
	12	31-Oct-22	1-28	1-Nov-22	2-22	2-Nov-22	N/A	3-Nov-22	1-29	4-Nov-22	2-23
		Lesson 28 Triple Integrals in Sphereical Coord		No Class		Study Day WPR 3		Lesson 29 Class Drop		No Class	
Block 3	13	7-Nov-22	1-30	8-Nov-22	2-24	9-Nov-22	1-31	10-Nov-22	2-25	11-Nov-22	N/A
		Lesson 30 Vector Fields		No Class PD Meeting		Lesson 31 Parametric Equations, Line Integrals		No Class		Veteran's Day No Class	
	14	14-Nov-22	1-32	15-Nov-22	2-26	16-Nov-22	N/A	17-Nov-22	1-33	18-Nov-22	2-27
		Lesson 32 Fundamental Theorem for Line Integrals		No Class PD Meeting		Study Day No Class		Lesson 33 Curl and Divergence		No Class	
	15	21-Nov-22	1-34	22-Nov-22	2-28	23-Nov-22	N/A	24-Nov-22	N/A	25-Nov-22	N/A
		Lesson 34 Green's Theorem Rumble 7		No Class		No Classes No Class		Thanksgiving No Class		Training Holiday No Class	
	16	28-Nov-22	1-35	29-Nov-22	2-29	30-Nov-22	1-36	1-Dec-22	2-30	2-Dec-22	1-37
		Lesson 35 Green's/ Parametric Surfaces		No Class PD Meeting		Lesson 36 Surface Area Rumble 8		No Class		Lesson 37 Surface Integrals	
	17	5-Dec-22	1-38	6-Dec-22	N/A	7-Dec-22	1-39	8-Dec-22	N/A	9-Dec-22 Modified	1-40
		Lesson 38 WPR 4		Study Day No Class PD Meeting		Lesson 39 Stokes' Theorem		Study Day No Class		Lesson 40 The Divergence Theorem	
TEE	18	12-Dec-22	N/A	13-Dec-22	N/A	14-Dec-22	N/A	15-Dec-22	N/A	16-Dec-22	N/A
		TEE Week		TEE Week		TEE Week		TEE Week		TEE Week	

KEY	No Class	Project	Course-wide Graded Event	Rumble	Assignment or Modified Sched.

MA205: Multi-Variable Calculus
AY23-01 Syllabus

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Textbook: Calculus - Ninth Edition, Early Transcendentals by James Stewart				
Lesson	***Date***	***Topic (Reading)***	***Homework (WebAssign)***	***Lesson Objectives***
		BLOCK 1: VECTORS AND GEOMETRY OF SPACE		
Lesson 1	8/15/2022	3D Coord Systems & Geometry of Vectors (12.1 - 12.2)	**12.1:** 7, 12, 37, 39 **12.2:** 13, 21, 25	1.1. Understand how to model position and distance in 3D. 1.2. Understand the properties of vectors: addition, scalar multiplication, length, magnitude, standard basis vectors, and position vectors.
Lesson 2	8/17/2022	Dot and Cross Product (12.3 - 12.4)	**12.3:** 8, 9, 11, 19, 41, 51 **12.4:** 1, 39, 41	1.3. Solve for and understand the definition and utility of the dot and cross products for practical applications between two vectors.
Lesson 3	8/19/2022	Eqns. of Lines & Planes/Quadric Surfaces (12.5-12.6)	**12.5:** 5, 27, 31, 71 **12.6:** 23-30 **12.3:** 14 **Rumble 1**	1.4. Model and understand relationships between lines and planes in three-dimensional space. 1.5. Determine the distance between different objects/points in three-dimensional space. 1.6. Model quadratic surfaces in three-dimensions. 1.13. Employ technology to visualize quadric surfaces.
Lesson 4	8/22/2022	Vector Functions & Space Curves (13.1)	**13.1:** 1, 22, 49, 25, 27, 29, 30	1.7. Model space curves in three-dimensions using vector functions.
Lesson 5	8/24/2022	Derivatives & Integrals of Vector Functions (13.2)	**13.2:** 10, 21, 37, 43, 44, 49 **Rumble 2**	1.8. Compute derivatives and integrals of vector functions. 1.9. Compute the tangent line to a space curve and understand its significance.
Lesson 6	8/26/2022	Arc Length & Curvature (13.3 - stop at Torsion p. 911)	**13.3:** 4, 17, 29, 51	1.10. Determine distance along a three-dimensional space curve. 1.11. Understand how the normal, binormal, and curvature of a space curve determine its shape.
Lesson 7	8/29/2022	Motion in Space: Vel. & Acc. (13.4 - stop at Kepler's laws p. 921)	**13.4:** 3, 11, 18, 23, 25	1.12. Model the motion of an object in three-dimensional space.
	8/31/2022	WPR1	**Study Day**	
Lesson 8	9/1/2022	Exam Comp Day	**No Class**	
		BLOCK 2: MULTIVARIABLE DIFFERENTIATION		
Lesson 9	9/7/2022	Multivariable Functions (10.3 - stop at Example 7, 14.1)	**10.3:** 12, 21, 24 **14.1:** 4, 21, 32, 34	2.1. Understand the definition of a function of two variables in cartesian and polar coordinates and how it can be represented algebraically, numerically, graphically, or verbally.
Lesson 10	9/9/2022	Partial Derivatives/Chain Rule (14.3 - stop at Partial Differential Equations p. 968; 14.5 - stop at implicit differentiation p. 990)	**14.3:** 20, 22, 47, 82 **14.5:** 5, 7, 16, 44	2.2. Given a function of several variables, calculate its partial derivatives. 2.3. Understand the relationship and differences between average rate of change and instantaneous rate of change of a multivariable function using partial derivatives. 2.6. Given a multivariable function and parameterizations for the function, use the chain rule to solve a rates of change problem.
Lesson 11	9/12/2022	Tangent Planes (14.4 - stop at Differentials p. 979)	**14.4:** 3, 17, 25, 27 **Rumble 3**	2.4. Given a function of two variables and a point, find the equation of the plane tangent to the surface at the given point. 2.5. Approximate the value of a multivariable function using a linear equation.
Lesson 12	9/15/2022	Gradient and Directional Derivative (14.6)	**14.6:** 1, 9, 13, 17, 30, 35	2.7. Understand what a directional derivative is in terms of a rate of change graphically, algebraically, and numerically. 2.8. Given a function f(x,y) of two variables, find the directional derivative of f(x,y) at a given point in any direction. 2.9. Given a function of two variables, find the gradient vector at a specified point. 2.10. Understand that the gradient vector gives the direction of the greatest increase in functional value at a given point for a differentiable function. 2.11. Understand that the magnitude of the gradient vector gives the maximum rate of change of a differentiable function at a given point. 2.12. Understand geometrically how the gradient vector is related to level curves. 2.13. Employ technology to calculate, visualize, and understand the Directional Derivative and Gradient of a multivariable function.
Lesson 13	9/19/2022	Optimization Project Assignment (In-class work)		
Lesson 14	9/21/2022	MV Optimization 1 (Min/Max) (14.7)	**14.7:** 3, 13*, 21*, 25*, 46, 47 *Technology Problems **Rumble 4**	2.14. Employ technology to calculate, visualize, and understand optimization problems. 2.15. Understand the definition of a critical point and be able to find the critical points of a function of two variables. 2.16. Use the Second Derivatives Test to classify a critical point as a local maximum, local minimum, or saddle point
Lesson 15	9/23/2022	MV Optimization 2 (Closed Interval Method) (14.7)	**14.7:** 38, 51, 54, 55	2.17. Find the absolute maximum and minimum of a given continuous function f(x,y) on a closed and bounded set using the Closed Interval Method for functions of two variables. 2.18. Model and solve unconstrained and constrained multivariable optimization problems.
Lesson 16	9/26/2022	Lagrange Multipliers (14.8)	**14.8:** 1, 5, 9, 33, 49	2.19. Solve optimization problems for functions of several variables subject to a single constraint using the Method of Lagrange Multipliers. 2.20. Solve optimization problems for functions of several variables subject to two constraints using the Method of Lagrange Multipliers.
	9/28/2022	WPR2	**Study Day**	
Lesson 17	9/29/2022	Exam Comp Day	**No Class**	
Lesson 18	10/3/2022	Project 1 presentations		Groups will be assigned a presentaiton time by their instructor during one of these class times
Lesson 19	10/5/2022	Project 1 presentations		

MA205: Multi-Variable Calculus
AY23-01 Syllabus

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Lesson	*Date*	*Topic (Reading)*	*Homework (WebAssign)*	*Lesson Objectives*
		BLOCK 3: MULTIVARIABLE INTEGRATION		
Lesson 20	10/7/2022	Volume Estimation/Iterated Integrals (15.1)	**15.1:** 1, 6, 7, 15, 19, 21, 42	3.1. Given a function f(x,y), use a double Riemann sum to estimate volume under a surface. 3.2. Understand the concept of volume under a surface as the limit of the sums of volumes of rectangular columns. 3.3. Use level curves to estimate the volume under a surface. 3.4. Estimate the average value of a function of two variables.
Lesson 21	10/12/2022	General Regions (15.2)	**15.2:** 5, 7, 8, 20, 39	3.5. For any region in the xy-plane, sketch the region, label the boundaries, and label the points of intersection. 3.6. Compute the iterated integral of a multivariable function over a specified domain.
Lesson 22	10/14/2022	Polar Regions (15.3)	**15.3:** 6, 11, 30, 33, 45	3.7. Understand and use the polar coordinate system to solve iterated integrals.
Lesson 23	10/17/2022	Center of Mass (8.3 - skim, 15.4 - stop at Moment of Inertia p. 1072)	**15.4:** 1, 3, 15	3.8. Find the mass, moments, and center of mass of any lamina whose area is a specified region in the xy-plane and whose density is a given function ρ(x,y). 3.9. Employ technology to find the mass, moments, and center of mass of any lamina whose area is a specified region in the xy plane and whose density is a given function ρ(x,y).
Lesson 24	10/20/2022	Probability (15.4 - start at Probability p. 1074)	**15.4:** 29, 31, 32 **Rumble 5**	3.10. Understand that a joint density function is a probability density function of multiple continuous random variables. 3.11. Given a joint density function of a pair of random variables X and Y determine the probability that f(x,y) lies in a region D. 3.12. Employ technology to create joint density functions and calculate probabilities. 3.13. Verify if multivariable functions are valid joint density functions.
Lesson 25	10/24/2022	Project 2 Work		
Lesson 26	10/26/2022	Triple Integrals (15.6)	**15.6:** 4, 5, 9, 13, 18	3.14. Evaluate a triple integral over a general region. 3.15. Employ technology to compute triple integrals.
Lesson 27	10/28/2022	Triple Integrals in Cylindrical Coordinates (15.7)	**15.7:** 10, 15, 22, 32 **Rumble 6**	3.16. Understand when and how to apply the cylindrical coordinate system to solve iterated integrals.
Lesson 28	10/31/2022	Triple Integrals in Spherical Coordinates (15.8)	**15.8:** 21, 23, 26, 27	3.17. Understand when and how to apply the spherical coordinate system to solve iterated integrals.
	11/2/2022	WPR3	**Study Day**	
Lesson 29	11/3/2022	Exam Comp Day	**No Class**	
		BLOCK 4: VECTOR CALCULUS		
Lesson 30	11/7/2022	Vector Fields (16.1)	**16.1:** 7*, 19, 11*, 27, 13, 15, 16, 31, 33 *Technology Problems **Project 2 presentation due**	4.1. Understand vector fields and their different applications. 4.14. Use technology to plot different types of vector fields.
Lesson 31	11/9/2022	Parametric Equations, Line Integrals (16.2)	**16.2:** 3, 7, 17, 22, 41	4.2. Understand the meaning and compute the line integral along a curve in space. 4.3. Understand different applications of line integrals. 4.15. Employ technology to pass a vector defined function to a function. 4.16. Employ technology to visualize a particle's path along a vector field.
Lesson 32	11/14/2022	Fundamental Theorem for Line Integrals (16.3 - stop at Conservation of Energy p1150)	**16.3:** 1, 3, 4, 7, 18, 23	4.4. Understand and apply the Fundamental Theorem for line integrals.
Lesson 33	11/17/2022	Curl and Divergence (16.5 - stop at Vector Forms of Green's Theorem p1167)	**16.5:** 1, 3, 7*, 11, 18 *Technology Problem	4.6. Understand the "del" operator and use it to compute curl and divergence. 4.7. Understand the physical representation of curl and divergence. 4.17. Employ technology to calculate curl and divergence of a vector defined function.
Lesson 34	11/21/2022	Green's Theorem (16.4)	**16.4:** 3, 5, 9, 17 **Rumble 7**	4.5. Understand and apply Green’s Theorem to general two-dimensional regions.
Lesson 35	11/28/2022	Green's/Parametric Surfaces (16.6 - stop at Surface Area p1177)	**16.4:** 7 **16.6:** 19, 20, 33, 35	4.8. Understand how vector functions can be used to model surfaces in three-dimensional space.
Lesson 36	11/30/2022	Surface Area (15.5, 16.6 - start at Surface Area p1177)	**16.6:** 40, 39, 45 **15.5:** 4 **Rumble 8**	4.9. Determine the area of a parametric surface.
Lesson 37	12/2/2022	Surface Integrals (16.7)	**16.7:** 5, 9, 23	4.10. Understand the meaning and compute surface integrals of a scalar function over a given surface. 4.11. Use the surface integral to determine the flux across a solid region.
Lesson 38	12/5/2022	WPR4		
Lesson 39	12/7/2022	Stokes' Theorem (16.8)	**16.8:** 9, 10, 19, 21	4.12. Understand and apply Stokes’ Theorem to general surfaces.

TAB E AY2022 COURSE ASSESSMENT

DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
West Point, New York 10996

MADN- **REPLY TO ATTENTION OF** MATH 15 February 2022

MEMORANDUM THRU

COL Yankovich, Service Program Director, Department of Mathematical Sciences, USMA, West Point, NY 10996-1786
COL Hartley, Professor and Head, Department of Mathematical Sciences, USMA, West Point, NY 10996-1786

FOR Office of the Dean, AARS, United States Military Academy, West Point, NY 10996-1926

SUBJECT: Annual Executive Summary for MA205, AY22-01

1. **Course Description.** MA205: Multivariable Calculus and Introduction to Differential Equations provides a foundation for the continued study of mathematics and for the subsequent study of the physical sciences, social sciences, and engineering. MA205 covers topics in multivariable differential and integral calculus, differential equations, and infinite series representations of functions. Throughout the course mathematical models motivate the study of topics such as optimization, accumulation, change in several variables, differential equations, and other topics from the natural, social, and decision sciences. An understanding of course material is enhanced through the use of the computer algebra system, MATLAB.

2. **Status of curricular change initiatives completed during the current academic year.** Each semester, beginning in AY19, the MA205 leadership reached out to external departments we support to gain perspective regarding their observations of cadets' abilities following the successful completion of MA205. Referencing the USMA Redbook, we reviewed which courses listed MA205 as a prerequisite, identifying 35 follow-on courses which require multivariable calculus skills. We regularly contact the course leadership from these courses soliciting feedback on MA205 cadets' abilities with multivariable calculus and differential equations concepts, and any programming languages used in their courses.

From AY 21-01 to 22-01, the MA205 course leadership met with course directors from the Departments of Electrical Engineering and Computer Science, Mechanical Engineering, Systems Engineering and Chemistry and Life Sciences. Feedback from previous semesters was the genesis for the transition from Mathematica to MATLAB that was completed in AY20, but neither department was overly concerned with the programming language introduced in MA205. The Department of Civil and Mechanical Engineering does not have a standard programming language, but the mechanical engineering program does use MATLAB for at least two of their classes. The Department of Systems Engineering uses a variety of programming languages and was more concerned with mapping the MA206 syllabus and course outcomes to systems engineering courses. MATLAB is a general programming language with direct application in a variety of STEM fields. For this reason, it continued to be the programming language for MA205 through AY22-01, but as LTC Lee Evans recommended in AY20-01, this decision must be revisited, and adjusted as necessary, in future semesters. In AY22-01, MA205 began its internal review of MATLAB use. Paragraph 5 below discusses the status of the on-going MATLAB review.

a. **Mode of Instruction-A New Normal.** In AY21-01, we were given the option to instruct in-person (with restrictions on numbers of cadets allowed in each classroom), full remote, or a combination of the two. The course leadership spent a considerable amount of time over the summer determining the best method of instruction. This deliberate decision-making process led to the determination that having cadets in the classroom was the best mode of instruction. Enclosure 1, published in Jan 2022 in the Journal of Humanistic Mathematics, captured the lessons learned from transitions from emergency remote teaching to a time-constrained class.

In AY21-02, MA205 returned to a 'new normal' during its off-cycle. Due largely in part to the smaller population of cadets, MA205 instructors taught in classrooms that accommodated more than 18 cadets while maintaining COVID protocols. The 'new normal consisted' of 38, 70-minute, in-person lessons. Based on the Dean's guidance, the Term End Exam (TEE) also remained 2.5 hours long instead of the normal 3.5 hours. In AY22-01, MA205 returned to a healthy population of 275 cadets completing the course, up from 199 in AY21-01. In addition, almost every COVID restrictions, testing requirement and COVID classroom protocols were lifted. Most notability, MA205 returned to 40, 75-minute, in-person lessons. The Term End Exam's (TEE) length returned to 3.5 hours.

b. **Interdisciplinary Course Projects (implemented in AY21-01).**

We continue to foster an interdisciplinary perspective throughout the course. Multivariable calculus is inherently interdisciplinary, and we incorporate problems related to a variety of STEM disciplines. In AY21-02 CPT Shane Smith coordinated with the Department of Civil and Mechanical Engineering and the Economics Program when designing the course project on the topics of structural integrity and production optimization. For AY22-01's course project, MA205 reached out to CPT Caspear Yi, from the Department of Chemistry and Life Sciences, and received a course project concept that is discussed further below.

c. **Increased MA104 Validations (implemented in AY20-02 and AY21-01, continued for AY21-02).** Unlike the past two off-cycle semesters, the number of Plebe MA104 validators decreased from 45 validators in AY21-02 and 50 MA104 validators in AY20-02 to only 16 in AY22-2. The reason for the steep decrease in MA104 validators was due to the Advanced Math Program (AMP) increasing its enrollment by 60 first-year cadets. It is unclear if the increased AMP enrollment is a permanent or temporary trend.

3. **Evaluation of Cadet Learning Outcomes.** MA205 maps to five Academic Program Goals (APGs). Enclosure 5 shows the six MA205 Course Goals mapped to the five APGs. The MA205 Course Goals can also be mapped to two Middle States Accreditation Standards of Design and Delivery of the Cadet Learning Experience and Educational Effectiveness Assessment. While the Course Learning Objective map more easily to the APGs that align with Educational Effectiveness Assessment, we have mapped the Course Learning Objectives to all five of the supported APGs, including assessment techniques for each as shown in Enclosure 6.

The assessments throughout the semester are planned around the Course Learning Objectives. This deliberate effort resulted in the evaluation of 70 of the 73 Course Learning Objectives (CLOs) during the semester. Of concern is the number of cadets who failed the TEE. In AY21-01, there were 65/195 (33.3%) who failed the TEE compared to 42/235 (18%) the previous year. Most recently, in AY22-1, there were 92/275 (33.5%) who failed the TEE. Overall, 23 cadets failed the course. It is important to note that the TEE administered in AY22-01 was nearly identical to AY19-01's TEE with the exact same grading rubric. It is believed that the deficient performance of many of the cadets is in large part due to when and how the cadets learned their fundamental mathematical skills. The overwhelming majority of these cadets took MA103 and MA104 in a sub-optimal classroom environment full of COVID distractions and two completely different teaching modalities. While MA103 utilized a delivery method like MA205 in AY21-1, MA104's Hy-Flex approach proved vastly different. Surely, an increase in cadet quantitative performance was not expected.

a. **APG 1: Graduates communicate effectively with all audiences.**

i. **Summary:** MA205 supports APG 1 through MA205 Course Goal 3.

ii. **Assessment: (NO CHANGE)** MA205 Course Goal 3 is "Communicate Effectively," which is primarily through the course project. The course project for AY21-01 consisted of an oral presentation on solving gradient descent problems in MATLAB, followed by a written executive summary, completed in groups of three cadets. cadets were required to submit their recommendations using a memorandum formatted in accordance with AR 25-50. This was an opportunity to combine concepts from the course with professional development, exposing the cadets to written communication standards that will be expected of them as officers. Many cadets commented in the mid-course feedback that they appreciated having to submit their write-up in memorandum format because they took time to properly format the document and now have a template for future memorandums.

b. **APG 2: Graduates think critically and creatively.**

i. **Summary:** MA205 supports APG 2 through MA205 Course Goals 2, 5, and 6.

ii. **Assessment:** MA205 is designed as a service course for science, technology, engineering, and mathematics (STEM) majors. As such, we seek to develop cadets who think critically and creatively in order to solve a myriad of STEM problems using the concepts taught in MA205. Course Goals 2, 5, and 6 are "Apply Technology, Build Competent and Confident Problem Solvers, and Foster Interdisciplinary Perspective," respectively.

In AY21-01, cadets did well on the MATLAB problem set, earning an average score of 84.28%, and relatively well on the validation exercise with a course average of 78.88%. However, when asked about how confident they were with MATLAB, cadet feedback indicated that 43% of the MA205 cadets were either *somewhat hesitant* or *very hesitant* in their ability to leverage MATLAB to solve problems outside of the course. This is an increase from the previous semester. While this could be explained by the fact that instructors dedicated less time to MATLAB in a 30-minute class period, we have built in eight lessons for AY21-02 where instructors will demonstrate MATLAB on specific topics in class.

Based on feedback received over the past few semesters, cadets felt blindsided by the amount of integration required on the course project. We cannot blame them considering the course was void of any formal MATLAB instruction on integration. As a result, in AY 22-01, CPT Shane Smith split the two MATLAB differentiation workdays into one differentiation workday during Block I, and one integration workday during Block II (See Enclosure 2 to review the Course Calendar). Figure 1 shows a comparison of MATLAB delivery between AYs 20-21, and AY 22. To supplement the MATLAB introduction during Block I, the course created four tutorial videos to flatten learning curves. The tutorial topics covered Live Editor, Matrix Operations, Basic Commands and Functions, Visualization (Plotting). This effort avoided the need to cover MATLAB during class. cadets could review MATLAB material on demand, outside of class, without instructor assistance. Cadets then used solved problem sets (Enclosure 3) to reinforce their elementary MATLAB skills for both Blocks and cover concepts learned during class.

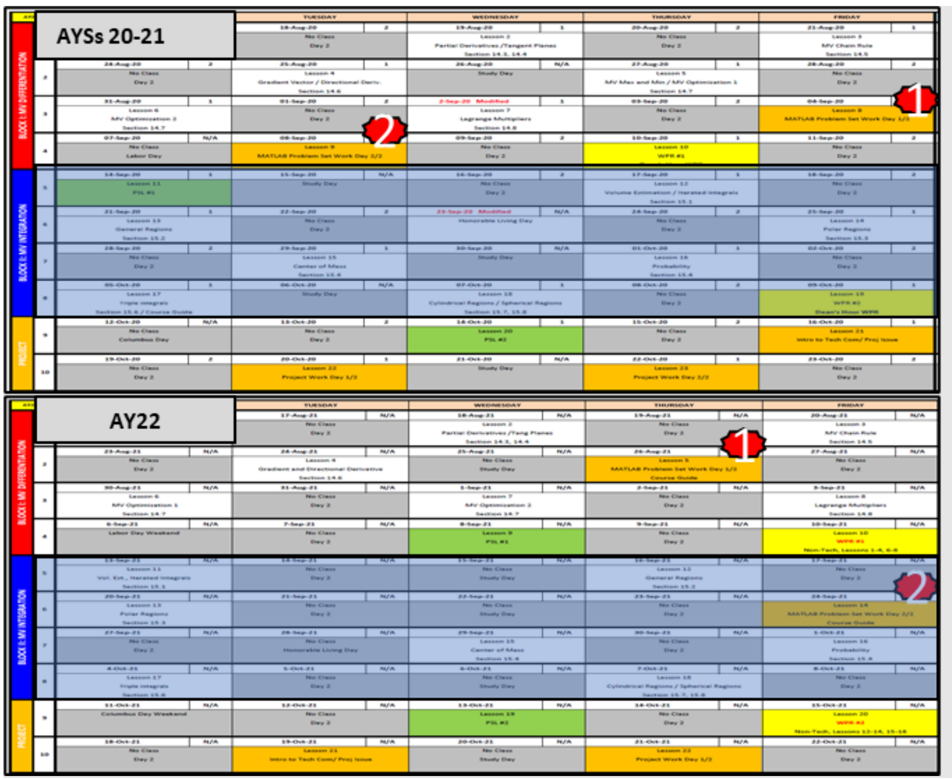

Figure 1. A Calendar comparison of MATLAB workdays.

In past semesters, cadets seemed uncomfortable with starting the course project. Not this past semester. Based on cadet observations, course leadership witnessed cadets take on the Course Project with a full head of steam. Perhaps additional resources and multiple repetitions at the same material reduced anxiety and improved expectation clarity. Still, as seen in Figure 2, when asked about how confident they were with MATLAB, cadet feedback indicated between 40-45% of the MA205 cadets had little to no confidence in their ability to leverage MATLAB. This is really no change from the previous semester. We had hoped to see a bigger shift in cadets' confidence in MATLAB over the course of Blocks I and II, and after completing the course project. In brief, three repetitions at MATLAB do not seem to result in the confidence swing MA206 experiences. MA205 continues to pursue finding its technology sweet spot.

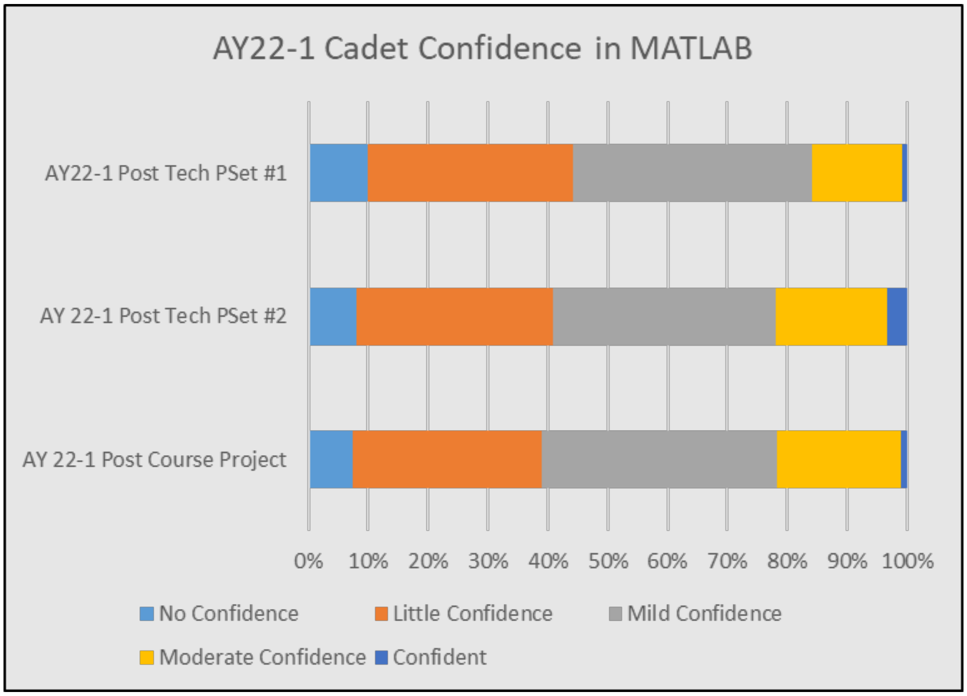

Figure 2. Self-Reported Cadet Confidence in MATLAB.

c. **APG 3: Graduates demonstrate the capability and desire to pursue progressive and continued intellectual development.**

i. **Summary:** MA205 supports APG 3 through MA205 Course Goals 4 and 5.

ii. **Assessment:** Cadets are undoubtedly spending more time preparing for MA205 as compared to three years ago but assessing the effectiveness of the daily preparation assignments continues to be difficult. Both graphs in Figure 3 show a positive correlation between the daily preparation assignments and the TEE scores, used as a proxy for cadet comprehension. The WebAssign scores are a better predictor of TEE scores, but we believe this is because cadets can take notes without truly understanding the material, whereas WebAssign forces cadets to show their understanding of the material. Chegg and Slater can be used as ways around this for WebAssign since the WebAssign problems are from the textbook, but we have used the WebAssign setting that alters the numbers in the problem for each cadet. At a minimum, cadets need to look at the solution and determine how to modify each step for the randomly assigned values displayed on their individually assigned WebAssign assignment, adding an additional layer that can lead to learning. For this reason, along with two weeks of AY21-02 being conducted in a virtual environment, more weight was placed on WebAssign for daily preparation grades.

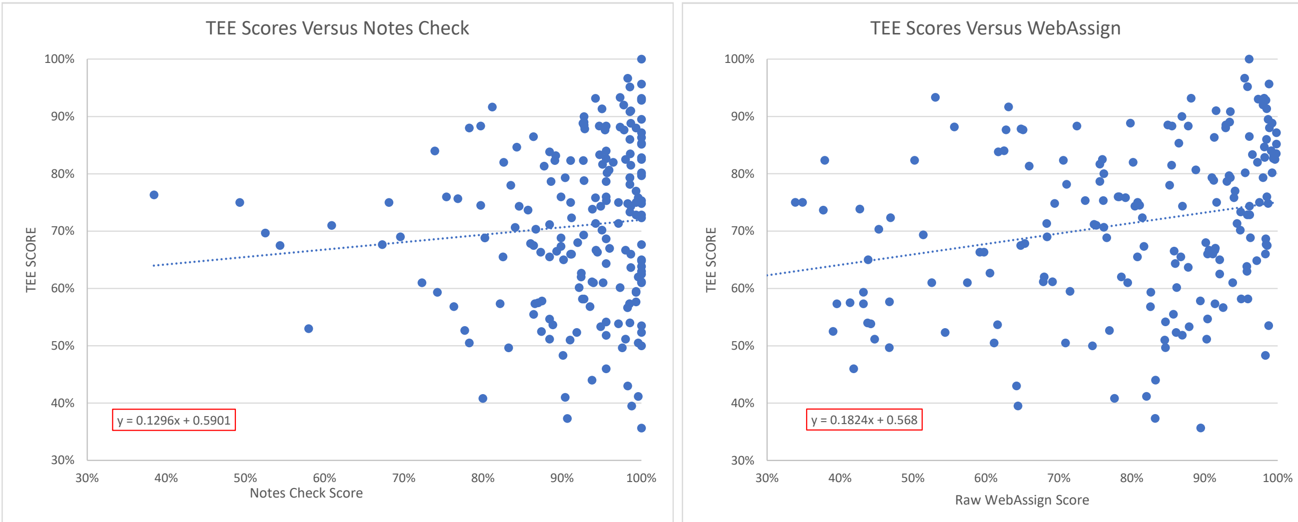

Figure 3. A comparison of TEE scores to the daily preparation assignments of notes checks and WebAssign.

Heading in AY22-01, we kept Daily Preparation quizzes (21/150 class preparation points), and WebAssign Homework (81/150 class preparation points), but dropped notes checks and replaced them by course quizzes we named "Rumbles." There are three main reasons for implementing Rumbles. First, course-wide, low-stakes, quizzes provide cadets and instructor feedback on learning objectives understanding before major graded events. Second, as discussed before, these quizzes combat the "Chegg effect" and "fluency illusion", ultimately requiring cadets to practice retrieval skills in a test-like environment. Lastly, we felt that we could review more material in the same amount of time. Since notes checks provided subjective feedback to 24 Learning Objectives (LO), we envisioned Rumbles would provide objective feedback to 34 LOs. We allotted 48/150 points preparation points and only 15 minutes class time to the Rumbles to minimize impacts to the flow of in-class instruction.

Mid-Course and End of Course analysis shows Rumbles (course quizzes) are the best predictor of TEE performance. Figure 2 shows an R^2 of .39 which dominates Daily Preparation and WebAssign's R^2 values of 0.18 and 0.13, respectively. This should come as no surprise since Rumbles were created to replicate an exam environment with limited time and resources. Although we only assessed 22 LOs out of the 34 LOs we wanted to assess, we felt that the quality of feedback and cadet preparations trumped the quantity of LOs recalled.

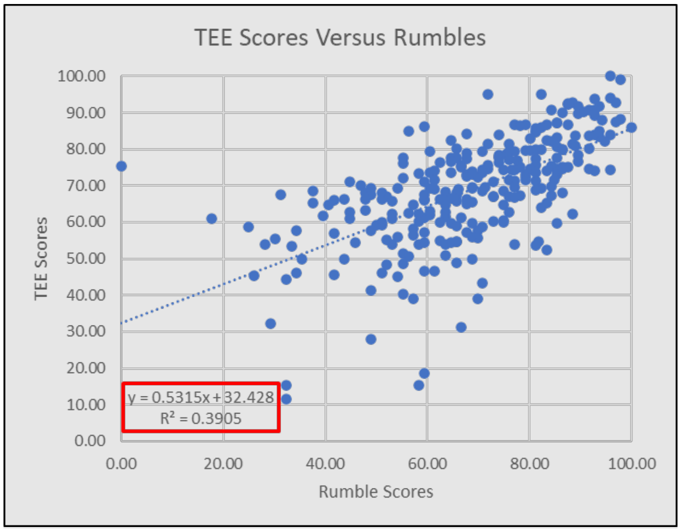

Figure 4. A comparison of TEE scores to Rumbles.

While some cadets may not see the value of the "Thayer Method". Coming to class prepared is the second highest indicator of exam performance. Helping cadets connect assigned homework problems (focused on Learning Objectives) to exam questions (assessing Learning Objectives) is a course challenge. Both graphs in Figure 5 show a positive correlation between the daily preparation assignments and the TEE scores.

Once again, daily preparation quizzes are better associated with TEE success when compared to WebAssign homework grades. Like the ongoing MATLAB review, we recommend exploring different options for requiring cadets to complete WebAssign Homework for points, but for different reasons. Luckly, at least hopefully, the impact of the pandemic is lessening. The need to hold cadets accountable for out-of-class time is nearly gone. Moreover, there are too many ways cadets receive Stuart's exercise answers online and expecting cadets to cite each source detracts from the time we want them to focus on reinforcing what they learned in class. A couple WebAssign alternatives include instructor-unique homework/problem sets, asking Cengage (WebAssign) to scramble different calculus book homework questions for cadets to answer, or only rely on Rumbles to assess cadet progress.

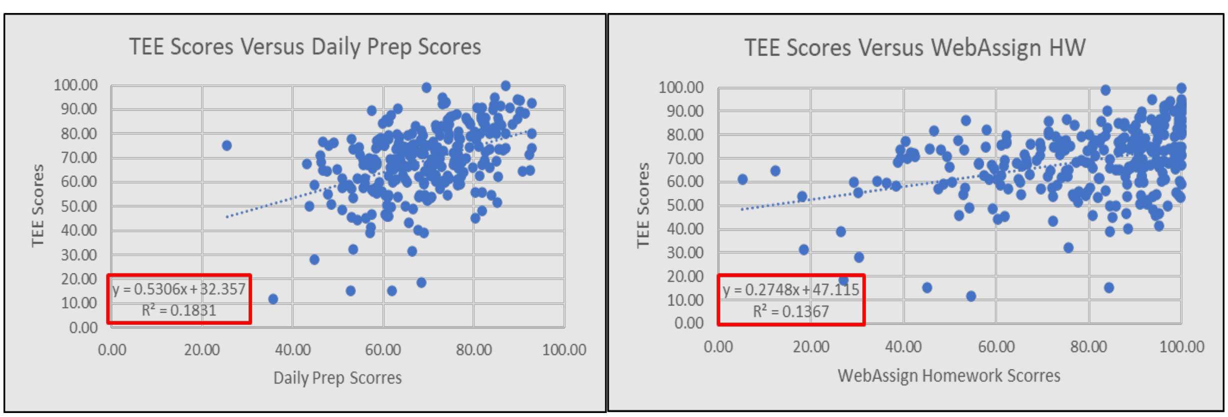

Figure 5. A comparison of TEE scores to the daily preparation assignments of notes checks and WebAssign.

d**. APG 5: Graduates apply science, technology, engineering, and mathematics concepts and processes to solve complex problems.**

i. **Summary:** MA205 supports APG 5 through MA205 Course Goals 2 and 5.

ii. **Assessment:** MA205 Course Goals 2 and 5 are "Apply Technology and Build Competent and Confident Problem Solvers." To assess the cadets' ability to apply technology to solve complex problems, we incorporated technology into the course project. Prior to AY20-02, technology was tested on the WPRs and the TEE. Since AY21-01 (excluding AY20-02), we have continued the practice of not testing technology on the WPRs and the TEE. Technology was tested on the course project which had cadets solve an unfamiliar problem using concepts discussed in the course. We continue to use the course project as an opportunity to expose cadets to STEM-related problems. The project topic used for AY21-01 was gradient descent, machine learning, and anomaly detection. The project in AY21-02 focused on structural integrity and optimization. In Fall of 2021, we teamed up with the Department of Chemistry and Life Science to calculate a dipole moment. We believe the project is the most appropriate forum for applying technology, as it should serve as a tool to build competent and confident problem solvers.

e**. APG 7: Graduates integrate and apply knowledge and methodological approaches gained through in-depth study of an academic discipline.**

i. **Summary:** MA205 supports APG 1 through MA205 Course Goals 1 and 6.

ii. **Assessment:** MA205 Course Goals 1 and 6 are "Acquire a Body of Knowledge and Foster Interdisciplinary Perspective." This is at the core of what we want cadets who have successfully completed MA205 to carry with them once they move on to their specific disciplines. One alarming statistic is the number of cadets who failed the TEE, as previously stated. As seen in Table 1 below, the TEE averages have dropped, then levelled off in the past few years. There are a couple of likely reasons. First, we believe the overarching reason for the drop in performance is the product of a different learning environment and the multiple COVID related distractions. The various learning environments (face-to-face, Hy-Flex, and hybrid) and distractions began the previous academic year spring, especially when academia made some tough decisions to protect the welfare of cadets and faculty. Based on the data, we hope the impact of the pandemic on cadet's quantitative reasoning skills have levelled off. Our second reason points to a widening gap of cadet aptitude entering West Point with the desire to major in a STEM field. A few concerning trends in Figure 7 in the following paragraph support this claim.

AY19-01	AY19-02	AY 20-01	AY20-02	AY21-01	AY21-02	AY22-01
77.8%	74.31%	75.7%	COVID	70.85%	68.04%	68.85%

Table 1. Term End Exam Scores since AY19.

4. **Data Driven Decisions to improve tracking Course Performance and Assessing Cadet Risk.**

During the past eighteen months, MA205 spearheaded the Department of Mathematical Science's efforts to make informed, data-based course decisions. CPT Shane Smith compiled data to analyze MA205's course performance at a macrolevel and microlevel. Both levels are equally important when we apply pedology methods to pending course changes; how might the change affect overall course objectives and be executed by instructors and cadets.

Although not as inclusive as envisioned, Figure 7 is just the beginning of what can be a powerful tool to help explain why, and when decisions were made and their impact on course facilitation and cadet performance. For example, the top blue line shows the course average over the past eight years. We can safely assume that the fluctuation of cadet performance depends on course design changes and even program director oversight highlighted by the yellow boxes. The gray line represents the lowest cadet grade by semester. While this data point may not be indicative of the overall trend to explain challenges, it begs the further analysis of at-risk students or at least points to cause for concern of the variability of skills for entering cadets. The yellow line highlights the percentage of D and F letter grades for each semester and the orange trend line tells a story of the standard deviation of cadet grades over the last eight years.

Data for this product is readily available and easy to consume. Future MA205 semesters can carry forward ideas that worked, look upstream to how other course decisions impact MA205, and evaluate how MA205 course decisions impact downstream courses.

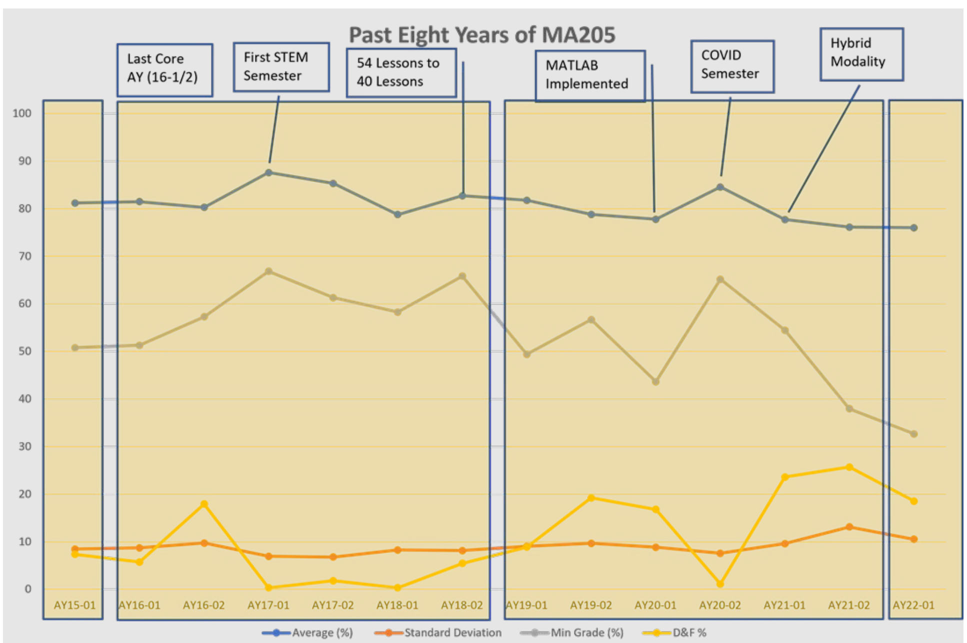

Figure 7. MA205 Longitudinal Data

Data can provide insight into a cadet's future potential by analyzing their past performance. For the past two years, MA205 used a simple linear regression model to help meet cadets at their entrance capability. The linear regression model, or "Objective and Neutral- Performance and Risk" (ON-PAR) Assessment is based on each cadet's past performance. The Linear Regression includes, Core APS, MA103/104/153 grades, and MA104 WRIT scores and "refresher quizzes" given the first day of each block of instruction. This tool assists course leadership with sectioning by cadet ability and determining a cadet's risk of failing the course before Lesson 1! ON-PAR has successfully flagged 95% of cadets deemed "At-Risk" at the beginning of the semester. These remarkable results also account for those flagged as possible course failures.

While powerful, data mining is VERY time intensive. Obtaining MA103/104/153 final grades, MA104 WRIT scores, and entering "refresher quizzes" is simple because it can be done with data collected within D/Math. However, scraping 350+ cadets' Core APS from AMS (Academy Management System) may not be an appropriate use of course leadership's time. In short, MA205 will continue to coordinate with Dean Staff to find an easier way to access and share data across USMA's faculty.

Admin Data:
Orange: Repeater
Green: Validator
Blue: MA153 Dropdown

WPR Predictor is based on each cadet's past performance. The Linear Regression includes, Core APS, MA103/104/153 grades, MA104 WRITs, or "refresher quizzes" given the first day of each block.

WPR Score and Error from Prediction

Reading Quizzes related to to Learning Objectives. Reading Quiz scores correlate to WPR Performance

Cadet Name	Co	Cadet ID	G158	Class	Core APS	MA 103/4 or MA 153 Grade	MA 104 WRIT 2 (DIFF)	DRQ P1	DRQ P2	DRQ P3	DRQ P4	Total %	DRQ WPR Predictor	WPR Score	Error +/-	Lesson 2 Prep Quiz	Lesson 3 Prep Quiz	Lesson 4 Prep Quiz	Lesson 4 Prep Quiz	Lesson 6 Prep Quiz	Lesson 7 Prep Quiz
KIM, SAMUEL	H4	C89584497	H159	23	2.284	3.67	100	1	1	1	2	100.00	69.23717605	81	11.76282395	2	5	4	5	4	3
LOPEZ, JOSE	G4	C43016185	H159	24	2.335	3.00		1	1	1	2	100.00	69.40014991	83	13.59985009	5	4	5	0	4	4
MAYS, BRANDON	D4	C50882319	H159	23	1.912	1.67	81	1	1	1	1	80.00	58.57230939	73	14.42769061	5	4	5	0	4	4
MILLER, JOSEPH	I2	C84612829	H159	23	2.113	2.67	70	1	1	1	2	100.00	65.34318726	81	15.65681274	5	5	5	0	5	5
RANDOLPH, PRESTON	D3	C46526177	H159	24	2.606	2.67	98	1	1	1	2	100.00	73.59691409	78	4.403085914	4	5	5	0	4	4
RIDEAU-WINDS, WES	F3	C56583186	H159	23	1.871	2.00	79	0	1	1	2	80.00	58.22616828	67	8.773831719	0	5	4	5	3	5
SMITH, NATHANIEL	A4	C15675857	H159	23	2.65	3.33	100	1	1	1	2	100.00	75.01410363	75	-0.014103629	5	3	4	5	4	3
TOLMAN, MARKUS	B4	C38938939	H159	23	2.183	3.00	91	1	1	1	0	60.00	62.10612254	74	11.89387746	5	3	5	0	4	5
WANG, PETER	C1	C17077237	H159	24	2.171	2.00		1	1	1	2	100.00	65.62335399	64	-1.623353992	2	5	5	5	4	1
WEST, BENJAMIN	I3	C92047859	H159	23	2.105	2.67	65	1	1	1	1	80.00	62.83461865	79	16.16538135	4	4	4	5	2	3
ZEPEDA, ABIGAYLE	E4	C54668462	H159	23	3.351	3.67	84	1	1	1	0	60.00	82.35145092	72	-10.35145092	0	4	4	0	3	2
BAZAN, ADAM	C1	C15074940	I159	23	2.429	2.00	53	0	0	0	0	0.00	58.06957911	68	9.930420886	5	5	5	0	5	5

<-15 -15 to -10.1 -10 to --5.01 -5 to 5 5.01 to 10 10.1 to 15 15<

Figure 8. Snapshot of MA205's Objective and Neutral- Performance And Risk (ON-PAR) Assessment.

5. **Significant planned or continuing-into-the-next-academic-year curricular change.** The research presented in Enclosure 1 is still relevant as we navigate through this COVID period. Lessons learned from face-to-face, Hy-Flex, and hybrid modalities will be retained and maintained as systems of record to be used when needed.

The most significant planned change for MA205 is a restructuring of topics to remove redundancy, build efficiency, and still meet the needs of STEM majors. As a two-course sequence for engineering majors, MA205 and MA364 (Engineering Mathematics) cover the topics of multivariable differential and integral calculus, sequences and series, vector calculus, and differential equations. MA205 used to be required for all cadets under the old core curriculum, and as such, it included a short block at the end of the course to introduce ordinary differential equations, as it was desirable for all cadets to have some exposure to this. Ordinary differential equations are not normally taught as part of a calculus course. MA364, which is required for all engineering majors, taught all the mathematics content required for engineering majors that was not included in the core math sequence, which was a mishmash of vector calculus (normally in a calculus course) and differential equations. In its current arrangement of material, the two-course sequence repeated all the material in the MA205 differential equations block at the beginning of MA364, since some cadets had a multi-year gap between the two courses.

We would like to make each course more of a cohesive set of content, instead of several topics lumped together. No content will be removed from the two-course sequence, but it will be re-arranged to make more sense. Each course could then also be taught out of its own single textbook. Since all cadets no longer take MA205, it is no longer necessary to include the differential equations content in MA205. Thus,

the differential equations and sequences and series block would be removed from MA205 and replaced with vector calculus. In MA364, we would then not need to teach vector calculus. Since the material on differential equations from MA205 is already in MA364, nothing needs to be added to MA364, but we would avoid the redundancy of teaching the same material twice, and we could then go more in depth in the topics currently in the course. MA205 would become a true Calculus 2/3 course and MA364 would be a differential equations course.
We will coordinate with all departments whose majors take these courses to ensure that the rearrangement is not an issue. We will also update the Redbook as necessary through the curriculum change process. Moreover, no new material is being developed and this has no impact on resourcing.

Given the cadet feedback on the comfort level with MATLAB, we are assessing both MATLAB and Mathematica in AY22-02 to decide which platform provides the best medium to learn multivariable calculus. The demand to introduce MATLAB is primarily from two departments: CME and EECS. Through surveys and feedback from cadets, MA205 instructors, and departments outside of D/Math we will facilitate future MA205 instruction through either MATLAB or Mathematica as the cost of employing both is taxing from both a managerial perspective as well as confusing to cadets. In AY22-02 we decided to run three sections with MATLAB and three with Mathematica ensuring that cadets who will see MATLAB in future course work will have the opportunity to use it. Early indicators suggest pedagogical benefits for using Mathematica. Since MA205 serves as a service course for STEM majors we will coordinate the technology platform with downstream courses and other departments to ensure cadets receive the proper exposure to solving complex problems using technology while employing the technology that best facilitates the learning objectives of the course and prepares cadets for future coursework.

6. **Proposed changes to the Cadet Learning Outcomes and the justification for doing so.** In AY22-01 we added the learning objective *Evaluate limits at infinity*. This additional learning objective exposed cadets to indeterminate forms and l'Hospital's Rule which are concepts often found in both undergraduate and graduate level coursework. We will maintain this learning objective for AY22-02. As discussed in the previous section, the restructuring of material between MA205 and MA364 will require a full analysis of learning objectives, thus no other changes will be implemented.

7. **Significant planned or implemented assessment process changes.** As previously discussed, the COVID environment drove changes in the assessment of class preparation through note checks and homework quizzes. As we have moved to full face-to-face instruction, we removed these assessments and will continue forward with timed, immediate feedback assessments described above as Rumbles. The positive feedback from cadets and high correlation between Rumble performance and major graded events provided enough evidence that these assessments forced a low stakes, high reward check on learning before major graded events that prepared cadets to both assess their understanding of the material and to learn the expectations of the course. The key that makes these assessments valuable to learning is immediate feedback. Cadets grade their own or another classmate's work using a course developed rubric upon completion, informing cadets of the concepts that are most heavily weighted and showing them a fully solved solution. In AY22-01, Rumbles assessed 34 learning objectives selected by the course leadership. The topics assessed were:

1. Calculate the Equation of the Tangent Place (LOs 1.1-1.5)
2. Evaluate the Directional Derivative at a point (LOs 1.7-1.11)
3. Optimize an unconstrainted function (LOs 2.1-2.2)
4. Integrate over a transformed region (LOs 3.8, 3.17-3.18)
5. Apply MV integration to STEM concepts (LOs 3.10-3.14)
6. Test a series for convergence (LOs 4.1-4.3)
7. Find an exact solution for first-order ODEs (LOs 5.1-5.3, 5.11-5.13)
8. Find an exact solution for second-order, Homogeneous ODEs (LOs 5.14-5.16, 5.21)

In AY22-02, we will continue to assess these eight topics with two additional Rumbles for a total of ten Rumbles in the semester. Additionally, we removed homework quizzes which were facilitated through WebAssign and redistributed points to allow instructors flexibility to conduct their own in-class quizzes as they see fit.

8. **Updated current assessment schedule/timeline.** None.

9. Point of contact is the undersigned at shane.smith@westpoint.edu, or phone 845-938-8137.

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9 Encls
1. Mode of Instruction Paper
2. AY22-01 Calendar
3. MATLAB Problem Sets
4. Course Project
5. APG/MA205 Assessment Technique Mapping
6. MA205 Learning Objective and Assessment Mapping

Block	LSN #	Title	Lesson Objectives	Readings Only Required for MATLAB Days	Homework Problems
BLOCK I: MV DIFFERENTIATION	1	Multivariable Functions	1.1	Stewart 10.3 p.684-688 Stop at Example 7 Skim Stewart 12.6 p875-880 (Review 3D Surfaces) Stewart 14.1 p933-946 Reference Course Guide p5-6 for MATLAB day (Lesson 5)	10.3 #10, 12, 21, 24 14.1 #4, 21, 32, 34
	2	Partial Differential Equations and Tangent Planes	1.2-1.5	Stewart 14.3 p961-968 (Stop at Partial Differential Equations) Stewart 14.4 p974-978 (Stop at Differentials) Reference Course Guide p7 for MATLAB day (Lesson 5)	14.3 #2, 6, 15, 22, 47, 82 14.4 #2, 3, 17, 25, 27
	3	Multivariable Chain Rule	1.6	Stewart 14.5 p985-989 (Stop at Implicit Differention)	14.5 #5, 7, 16, 39, 44
	4	Gradient Vector / Directional Derivative	1.7-13	Stewart 14.6 p994-1104 Reference Course Guide p8-10 for MATLAB day (Lesson 5)	14.6 #1, 9. 13, 17, 30, 35, 40
	5	MATLAB Problem Set Work Day 1/2	1.1-2.6	MATLAB Problem Set Handout MATLAB TUTORIAL VIDEOS	MATLAB Problem Set
	6	Multivariable Max and Min / Optimization 1	2.1-2.4	Skim Stewart 4.1 p284 (Review Closed Interval Method) Stewart 14.7 p1008-1016 Reference Course Guide p11-12 for MATLAB day (Lesson 5)	14.7 #3, 11, 13, 19, 38
	7	Multivariable Optimization 2	2.4	Stewart 14.7 p1008-1016	14.7 #46, 47, 51, 54, 55
	8	Lagrange Multipliers	2.5-2.6	14.8 p1020-1026 Course Guide p13	14.8 #1, 5, 9, 33, 49
	9	Problem Solving Lab #1	1.1-2.6	Per Instructor	-
	10	Written Partial Review #1	1.1-2.6	WPR #1 Coversheet	-
BLOCK II: MV INTEGRATION	11	Volume Estimation / Iterated Integrals	3.1-3.6	Stewart 15.1 p1038-1048 Reference Course Guide p14 for MATLAB day (Lesson 14)	15.1 # 6, 7, 15, 19, 21, 22, 41
	12	General Regions	3.7-3.8	Stewart 15.2 p1051-1059	15.2 #5, 8, 10, 21, 39
	13	Polar Regions	3.9	Stewart 15.3 p1062-1066 Reference Course Guide p15-16 for MATLAB day (Lesson 14)	15.3 #3, 11, 30, 33, 45
	14	MATLAB Problem Set Work Day 2/2	3.1-3.9	MATLAB Problem Set Handout MATLAB TUTORIAL VIDEOS	MATLAB Problem Set
	15	Center of Mass	3.10-3.11	Skim Stewart 8.3 p578-583 (Review Moments and Centers of Mass) Stewart 15.4 p1069-1072 (Stop at Moment of Inertia) Reference Course Guide p17-18 for MATLAB day (Lesson 14)	15.4 #1, 4, 15
	16	Probability	3.12-3.14	Stewart 15.4 p1074-1077 Reference Course Guide p19 for MATLAB day (Lesson 14)	15.4 #29, 32, 34 (MATLAB)
	17	Triple Integrals	3.15-3.16	Stewart 15.6 p1082-1092 Reference Course Guide p20-21 for MATLAB day (Lesson 14)	15.6 #2, 4, 5, 6 , 11, 24
	18	Cylindrical and Spherical Regions	3.17-3.18	Stewart 15.7 p1095-1099 Stewart 15.8 p1102-1105 Reference Course Guide p22 for MATLAB day (Lesson 14)	15.7 #10, 15, 32, 22 (In order) 15.8 #17, 19, 20, 27
	19	Problem Solving Lab #2	3.1-3.18	Per Instructor	-
	20	Written Partial Review #2	3.1-3.18	WPR #2 Coversheet	
PROJECT	21	Intro to Tech Com/ Proj Issue	1.1-3.18	Project Handout	-
	22	Project Work Day 1/2	1.1-3.18	Project Handout	-
	23	Project Work Day 2/2	1.1-3.18	Project Handout	-
	24	Project Briefing 1/2	1.1-3.18	Project Handout	-
	25	Project Briefing 1/2	1.1-3.18	Project Handout	
	26	Limits at Infinity and Infinite Sequences	4.1 - 4.4	Review Stewart 2.3 Stewart 2.6 Infinite Limits at Infinity, p130 -p133 (Stop at Example 12) Stewart 4.4 p309-313 (Stop after Example 5) Stewart 11.1 p724-731 (Stop after Example 10)	2.6 #18, 20, 36, 27 4.4 # 20, 26, 43, 47 11.1 #1, 15, 21 , 31, 73
	27	Infinite Series the Integral Test	4.3-4.4.4	Stewart 11.2 p738-747 Stewart 11.3 p751-758 (Skip Example 1) Reference Course Guide p23 for Lesson 27's MATLAB Code	11.2 #1, 15, 16, 28, 33, 36 11.3 #7, 17, 25
BLOCK III: INFINITE SEQUENCES AND SERIES / ODES	28	Ratio Test	4.4.5	Skim 11.5 p769-771 (Absolute Convergence) Stewart 11.6 p774- 776 (Stop after Example 3) Reference Course Guide p24 for Lesson 28's MATLAB Code	11.6 #1, 2, 3, 11, 17
	29	Strategy for Testing Series / Power Series	4.3-4.5	Stewart 11.7 p779-780 (Read paragraphs 1, 2, 3, 6, and 8 plus Examples 1, 3, 5) Stewart 11.8 p781-785	11.7 #13, 15, 18, 19 , 21 11.8 #4, 13, 15 , 22
	30	Taylor Series	4.6-4.8	Skim Stewart 11.9 Pages 787-792 Stewart 11.10 p795-802 (Stop after Example 7) Reference Course Guide p25 for Lesson 30's MATLAB Code	11.10 #11, 14, 24, 27
	31	BLOCK IIIa WRIT	4.1-4.8	BLOCK IIIa WRIT Coversheet	
	32	Intro to ODEs / Slope Fields	5.1-5.8	Stewart 9.1 p606-610 Stewart 9.2 p612-616 Reference Course Guide p26-30 for Lesson 32's MATLAB Code	9.1 #13, 17 9.2 #1, 3-6
	33	Euler's Method / Separation of Variables	5.9-5.11	Stewart 9.2 p616-618 Stewart 9.3 p621-624 (Stop at Orthogonal Trajectories) Reference Course Guide p31-32 for Lesson 33's MATLAB Code	9.2 #21, 25 9.3 #2, 11, 16
	34	Integrating Factor	5.12-5.13	Stewart 9.5 p641-646	9.5 #6, 9, 20, 21
	35	Method of Characteristic Equations	5.14-5.16	Stewart 17.1 p1154-1159 Chapter 17 is omitted from Stewart 9E, **MUST USE Online Text**	17.1 #2, 3, 11, 17, 19, 23
	36	Population Growth	5.17-5.18	Stewart 9.4 p631-638	9.4 #3, 4
	37	Heating/Cooling and Mixing Problems	5.19-5.20	Stewart 3.8 p242 (Review Newton's Law of Cooling) Stewart 9.3 p604-605 Reference Course Guide p33 for Lesson 34's MATLAB Code	3.8 #17, 18 9.3 #46, 47
	38	Spring-Mass	5	Chapter 17 is omitted from Stewart 9E, MUST USE Online Text	Cengage 17.3 #3, 4, 6
	39	Systems of ODEs	5.22-5.28	Stewart 9.6 p649-653 Reference Course Guide p34-39 for Lesson 39's MATLAB Code	9.6 #1, 3, 5, 11
	40	BLOCK IIIb WRIT	5.1-5.8	BLOCK IIIb WRIT Coversheet	

TAB F AY2021 COURSE ASSESSMENT

MA205 Assessment

AY21-2 MA205's Off-Cycle Scheme of Maneuver.

This semester, MA205 looks and feels more "normal" for the first time since 2019, but with minor changes! With 70-minute classes with all 18 cadets participating at once, we are excited to be back in the classroom after receiving the majority of Block I in a remote setting. Block I (Differentiation), ends next Monday with WPR1. Like Block I, Block II (Integration) will not afford WPR Test Corrections. Before moving into Block IIIa (Sequences and Series), we'll introduce a new interdisciplinary project with input from CME, SE, and SOSH (Economics and MA367). We'll assess cadet understand of Block IIIa and Block IIIb (ODEs) with WRITs. Finally, after Block III, we'll offer a TEE review (PSL#3) before the start of TEE Week.

MA205 Assessment

Key Performance Parameters (KPP)

- Maintaining course goal of testing overwhelming majority of lesson objectives (threshold: 5%, obj: 100%)
  - Can drop 4 (%5) lesson OBJs, but want to keep 100%
  - Split WPR3 into two writs (Kept Change for AY21-2)
- Keep TEE the same from year to year (threshold: minor changes *within* questions, obj: all questions intact and unchanged)
  - Minor Changes to the TEE (IBP, # of Questions)
- One 'benchmark' question per WPR the same across each semester
  - Track one question from the first WPR (WPRs AY19-2)

Key System Attributes (KSA)

- TEE worth 300 points (30% of course grade)
- Maintain assessment via WPR, and WRITs
- Service STEM courses (customers) through implementing technology; MATLAB in this case.
  - Surveyed and met with Systems, CME regarding technology requirements and recommendations for tying in problem sets and/or projects

Quantitative Measurements (MOP)

- Keep TEE the same and observe trends among on-cycle to on-cycle classes as well as off-cycle to off-cycle
- Keep one question the exact same on each WPR, and the two BL3 writs, to observe trends among on-cycle to on-cycle classes as well as off-cycle to off-cycle
  - WPR1 PG2:
    - AY20-1: 76.85%
    - AY21-1: 81.48%
  - WPR2 PG2:
    - AY20-1: 76.84%
    - AY21-1: 73.12%

Qualitative Measurements (MOE)

- "MA205 Learning Objective Assessment Plan" highlighting 71/73 lesson objectives assessed during the term
- "Academic Program Goals – MA205 Assessment Techniques" crosswalk showing nested objectives
- Pre/Post-assessment surveys for cadet feedback regarding course approach and outcomes

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MA205 Assessment

Academic Program Goals → MA205 Assessment Techniques						
Supported Academic Program oals (from *Educating Army Leaders*)	1. Graduates communicate effectively with all audiences	2. Graduates think critically and creatively	3. Graduates demonstrate the capability and desire to pursue progressive and continued intellectual development	5. Graduates apply science, technology, engineering, and mathematical concepts and processes to solve complex problems	7. Graduates integrate and apply knowledge and methodological approaches gained through in-depth study of an academic discipline
MA205 Course Goals (align with Core Math Goals)	3. Communicate Effectively	2. Apply Technology; 5. Build Competent and Confident Problem Solvers; 6. Foster Interdisciplinary Perspective	4. Develop Habits of Mind; 5. Build Competent and Confident Problem Solvers	2. Apply Technology; 5. Build Competent and Confident Problem Solvers	1. Acquire a Body of Knowledge; 6. Foster Interdisciplinary Perspective
Middle States Accreditation Standards	3. Design and Deliver of the Student Learning Experience			5. Educational Effectiveness Assessment	
MA205 Course Learning Objectives*	1. Perform multivariable differentiation; 2. Model and solve problems through multivariable optimization; 3. Integrate multivariable functions	5. Produce solution equations for standard ODEs	1. Perform multivariable differentiation; 3. Integrate multivariable functions	4. Model functions using infinite series; 5. Produce solution equations for standard ODEs	2. Model and solve problems through multivariable optimization; 4. Model functions using infinite series
MA205 Assessment Techniques	Project; Instructor Assigned Evaluations; Surveys	Project; WPR3; Surveys; Tech Week Assessment; Shortened class requires cadets to prepare for lecture prior to class and utilize avaialble class period	Daily notes check for preparedness; Surveys	Tech Week Assessment; Project; TEE;	WPR1; WPR2; WPR3; TEE; Outreach to Supported Departments

* Course Learning Objectives further broken down into 73 Lesson Objectives

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MA205 Assessment

MA 206, MA3XY, MA4XY

<u>On the horizon:</u>
> MA205 implementation of vector calculus to support MC 300, MA371 and MATLAB instruction

MA205 Curriculum Complements:

MA371, MA206

<u>MA205 Course Assessment</u>:

Daily Assessment:
>Reading Quizzes and Homework Grades (Change from Daily Notes Checks)

Block Assessment:
>Use a benchmark WPR/WRIT question for each block

Technology Assessment:
>More frequent MATLAB demonstrations during the differentiation and integration blocks (Change)

End of Course Assessment:
>Use similar TEE questions from semester to semester (*Assume 2.5 TEE Duration)

MA205 Curriculum Supports: Engineer Foundational Classes, SE302, MC300, EE301/2. NE300

Cadet Experience before MA205:
MA103, MA104, CH101, PH205, IT105

<u>Track Individual Performance:</u>
Hypothesis: Sectioning based on CQPA (3.0 split) improves cadet MA205 performance. Cadets are better prepared for follow-on STEM courses.

Use MA103 and/or MA104 performance as baseline.
Assess by tracking WPR Questions and next STEM course.

AY21-1 MA205 Learning Objectives Evaluation Plan

		Block I			
		WPR 1	PSet	Project	TEE
MV Differentiation	1.1	MC, WPR 2	1, 2	X	
	1.2	MC, 2, 3, 4	1, 2	X	5, 6, 7
	1.3	4			7
	1.4	6			9
	1.5	6			9
	1.6	MC, 3			5
	1.7		1		6,9
	1.8	MC, 4	1		7
	1.9	4	1, 2	X	7
	1.10	MC, 4	1	X	7
	1.11	4	1		7
	1.12		1		6
	1.13		1		6
Optimization	2.1	5		X	
	2.2	MC, 5		X	9
	2.3		2	X	
	2.4		2	X	
	2.5	2			11
	2.6		2		

Notes:
-Create a second PSet such that:
PSet1: Gradient & Directional Derivative
PSet2: Closed Interval Method and Lagrange Opt.
-Assign PSet and compare to WPR1 Performance.

		Block II		
		WPR 2	Project	TEE
MV Integration	3.1	2	X	
	3.2	MC	X	
	3.3	2	X	
	3.4	2	X	
	3.5	2	X	
	3.6	4	X	
	3.7	3, 6	X	8
	3.8	3	X	8
	3.9	MC		
	3.10	6		9,12
	3.11	6		
	3.12	4	X	9
	3.13	4	X	9
	3.14	4	X	
	3.15			7
	3.16	3		8
	3.17	5		
	3.18	5		

		Block III		
		WPR 3	Project	TEE
Sequences & Series	4.1	MC		
	4.2	MC		
	4.3	MC,3,6		
	4.4	6		
	4.5	6		
	4.6			1
	4.7	WRIT 3A		1
	4.8			1
Intro to ODEs	5.1		X	2,3,4
	5.2		X	
	5.3	MC,5	X	2,3,4
	5.4			10
	5.5			10
	5.6	2		
	5.7	2	X	
	5.8	2		
	5.9	7	X	
	5.10	7	X	8
	5.11	MC		2,3,8
	5.12	5		2,3
	5.13	5		2,3
	5.14	4		4
	5.15	4		4
	5.16	4		4
	5.17		X	
	5.18		X	
	5.19			3
	5.20			2
	5.21			4,8
	5.22			10
	5.23			10
	5.24			10
	5.25	Euler to solve system of DE		
	5.26	plot phase portrait using tech		
	5.27	WRIT 3B	X	
	5.28		X	

Denotes a Previously Unevaluated L.O. during AY20-1

Denotes an Unevaluated L.O. during AY20-2

AY19-2 Situation
-105 L.Os (88 prior to WPR 3 and 17 after)
-35 L.Os untested in AY19-2
-18 WPR pages (~3.5 L.Os per page)

AY20-1 Changes
-73 L.Os (61 prior to WPR 3 and 12 after)
-6 L.Os untested in AY20-1 (3 of those covered in PSL)
-19 WPR pages (~3.2 L.O. per page)

AY20-2 Changes
-No "Tech" portions to examinations
-Heavier focus on MATLAB throughout course
-Focus on consistency of Grad/DD and Optimization for remediation study (on hold due to COVID)

AY21-1 Changes
-55 min WPR reduces number of problems during exam from 6 to 4
-30 min lecture perid with 1/2-size class (8-9 cadets)
-Pairing cadets on project (one Optimization from PSet, the other had Gradient/DD)

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Mapping Lesson Objectives

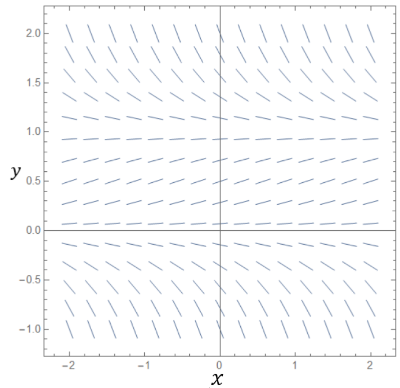
Cadet____________________

Total Weight: 100 points

Section____________

MA205 20-1 WPR III (Ver. ζ - nontech)

(10 Points) Use the figure to answer the following questions:



a. (2 points) Find all the equilibrium solutions.

b. (2 points) Sketch the graph of the particular solution that satisfies the initial condition $y(-1) = 0.25$.

c. (3 points) Determine the long-term behavior of the particular solution that satisfies the initial condition $y(-1) = 0.25$.

d. (3 points) Circle the differential equation that matches with the depicted slope field.

i. $y' = 3y + 6$

ii. $y' = y^2 - y - 2$

iii. $y' = \sin(x) + 2y$

iv. $y' = y - y^2$

LO: 5.6, 5.7, 5.8

- 5.6 Create a slope field for a given first order differential equation.
- 5.7 Given an initial condition, approximate a solution to a first order ODE using a slope field.
- 5.8 Evaluate the long-term behavior and equilibrium solution of a first order differential equation based on the slope field.

Survey Questions

- How much time do you spend preparing ***prior to class*** for a typical MA205 lesson?
- How much time do you spend preparing ***after class*** for the same MA205 lesson you prepared for prior to class?
- How confident are you in your ability to use MATLAB?
- Rank the following methods of instruction you think work best for a math class.
- Provide general comments concerning the instruction, material, administration, and assessments in MA205 during Block I. Potential topics of discussion include: reflection on personal preparation, quality of course resources, quality of instruction, course organization, and personal best practices.

GOALS:

- Frequency: before and after each assessment:
- Track across semesters self-reported time allotted to preparation
- Pre/Post-assessment self-reported technology comfort level
- Feedback on our chosen method of delivery vs. other options cadets have likely experienced

DEPARTMENT OF MATHEMATICAL SCIENCES
UNITED STATES MILITARY ACADEMY
WEST POINT, NY 10996-1786

MADN-MATH 24 July 2020

MEMORANDUM FOR MA205 Students, Department of Mathematical Sciences, USMA

SUBJECT: AY 21-01 MA205 Multivariable Calculus and Introduction to Differential Equations

1. **Purpose:** This memorandum specifies course material, describes the goals, and outlines policy and procedures for MA205 for AY 21-01.

2. **Text, Software and Supporting Materials:**

a. Calculus Early Transcendentals, Eighth Edition, James Stewart, Brooks/Cole, 2016

b. MA205 Blackboard: https://usma.blackboard.com/

i. MA205 Course Guide

ii. Formula Reference Cards: Core Math & MA205

c. WebAssign: http://www.webassign.net

d. Wolfram Mathematica

e. MATLAB

f. TI-30X Issued Calculator

g. Microsoft Office

3. **Goals:** This course is designed to help you become more confident in learning and applying mathematics to solve problems with an emphasis on the goals below.

a. Acquire a base of knowledge. Acquire a body of knowledge to support the rigorous course work in science, technology, engineering, and mathematics (STEM) disciplines. MA205 will prepare you to understand the concepts of multivariable calculus (differentiation and integration) and apply these concepts to solve problems. MA205 will also prepare you to model and solve problems using ordinary differential equations.

b. Leverage technology. You will have numerous opportunities to use MATLAB to enhance your capability to visualize, solve, analyze, and experiment with a myriad of mathematical functions and models that you will develop in the course. Specifically, you will become confident and competent in using technology to leverage your ability to solve complex problems. Additionally, you will have the opportunity to check your understanding of practice problems from the Stewart textbook and receive immediate feedback through WebAssign.

c. Improve communication skills. Being a good communicator is an important characteristic of being an effective leader. The fundamentals of successfully conveying your intent to the soldiers under your command and describing your thought process in solving a mathematics problem are the same. All leaders must be able to clearly articulate their thoughts. You will have many opportunities to improve your communication skills both verbally and in writing. These opportunities include board presentations, preparation of a technical report, and an oral presentation.

d. Build confident and competent problem solvers. You will develop modeling and problem-solving abilities through in-class experiences, homework exercises, and projects. These events will require you to analyze real world problems, make critical assumptions, model the problem, solve the problem, and then interpret your results.

e. Develop habits of mind. Key habits of mind that you should develop are: time management, creativity, work ethic, thinking interdependently, critical thinking, lifelong learning, and curiosity. You can certainly reach a higher potential if all of the elements are incorporated and pursued simultaneously. This course implements strategies that promote and develop each of these for you. Ultimately, you will be introduced to the importance of lifelong learning.

f. Foster interdisciplinary perspective. MA205 seeks to expose cadets to problems with an interdisciplinary perspective. The intent is for cadets to transfer knowledge across disciplinary boundaries to solve complex problems.

4. **Evaluation:** Your understanding of the mathematics presented in this course will be assessed through written exercises, an oral presentation, and exams to include a comprehensive final examination. Adequate daily preparation on your part is necessary to be successful in this course and future mathematics, science, and engineering courses.

a. The following is the distribution of points for the course:

Table 1: Point Distribution

Event	Points
Instructor Points	100
Daily Preparation (Reading Quizzes, Notes Checks)	150
WPRs (3 x 100 pts)	300
MATLAB Problem Set	50
Project	100
TEE	300
Total	**1000***

*Each instructor has the discretion to offer up to 10 bonus points (1% of total grade) for the semester.

b. The authorized references for in-class graded event are as follows:

i. The MA205 Formula Reference Card (FRC) will be issued as a document to be filled out by students as the course progresses. Students are permitted to record reference formulations relevant to the exact section enumerated on the MA205 FRC document.

ii. The Core Math FRC will be issued as a document prepopulated with key concepts and formulas from MA103 and MA104. Cadets are not authorized to add formulas or notes to the Core Math FRC.

iii. Technology. Issued calculator and both the Core Math and student-completed MA205 FRCs are allowed on all graded events. Computers, mobile devices, and any other forms of communication are strictly prohibited on any in-class graded event unless your instructor explicitly states otherwise.

c. Daily Preparation. As in previous math courses you have taken at USMA, MA205 continues the use of WebAssign for daily preparation and understanding of course content. You are to use your pre-existing WebAssign account set up with the original access code packaged with your Stewart Calculus textbook issued during your first academic semester. You will be graded on completion of the WebAssign daily practice problem. Additionally, you will receive a grade from you instructor for a daily notes check.

i. Concept. Each class will begin with a notes check. The notes you take for each lesson should summarize the reading, list any questions you may have on topics that are unclear, and your work shown for the daily practice problems. If you get stuck on any of the practice problems and cannot get help to solve them, clearly identify in your notes the step where you confused and a question that, if answered, would have allowed you to complete the problem. The standards for the daily notes check grading can be found on the MA205 Blackboard site under Daily Preparation Assignments -> Overview and Guidance.

ii. Submissions & Extensions. In the case of a known absence, you are required to submit your notes to your instructor electronically. In the event of an unplanned absence, contact your instructor and propose a time that you can submit your notes and complete the practice problems.

iii. Scoring. Notes check scores will be averaged for each block and entered into CIS prior to each WPR. Scoring is based on the tiered system in Table 2.

Table 2: Tiered Notes Check Scoring

Notes Check Rating	Score (%)
Plus “+”	100
Check “✓”	85
Minus “-“	70
Zero “0”	As-is

d. Final course grades will follow the Department of Mathematical Sciences’ standing guidelines. Cadets can view their performance on all graded events in CIS. Grade cut-offs follow Table 3.

Table 3: Guidelines for Assigning Grades

Letter Grade	Numerical Grade (%)	Description
A+	97-100	Beyond Expectations
A	93-97	Dominates the material
A-	90-93	Mastery
B+	87-90	Excellent performance
B	83-87	Good understanding
B-	80-83	Proficient; aptitude for the subject
C+	77-80	Can build on foundations
C	73-77	Passing; Proficient
C-	70-73	Short-range understanding
D	65-70	Marginal performance/some learning
F	< 65	Failed to demonstrate understanding

e. A final course average of less than 65% or a score of less than 50% on the TEE (regardless of final course average) may indicate insufficient knowledge or ability and will likely result in course failure.

5. **Professional Development:** To help you achieve the student growth goals stated earlier, we have the following expectations for your efforts.

a. Success in this course depends heavily on your daily preparation. Dedicate the time required for success - we have designed this course so that the average student can build a proficient aptitude in multivariable calculus and ordinary differential equations with an average of 2 hours of daily preparation.

b. Come prepared for class: Review the lesson in the course guide, read the lesson objectives, complete the assigned reading, take thorough notes, and complete the WebAssign problems. The goal is to develop a deeper understanding of the lesson objectives and supporting concepts at the conclusion of your preparation. Come to class prepared to ask questions on any unclear materials from the assigned reading and practice problems. Participate in the instruction and discussion. When preparing for a new lesson, you should also spend time reviewing the previous lesson as material tends to build off previous lessons.

c. Seek assistance when needed - from the text, online sources, your classmates, or your instructor. Additional instruction (AI) may be scheduled with any instructor and is highly encouraged regardless of class standing. The primary purpose of AI is to address specific questions that you may have. You should seek AI as soon as you need it – do not wait until you are completely lost in the course.

d. Submit your assignments on time. You are expected to complete and submit all assignments by the advertised deadline. If you are unable to submit an assignment when it is due, be professional and inform your instructor. You may request an extension but be prepared for a grade cut on all late submissions. The standard penalty for late work in this course is 10% per day.

6. **Classroom Policies:** All cadets are expected to maintain proper military bearing and appearance during instruction in accordance with appropriate regulations.

a. Respect others in the classroom: No profanity, unprofessional jokes, or unprofessional media (audio or visual).

b. No electronic forms of communication during class (email, social media, etc.) unless authorized by your instructor.

c. No videotaping, audio recording, or cell phone use during class (with exception of taking pictures of board work with instructor approval).

d. Food and gum are not permitted in the classroom. Cadets are only allowed beverages from a sealable container. Containers with openings, such as the ones sold at coffee shops, are not approved for use in the classroom.

e. No hats, jackets, overshoes, or backpacks should be brought into the classroom unless approved by instructor. Only materials needed for this course are allowed on your desk.

f. Cadets are not authorized to de-blouse under any circumstances unless approved by the Department Head. Outer garments (e.g., black jacket) may be worn if approved by instructor.

g. Be logical, neat, and organized in all work. Board work will include the names of cadet(s) boxed in the upper right corner and cadets should be prepared to recite their work.

h. Recitation procedures:

i. When called upon a cadet will stand on the unobstructed side of the blackboard with a serviceable pointer in hand closest to problem.

ii. Addressing the section members, the reciting cadet will introduce themselves and clearly state the problem and how they solved it.

7. **COVID-19 Adjustments:** The MA205 leadership team has made numerous adjustments to ensure AY21-01 can be conducted in a safe manner. While the purpose of the course is to prepare cadets for their follow-on STEM courses, the safety of cadets, staff, and faculty will remain the top priority. The following precautions are being implemented to ensure a safe classroom environment to the greatest extent possible.

a. Masks must be worn in the classroom, maintain social distancing to the greatest extent possible, and the classroom door will remain open.

b. Instructors will clean desks and chairs between use.

c. Hand sanitizer and cleaning supplies will be available in every classroom.

d. Classes will be conducted with a maximum of nine students. For a 70-minute class with 18 students, nine students will be in the classroom for the first 30 minutes, there will be a 10-minute transition period to clean the classroom, then the remaining nine students will be in-class for the remaining 30 minutes.

e. Each class will follow the Read-Watch-Do-Discuss format. This is a minor modification from traditional classes. Since students will only be in the classroom for approximately half the time of a traditional class, the traditional lecture portion of the classroom will be replaced with videos that cadets will watch prior to class. The time in class will be focused on solving problems related to the lesson material.

8. **Scholarship:** This course will not only teach you new concepts in mathematics, but it will build upon what you have already learned. Therefore, it may be necessary and appropriate to refer to sources of

information beyond the course text. Adhere to the standards of scholarly work: documentation of your work should follow the guidance contained in the most recent versions of *Documentation of Academic Work* (Office of the Dean) and The Little Brown Handbook. If you are unsure about how or when to document a source, consult your instructor before submitting your work.

9. **Supervisory Chain:** Table 4 depicts the supervisory chain for MA205. The Department of Mathematical Sciences has an open door policy. Your initial attempt to address problems or concerns should go through the chain of command. However, if you are uncomfortable addressing an issue with your instructor, please do not hesitate to approach any individual in the chain of command.

Table 4: MA205 Chain of Command

	Faculty Member	Room	Phone
My instructor:			
MA205 Course Director:	MAJ Walsh	Th 221B	938-8137
MA205 Program Director:	LTC Evans	Th 223	938-3605
Deputy Head of Department	COL Scioletti	Th 238A	938-5870
Head of the Department:	COL Hartley	Th 238	938-5285

10. **Your Learning:** This is your mathematics education. You must take responsibility for your own learning and participate as an active learner. Your instructor is here as a facilitator to help guide you through the educational experience this course offers. The quantitative reasoning and problem-solving skills that you continue to develop in this course will serve you in the future regardless of your major as a cadet or your branch as an officer. Use this opportunity to develop confidence, competence, and good habits of mind. I wish you the best in this course, in the rest of your cadet experience, and in your future service to our nation.

LEE A. EVANS
LTC, AV/49
Program Director

SECTION II

EXAMINATIONS

UNITED STATES MILITARY ACADEMY
WEST POINT

TAB G WPR 1

Instructor Solution

First and Last Name: Solution

Total Weight: 100 points

Cadet ID:

Time Limit: 55 Minutes

Section and Instructor:

MA205 AY26–1 WPR I (v1)

Number of Extra Sheets: ______

READ THESE INSTRUCTIONS CAREFULLY BEFORE STARTING WORK

1. **SHOW ALL WORK.** Sufficient work is required to indicate the method of reasoning and the operation performed. Clearly indicate your final answer.

2. If you need more space, **use a separate sheet for each problem**. Clearly indicate which problem is continued at the top of the sheet. Print your name and section on every extra sheet used.

3. **You are authorized your issued calculator and one handwritten 8.5 x 11 note-sheet (front and back). You may NOT receive assistance from anyone else.**

4. This examination consists of five graded pages and one cover sheet.

5. Define any variables that you use which are not explicitly defined in the problem. If you make assumptions to simplify a problem, clearly state them.

6. All work written on the WPR will be graded unless marked through or explicitly marked with words to the effect of "do not grade." **BE SURE TO BOX OR DOUBLE UNDERLINE YOUR ANSWERS.**

7. **Assume angles are in radians unless specified as degrees.**

8. Ensure your answers are written using proper mathematical notation. When appropriate, round all answers to 2 decimal places. Include units when necessary.

Question	Points	Score
1	25	
2	25	
3	20	
4	30	
Total:	100	

"I did not use any unauthorized sources nor did I receive any assistance while completing this assessment. I will not discuss any aspects of this assessment with anyone except an MA205 instructor until it is released from academic security on 17 September 2025."

Sign and date here to show acknowledgement ______________________

1. **Vector Operations:** Let $O(1,-2,3)$, $P(3,4,1)$, and $Q(7,6,2)$.

(a) (10 points) Find the angle between the vectors $\overrightarrow{OP}$ and $\overrightarrow{OQ}$.

$\overrightarrow{OP} = \langle 2, 6, -2 \rangle$
$\overrightarrow{OQ} = \langle 6, 8, -1 \rangle$

$$\cos^{-1}\left(\frac{12+48+2}{\sqrt{44}\sqrt{101}}\right) = \boxed{.38 \text{ radians}} = \boxed{21.56^\circ}$$

(b) (10 points) Compute the cross product $\overrightarrow{OP} \times \overrightarrow{OQ}$ and select the best interpretation of its geometric significance:

$$\begin{vmatrix} i & j & k \\ 2 & 6 & -2 \\ 6 & 8 & -1 \end{vmatrix} = \langle -6-(-16), -12-(-2), 16-36 \rangle$$
$$= \boxed{\langle 10, -10, -20 \rangle}$$

Circle the best answer.

A. A vector representing the shortest path between P and Q.
B. A vector perpendicular to the plane containing O, P, and Q.
C. A vector whose magnitude equals the distance between P and Q.
D. A vector pointing from O to the midpoint of P and Q.

(c) (5 points) Which of the following best describes the **vector projection of $\overrightarrow{OQ}$ onto $\overrightarrow{OP}$**?
Circle the best answer.

A. It is a scalar equal to $|\overrightarrow{OQ}|\cos\theta$, where θ is the angle between $\overrightarrow{OP}$ and $\overrightarrow{OQ}$.
B. It is a vector lying along $\overrightarrow{OP}$, whose length equals the component of $\overrightarrow{OQ}$ in the direction of $\overrightarrow{OP}$.
C. It is a vector lying along $\overrightarrow{OQ}$, whose length equals the component of $\overrightarrow{OP}$ in the direction of $\overrightarrow{OQ}$.
D. It is always orthogonal to $\overrightarrow{OP}$ and represents the "remainder" of $\overrightarrow{OQ}$ after subtracting $\overrightarrow{OP}$.

2. **Equations of Lines and Planes:**

(a) (10 points) Write the vector equation for the line segment from $P(3,4,1)$ to $Q(7,6,2)$. Express in terms of the parameter t, and restrict the domain accordingly.

$$r(t) = (1-t)(3,4,1) + t(7,6,2)$$

$$x = 3-3t + 7t = 3+4t$$

$$y = 4-4t + 6t = 4+2t$$

$$z = 1-t + 2t = 1+t$$

$$\boxed{r(t) = \langle 3+4t, 4+2t, 1+t \rangle \quad 0 \le t \le 1}$$

$$\boxed{r(t) = \langle 3,4,1 \rangle + t\langle 4,2,1 \rangle \qquad 0 \le t \le 1}$$

(b) (10 points) Find the scalar equation of the plane containing $O(1,-2,3)$, $P(3,4,1)$, and $Q(7,6,2)$.

$$\vec{n} = \langle 10, -10, -20 \rangle$$

$$P_0 = (1, -2, 3)$$

$$a(x-x_0) + b(y-y_0) + c(z-z_0) = 0$$

$$\boxed{10(x-1) - 10(y+2) - 20(z-3) = 0}$$

$$10x - 10 - 10y - 20 - 20z + 60 = 0$$

$$10x - 10y + 20z + 30 = 0$$

$$\boxed{10x - 10y + 20z = -30}$$

(c) (5 points) Determine whether the point $R(2,0,5)$ lies on the plane you found in part (b). Justify your answer.

$$10(2) - 10(0) + 20(5) = -30$$

$$20 - 0 + 100 = -30$$

$$\boxed{80 \neq -30}$$

$\therefore$ Not on the plane.

3. **Vector Functions and Space Curves:**

(a) (10 points) Find the arc length of the curve

$$\mathbf{r}(t) = \langle \frac{2}{3}t^3,\ t^2 - 2,\ 7\rangle, \qquad 0 \le t \le 2.$$

$\int_a^b |r'(t)|dt$

$r'(t) = \langle 2t^2, 2t, 0\rangle$

$|r'(t)| = \sqrt{4t^4 + 4t^2}$

$= \sqrt{4t^2(t^2+1)}$

$= \int_0^2 2t\sqrt{t^2+1}\,dt$

$u = t^2+1$

$du = 2t\,dt$

$\int_0^2 u^{\frac{1}{2}}du = \frac{2}{3}u^{\frac{3}{2}}\Big|_0^2 = \frac{2}{3}(t^2+1)^{\frac{3}{2}}\Big|_0^2$

$= \frac{2}{3}(2^2+1)^{\frac{3}{2}} - \frac{2}{3}(0+1)^{\frac{3}{2}} = \boxed{\frac{2}{3}(5^{\frac{3}{2}}-1) \approx 6.79 = L}$

(b) (10 points) Determine whether the curve

$$\mathbf{r}_{\text{alt}}(t) = \langle 7 - t,\ 6 - 2t,\ 2 - t\rangle, \qquad 0 \le t \le 3,$$

intersects the plane $2x - y + z = 5$. If so, find the value of t at the intersection.

$2(7-t) - (6-2t) + (2-t) = 5$

$14 - \cancel{2t} - 6 + \cancel{2t} + 2 - \cancel{t} = 5$

$-t + 10 = 5$

$-t = -5$

$t = 5$

$\boxed{\text{Outside } 0 \le t \le 3 \ \therefore \text{ No intersection.}}$

4. **Motion in Space:** A particle's velocity is given by

$$\mathbf{v}(t) = \langle 2t, 3, 3 - 4.9t \rangle.$$

(a) (10 points) Find the **acceleration vector** $\mathbf{a}(t)$.

$$\boxed{a(t) = \langle 2, 0, -4.9 \rangle}$$

(b) (15 points) Using the initial condition $\mathbf{r}(1) = \langle 5, 6.5, 1.5 \rangle$, determine the specific **position vector** $\mathbf{r}(\mathbf{t})$.

$$r(t) = \langle t^2 + C_1, 3t + C_2, 3t - \frac{4.9}{2}t^2 + C_3 \rangle$$

$$= \langle 1^2 + C_1 = 5, 3(1) + C_2 = 6.5, 3(1) - \frac{4.9}{2}(1)^2 + C_3 = 1.5 \rangle$$

$$\boxed{r(t) = \langle t^2 + 4, 3t + 3.5, 3t - 2.45t^2 + .95 \rangle}$$

(c) (5 points) Determine the value(s) for t when the particle intercepts the xy-plane.

$$3t - 2.45t^2 + .95 = 0$$

$$\boxed{t = -.26, 1.49}$$

–USE AS A CONTINUATION SHEET IF NECESSARY.–

TAB H WPR 2

Instructor Solution

Total Weight: 100 points

Time Limit: 55 Minutes

First and Last Name: Solution

Cadet ID: ____________

Section and Instructor: ____________

MA205 AY26–1 WPR II (v1)

Number of Extra Sheets: ______

READ THESE INSTRUCTIONS CAREFULLY BEFORE STARTING WORK

1. **SHOW ALL WORK.** Sufficient work is required to indicate the method of reasoning and the operation performed. Clearly indicate your final answer.

2. If you need more space, **use a separate sheet for each problem**. Clearly indicate which problem is continued at the top of the sheet. Print your name and section on every extra sheet used.

3. **You are authorized your issued calculator and one handwritten 8.5 x 11 note-sheet (front and back). You may NOT receive assistance from anyone else.**

4. This examination consists of five graded pages and one cover sheet.

5. Define any variables that you use which are not explicitly defined in the problem. If you make assumptions to simplify a problem, clearly state them.

6. All work written on the WPR will be graded unless marked through or explicitly marked with words to the effect of "do not grade." **WRITE YOUR FINAL ANSWERS IN THE ASSOCIATED BOX.**

7. **Assume angles are in radians unless specified as degrees.**

8. Ensure your answers are written using proper mathematical notation. When appropriate, round all answers to 2 decimal places. Include units when necessary.

Question	Points	Score
1	25	
2	35	
3	20	
4	20	
Total:	100	

"I did not use any unauthorized sources nor did I receive any assistance while completing this assessment. I will not discuss any aspects of this assessment with anyone except an MA205 instructor until it is released from academic security on 8 October 2025."

Sign and date here to show acknowledgement ____________

1. **Problem 1**

(a) (15 points) Find the equation of the plane tangent to the surface $f(x,y) = x^2y + \sin(xy)$ at the point $P(2, \pi, 4\pi)$.

$\boxed{z = 5\pi x + 6y - 12\pi}$

$f_x = 2xy + y\cos(xy)$

$f_y = x^2 + x\cos(xy)$

$f_x(2,\pi) = 4\pi + \pi\cos(2\pi) = 5\pi$

$f_y(2,\pi) = 4 + 2\cos(2\pi) = 6$

$z - 4\pi = 5\pi(x-2) + 6(y-\pi)$

$z = 5\pi x - 10\pi + 6y - 6\pi + 4\pi$

$z = 5\pi x + 6y - 12\pi$

(b) (5 points) Use the tangent plane equation derived above to **approximate** the value of $z = f(x,y)$ at the point $(1.95, 3.19)$.

$L(x,y) = f(x_0,y_0) + f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0)$

$\boxed{L(1.95, 3.19) \approx 12.07}$

$= 4\pi + 5\pi(x-2) + 6(y-\pi)$

$= 5\pi x + 6y - 12\pi$

$L(1.95, 3.19) = 5\pi(1.95) + 6(3.19) - 12\pi \approx 12.07$

(c) (5 points) **Multiple Choice:** Suppose $f(x,y)$ models the temperature distribution (in degrees Celsius) for a given coordinant position. Why is the tangent plane useful near P? **Write your choice in the box provided.**

(A) It gives the exact temperature everywhere.
(B) It provides a local linear approximation of temperature near P.
(C) It guarantees the maximum temperature occurs at P.
(D) It eliminates the need for partial derivatives.

$\boxed{B}$

2. **Problem 2**

(a) (5 points) **Check ALL That Apply:** Which of the following statements about the gradient $\nabla f(x, y, z)$ are true?

- [x] The gradient ∇f is always perpendicular to the level curve or level surface of f.
- [x] The gradient ∇f points in the direction of greatest rate of increase of f.
- [ ] The gradient ∇f is always a unit vector.
- [ ] The gradient ∇f at a critical point must be a nonzero vector.
- [x] The gradient ∇f can be used to compute directional derivatives in any given direction.

(b) (5 points) **Check ALL that Apply:** The **second derivative test** for functions of two variables provides information about:

- [x] Whether a critical point is a local maximum.
- [x] Whether a critical point is a local minimum.
- [x] Whether a critical point is a saddle point.
- [x] Whether the test is inconclusive at a given point.
- [ ] The absolute maximum of the function over a closed region.

(c) (5 points) **Multiple Choice:** When using the Closed Interval Method in multivariable calculus to determine absolute extrema on a bounded region, which steps are necessary? **Write your choice in the box provided.**

(A) Compute ∇f, find all inflection points, then compare function values at those points (e.g., at (x,y)).
(B) Compute ∇f and select the largest gradient vector as the absolute maximum.
(C) Compute ∇f, find critical points, evaluate f at those points (e.g., at (x,y)), and all boundary points, then compare function values.
(D) Compute ∇f, find critical points, and classify them using the second derivative test.

$\boxed{C}$

Now consider the function

$$f(x,y) = (x-1)^3 - 3xy^2.$$

(d) (20 points) Using the **Second Derivative Test**, Find **AND** Classify the **critical point(s)** of $f(x,y)$.

Critical Point	Classification
Ex: (a,b)	*Inconclusive*
$(0,1)$	Saddle Point
$(0,-1)$	Saddle Point
$(1,0)$	Inconclusive

$\nabla f = \langle 3(x-1)^2 - 3y^2, -6xy \rangle$

$3(x-1)^2 - 3y^2 = 0$

$-6xy = 0$

$x = 0$ or $y = 0$

when $x=0$:

$3(0-1)^2 - 3y^2 = 0$

$3 - 3y^2 = 0$

$-3y^2 = -3$

$y^2 = 1$

$y = \pm 1$

Critical Points: $(0,1)$ and $(0,-1)$

When $y=0$:

$3(x-1)^2 - 3(0)^2 = 0$

$3(x-1)^2 = 0$

$x - 1 = 0$

$x = 1$

Critical Point: $(1,0)$

$f_{xx} = 6(x-1)$, $f_{yy} = -6x$, $f_{xy} = f_{yx} = -6y$

For $(0,1)$

$D = (-6)(0) - (-6)^2 = -42 < 0$ Saddle Point

For $(0,-1)$

$D = (-6)(0) - (6)^2 = -42 < 0$ Saddle Point

For $(1,0)$

$D = (0)(-6) - (0)^2 = 0$ inconclusive

3. **Problem 3** Consider the objective function

$$f(x, y, z) = 4x + 2y + z \quad \text{subject to} \quad x^2 + y + z^2 = 1$$

(a) (20 points) Use Lagrange multipliers to find the extreme value(s) of the function $f(x, y, z)$ subject to the given constraint.

$\boxed{\text{Absolute MAX value } 4.13 \text{ at } (1, -\frac{1}{16}, \frac{1}{4})}$

$\nabla f = \lambda \nabla g$

① $4 = \lambda(2x)$

② $2 = \lambda$

③ $1 = \lambda(2z)$

④ $x^2 + y + z^2 = 1$

① $4 = 2(2x)$

$4 = 4x$

$1 = x$

③ $1 = 2(2z)$

$1 = 4z$

$\frac{1}{4} = z$

④ $1^2 + y + (\frac{1}{4})^2 = 1$

$1 + y + \frac{1}{16} = 1$

$y = -\frac{1}{16}$

$(1, -\frac{1}{16}, \frac{1}{4})$

$f(1, -\frac{1}{16}, \frac{1}{4}) = 4 + 2(-\frac{1}{16}) + \frac{1}{4}$

$= \frac{64}{16} - \frac{2}{16} + \frac{4}{16}$

$= \frac{66}{16} = \frac{33}{8} \approx 4.13$

4. **Problem 4**

(a) (10 points) Determine the direction of greatest increase of $f(x, y, z) = x^2y + x\sqrt{1+z}$ at the point $(1, 2, 3)$. **Express your answer as a vector.**

$$\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle$$

$$\boxed{\nabla f(1,2,3) = \left\langle 6, 1, \frac{1}{4} \right\rangle}$$

$$\frac{\partial f}{\partial x} = 2xy + \sqrt{1+z}$$

$$\frac{\partial f}{\partial y} = x^2$$

$$\frac{\partial f}{\partial z} = \frac{x}{2\sqrt{1+z}}$$

$$\nabla f(1,2,3) = \left\langle 2(1)(2) + \sqrt{1+3},\ 1^2,\ \frac{1}{2\sqrt{1+3}} \right\rangle$$

$$= \left\langle 6, 1, \frac{1}{4} \right\rangle$$

(b) (10 points) Find the rate of change of $f(x, y, z) = x^2y + x\sqrt{1+z}$ at $(1, 2, 3)$ in the direction toward $(3, 3, 1)$.

$$\boxed{\frac{25}{6} \approx 4.17}$$

$$D_{\vec{u}} f = \nabla f \cdot \vec{u}$$

From (a) $\nabla f(1,2,3) = \left\langle 6, 1, \frac{1}{4} \right\rangle$

$$\vec{v} = \langle 3, 3, 1 \rangle - \langle 1, 2, 3 \rangle = \langle 2, 1, -2 \rangle$$

$$\|\vec{v}\| = \sqrt{2^2 + 1^2 + (-2)^2} = \sqrt{9} = 3$$

$$\vec{u}_{\vec{v}} = \left\langle \frac{2}{3}, \frac{1}{3}, \frac{-2}{3} \right\rangle$$

$$\left\langle 6, 1, \frac{1}{4} \right\rangle \cdot \left\langle \frac{2}{3}, \frac{1}{3}, -\frac{2}{3} \right\rangle = 6\left(\frac{2}{3}\right) + 1\left(\frac{1}{3}\right) + \frac{1}{4}\left(-\frac{2}{3}\right)$$

$$= 4 + \frac{1}{3} - \frac{1}{6}$$

$$= \frac{24}{6} + \frac{2}{6} - \frac{1}{6} = \frac{25}{6}$$

TAB I WPR 3

Instructor Solution

Total Weight: 100 points

Time Limit: 55 Minutes

First and Last Name: Solution

Cadet ID: ____________

Section and Instructor: ____________

MA205 AY26–1 WPR III (v1)

Number of Extra Sheets: ______

READ THESE INSTRUCTIONS CAREFULLY BEFORE STARTING WORK

1. **SHOW ALL WORK.** Sufficient work is required to indicate the method of reasoning and the operation performed. Clearly indicate your final answer.

2. If you need more space, **use a separate sheet for each problem**. Clearly indicate which problem is continued at the top of the sheet. Print your name and section on every extra sheet used.

3. **You are authorized your issued calculator and a handwritten 8.5 x 11 note-sheet (front and back). You may NOT receive assistance from anyone else.**

4. This examination consists of four graded pages and one cover sheet.

5. Define any variables that you use which are not explicitly defined in the problem. If you make assumptions to simplify a problem, clearly state them.

6. All work written on the WPR will be graded unless marked through or explicitly marked with words to the effect of "do not grade." **WRITE YOUR FINAL ANSWERS IN THE ASSOCIATED BOX, IF PROVIDED.**

7. **Assume angles are in radians unless specified as degrees.**

8. Ensure your answers are written using proper mathematical notation. When appropriate, round all answers to 2 decimal places. Include units when necessary.

Question	Points	Score
1	30	
2	25	
3	25	
4	20	
Total:	100	

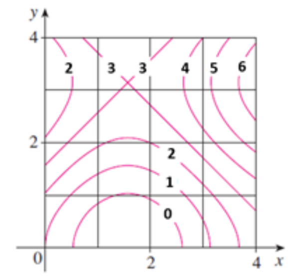
"I did not use any unauthorized sources nor did I receive any assistance while completing this assessment. I will not discuss any aspects of this assessment with anyone except an MA205 instructor until it is released from academic security on 11 November 2025."

Sign and date here to show acknowledgement ____________

1. (30 points) Multiple Choice:

(a) (10 points) **Double Riemann Sum**

A contour map is shown below for a function f on the square $R = [0, 4] \times [0, 4]$. Using the Midpoint Rule with $m = n = 2$, the value of $\iint_R f(x, y)\, dA$ is: **Note: Show your work for full credit.**



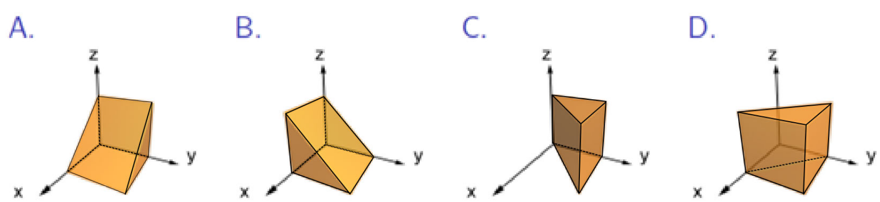
$(.25 + 1.6 + 2.6 + 4.6) \cdot 4 = 36.2$

A. Approximately 48.
B. Approximately 25.
C. Approximately 9.
D. Approximately 35.

$\boxed{D}$

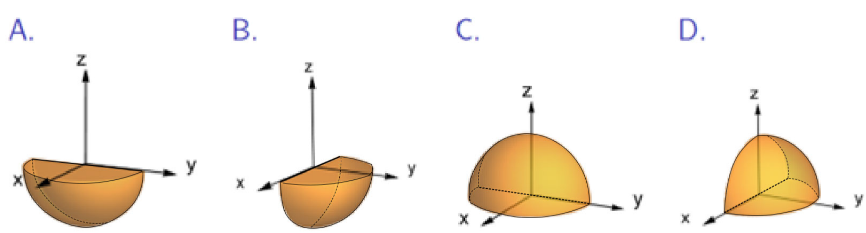
(b) (10 points) Identify the region of integration of the integral

$$\int_0^1 \int_x^1 \int_0^1 f(x, y, z)\; dz\; dy\; dx$$



$\boxed{C}$

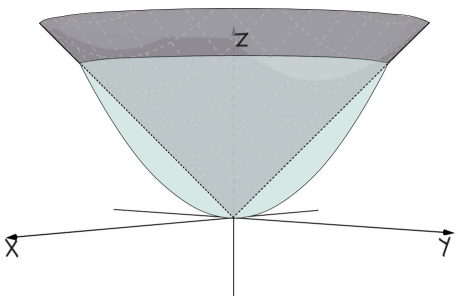
(c) (10 points) Which of the following quarter spheres is described by the inequalities $0 \le \rho \le 1$, $0 \le \theta \le \pi$, and $\frac{\pi}{2} \le \phi \le \pi$ in spherical coordinates?



$\boxed{B}$

2. (25 points) **Triple Integral Conversion**

Find the volume contained above $z = x^2 + y^2$ and below $z = \sqrt{x^2 + y^2}$.



$x^2 + y^2 = \sqrt{x^2+y^2}$

$r^2 = r$

$r^2 - r = 0$

$r(r-1) = 0$

$r = 0, 1$

$$\int_0^{2\pi} \int_0^1 \int_{r^2}^{r} 1 \, r \, dz \, dr \, d\theta$$

$$zr \Big|_{r^2}^{r} = r^2 - r^3$$

$$\frac{1}{3}r^3 - \frac{1}{4}r^4 \Big|_0^1 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$

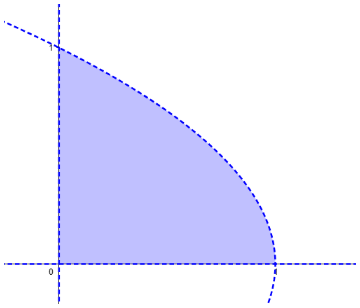
$$\int_0^{2\pi} d\theta = \theta \Big|_0^{2\pi} = 2\pi$$

$$\frac{1}{12} \cdot 2\pi = \frac{\pi}{6}$$

$$\boxed{\frac{\pi}{6}}$$

3. (25 points) **Applications**

(a) (15 points) Find the mass of the lamina that occupies the region D bounded by the parabola $x = 1 - y^2$ and the coordinate axes in the first quadrant with density function $\rho(x, y) = y$.



$$\int_0^1 \int_0^{1-y^2} y \, dx \, dy$$

$$\boxed{m = \frac{1}{4}}$$

$$xy \Big|_0^{1-y^2} = (1-y^2)y - 0$$

$$\int_0^1 y - y^3 \, dy$$

$$\frac{1}{2}y^2 - \frac{1}{4}y^4 \Big|_0^1 = \frac{1}{2} - \frac{1}{4} - 0 = \frac{1}{4}$$

(b) (10 points) The joint density for random variables X and Y is

$$f(x, y) = \begin{cases} \frac{1}{15}(x + y), & \text{if } 0 \le x \le 3,\ 0 \le y \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

Setup the integral to find $P(X + Y \le 2)$. **DO NOT SOLVE!**

$$\boxed{\int_0^2 \int_0^{2-x} \frac{1}{15}(x+y) \, dy \, dx}$$

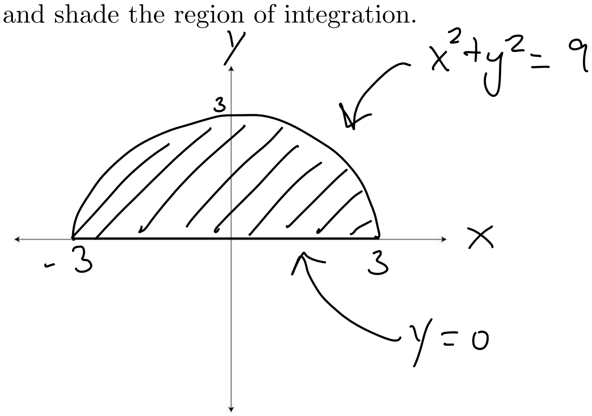
4. (20 points) **Double Integral over a General Region**

Consider the double integral:

$$\iint_R \sqrt{1+x^2+y^2}\, dA$$

where R is the region that lies above the x-axis and within the circle $x^2+y^2=9$.

4a. On the axis provided, sketch the region R in the xy-plane. Clearly label the coordinate axes, each bounding curve, and all intersection points, and shade the region of integration.



4b. Evaluate the double integral over the region R, rounding your final answer to two decimal places. Show all calculus steps, including the correct setup with explicit limits, antiderivatives, and their evaluation at the bounds. Use your calculator only for final arithmetic; answers without full work will not receive full credit.

$$\boxed{\frac{\pi(10^{3/2}-1)}{3} \approx 32.07}$$

$$\int_0^{\pi}\int_0^3 \sqrt{1+r^2}\, r\, dr\, d\theta$$

$$u = 1+r^2$$
$$du = 2r\,dr$$
$$\frac{1}{2}du = r\,dr$$

$$\frac{1}{2}\int_0^3 u^{\frac{1}{2}}\,du$$

$$\frac{1}{2}\left(\frac{2}{3}u^{\frac{3}{2}}\Big|_0^3\right)$$

$$\frac{1}{3}(1+r^2)^{\frac{3}{2}}\Big|_0^3 = \frac{1}{3}\left(10^{\frac{3}{2}}-1\right)\cdot\pi = 32.068$$

$$\int_0^{\pi} d\theta = \theta\Big|_0^{\pi} = \pi$$

TAB J WPR 4

Instructor Solution

First and Last Name: Solution

Total Weight: 100 points **Cadet ID:** ______

Time Limit: 55 Minutes **Section and Instructor:** ______

MA205 AY26–1 WPR IV (v1)

Number of Extra Sheets: ______

READ THESE INSTRUCTIONS CAREFULLY BEFORE STARTING WORK

1. **SHOW ALL WORK.** Sufficient work is required to indicate the method of reasoning and the operation performed. Clearly indicate your final answer.
2. If you need more space, **use a separate sheet for each problem**. Clearly indicate which problem is continued at the top of the sheet. Print your name and section on every extra sheet used.
3. **You are authorized your issued calculator and one handwritten 8.5 x 11 note-sheet (front and back).. You may NOT receive assistance from anyone else.**
4. This examination consists of five graded pages and one cover sheet.
5. Define any variables that you use which are not explicitly defined in the problem. If you make assumptions to simplify a problem, clearly state them.
6. All work written on the WPR will be graded unless marked through or explicitly marked with words to the effect of "do not grade." **WRITE YOUR FINAL ANSWERS IN THE ASSOCIATED BOX, IF PROVIDED.**
7. **Assume angles are in radians unless specified as degrees.**
8. Ensure your answers are written using proper mathematical notation. When reporting answers as decimal values, round to 2 decimal places. Include units when necessary.

Question	Points	Score
1	10	
2	35	
3	25	
4	30	
Total:	100	

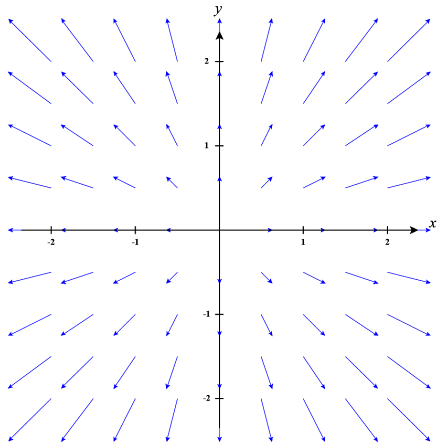
"I did not use any unauthorized sources nor did I receive any assistance while completing this assessment. I will not discuss any aspects of this assessment with anyone except an MA205 instructor until it is released from academic security on 1 December 2025."

Sign and date here to show acknowledgment ______

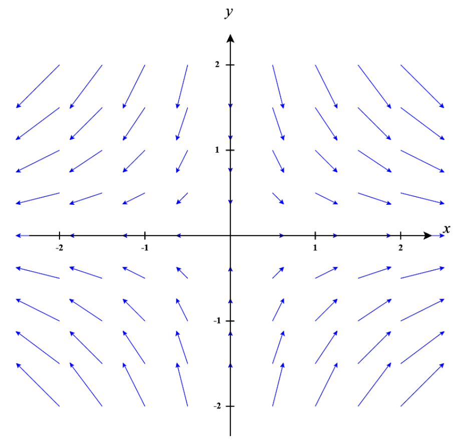
1. (10 points) **Vector Fields**

   Match each plot with one of the listed vector fields below.

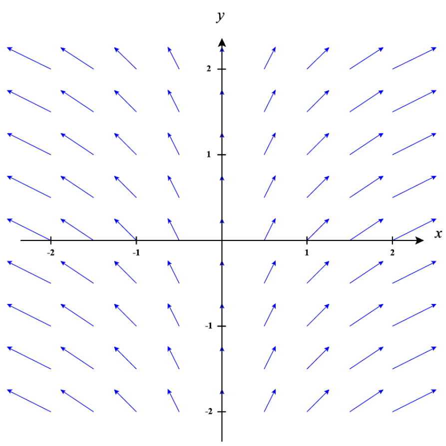
   A. $\mathbf{F} = \langle y, x \rangle$
   B. $\mathbf{F} = \langle x, y \rangle$
   C. $\mathbf{F} = \langle x, -y \rangle$
   D. $\mathbf{F} = \langle x, 1 \rangle$



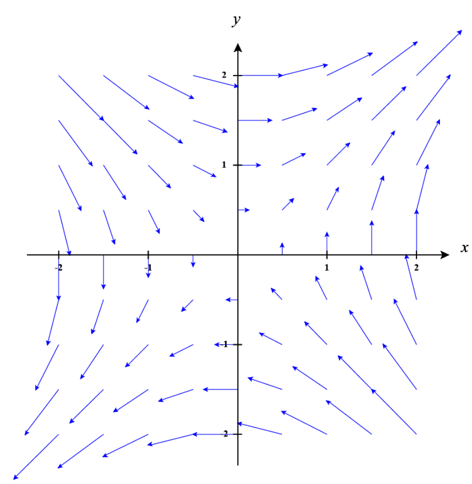
B



C



D



A

2. (35 points) **Line Integrals**

(a)(10 points) Create a parametric curve $\mathbf{r}(t)$ for the curve $y^2 = x - 1$ from point $A(2, 1)$ to $B(10, 3)$.

$x = t$

$y = f(t)$

$\vec{r}(t) =$	$\langle t, \sqrt{t-1} \rangle$, or $\langle t^2+1, t \rangle$
	$2 \le t \le 10$
	$\le t \le$
	$1 \le t \le 3$

(b)(25 points) Calculate the work done by the vector field $\mathbf{F}(x, y) = \langle xy, x^2 - y \rangle$ on a particle moving along the curve C represented by the vector function $\mathbf{r}(t) = \langle t, 2t - 1 \rangle$ where $0 \le t \le 1$.

$$\int_C F \cdot dr = \int_a^b F(r(t)) \cdot r'(t)\, dt$$

$$\int_0^1 \langle (t)(2t-1), (t)^2 - (2t-1) \rangle \cdot \langle 1, 2 \rangle\, dt$$

$$\int_0^1 \langle 2t^2 - t, t^2 - 2t + 1 \rangle \cdot \langle 1, 2 \rangle\, dt$$

$$\int_0^1 2t^2 - t + 2t^2 - 4t + 2\, dt$$

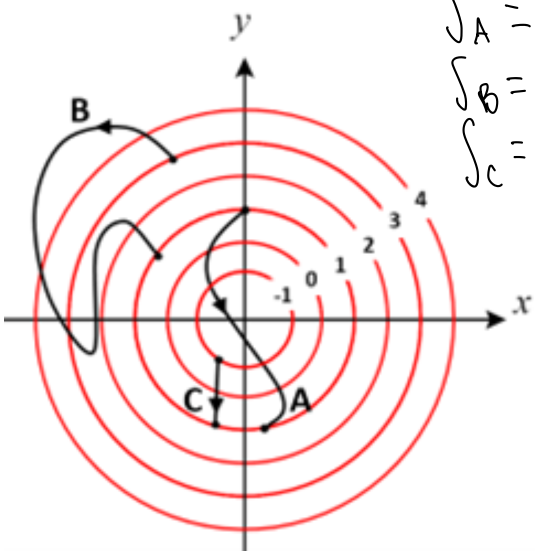
$$\int_0^1 4t^2 - 5t + 2\, dt$$

$$\left[\frac{4}{3}t^3 - \frac{5}{2}t^2 + 2t \right]_0^1$$

$$= \frac{4}{3} - \frac{5}{2} + 2 - 0 = \frac{5}{6}$$

$$\boxed{\frac{5}{6}}$$

3. (25 points) **(a)(10 points)** The contour map of a function $f(x, y)$ is shown below, along with the oriented curves A, B, and C. If $\vec{F} = \nabla f$, which of the following is true? **Write your choice in the box provided.**



$\int_A = 1 - 1 = 0$

$\int_B = 1 - 3 = -2$

$\int_C = 1 - -1 = 2$

I $\int_C \vec{F} \cdot d\vec{r} < \int_A \vec{F} \cdot d\vec{r}$

II $\int_B \vec{F} \cdot d\vec{r} = \int_C \vec{F} \cdot d\vec{r}$

III $\int_A \vec{F} \cdot d\vec{r} = 0$

IV $\int_A \vec{F} \cdot d\vec{r} = \int_B \vec{F} \cdot d\vec{r}$

$\boxed{\text{III}}$

(b)(5 points) Show that the vector field $\vec{F}(x, y, z) = \langle 2xy, x^2 - e^y \rangle$ is conservative.

P Q

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

$2x = 2x$ $\therefore$ Conservative

(c)(10 points) Given that the vector field $\vec{F}$ in part (b) is conservative, find a function f such that $\mathbf{F} = \nabla f$.

$\int 2xy\,dx$ $\quad$ $\int x^2 - e^y\,dy$

xy^2 $\quad$ $xy^2 - e^y$

$$\boxed{f(x,y) = xy^2 - e^y + K}$$

4. (30 points)

(a)(5 points) Which of the following formulas is appropriate to solve the following problem?

Calculate the total work done by the vector field

P Q

$$\tilde{\mathbf{F}}(x,y) = (x^2\sin(x) + xy^2)\hat{\imath} + (2x^3 + e^y)\hat{\jmath}$$

on a particle moving from the origin along a straight line to the point $(2,4)$, then straight left to the point $(0,4)$, and straight down to the origin.

A $\int_C \vec{F}\cdot d\vec{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$

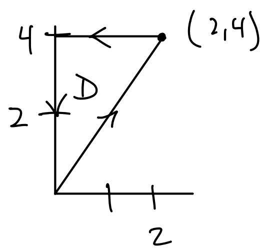
B $\int_C \vec{F}\cdot d\vec{r} = \iint_D \Big(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\Big)dA$

C $\int_C f(x,y)ds = \int_a^b f(x(t),y(t))\sqrt{\Big(\frac{dx}{dt}\Big)^2 + \Big(\frac{dy}{dt}\Big)^2}dt$

D None of the above.

$\boxed{B}$

(b)(25 points) Evaluate the line integral.



$$\iint_D\Big(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\Big)dA$$

$$\int_0^2\int_{2x}^4\big(6x^2 - 2xy\big)\,dy\,dx$$

$$\int_0^2\Big(6x^2y - xy^2\Big|_{2x}^4\Big)dx$$

$$6x^2(4) - x(4)^2 - \big(6x^2(2x) - x(2x)^2\big)$$

$$24x^2 - 16x - 12x^3 - 4x^3$$

$$\int_0^2 -8x^3 + 24x^2 - 16x\,dx$$

$$\Big[-2x^4 + 8x^3 - 8x^2\Big]_0^2 = -2(2)^4 + 8(2)^3 - 8(2)^2$$

$$\boxed{0}$$

TAB K TERM END EXAMINATION

Instructor Solution

First and Last Name: Solution

Total Weight: 300 points

Cadet ID: ____________

Time Limit: 210 Minutes

Section and Instructor: ____________

Number of Extra Sheets: ______

READ THESE INSTRUCTIONS CAREFULLY BEFORE STARTING WORK

1. **SHOW ALL WORK.** Sufficient work is required to indicate the method of reasoning and the operation performed. Clearly indicate your final answer.
2. If you need more space, **use a separate sheet for each problem**. Clearly indicate which problem is continued at the top of the sheet. Print your name and section on every extra sheet used.
3. You are authorized your issued calculator and four hand-written note sheets (front and back). You may NOT receive assistance from anyone else.
4. This examination consists of eleven graded pages and one cover sheet.
5. Define any variables you use which are not explicitly given in the problem. If you make assumptions to simplify a problem, clearly state them.
6. All work written on the WPR will be graded unless marked through or explicitly marked with words to the effect of "do not grade."
7. **Assume angles are in radians unless specified as degrees.**
8. Ensure your answers are written using proper mathematical notation. When appropriate, round all answers to 2 decimal places. **WRITE YOUR FINAL ANSWERS IN THE ASSOCIATED BOX, IF PROVIDED.**
9. **This exam will not be released from academic security. You may not discuss any aspects of this exam with anyone except an MA205 instructor.**

Question	Points	Score
1	30	
2	30	
3	30	
4	30	
5	30	
6	30	

Question	Points	Score
7	30	
8	30	
9	30	
10	30	
Total:	300	

1. Space Curves and Planes

   (a) (20 points) Find a vector equation or a set of parametric equations for the tangent line to the curve $\mathbf{r}(t) = \langle t,\ t^2,\ t^3\rangle$ at time $t = 2$.

$r(2) = \langle 2, 4, 8 \rangle$

$r'(t) = \langle 1, 2t, 3t^2 \rangle$

$r'(2) = \langle 1, 4, 12 \rangle$

$$\boxed{r_2(t) = \langle 2+t, 4+4t, 8+12t \rangle}$$

   (b) (10 points) At what point (x, y, z) does the line $\mathbf{r}(t) = \langle 2+t,\ 4+4t,\ 8+12t\rangle$ intersect the plane $x + 2y + 3z = 169$?

$(2+t) + 2(4+4t) + 3(8+12t) = 169$

$34 + 45t = 169$

$45t = 135$

$t = 3$

$r(3) = \langle 5, 16, 44 \rangle$

$$\boxed{(5, 16, 44)}$$

2. Consider the function $f(x, y) = xy^3 - x^2$.

(a) (10 points) Calculate the **direction of the greatest rate of increase** at the point $(2, -1)$.

$$\langle y^3 - 2x, 3xy^2 \rangle$$

$$\langle (-1)^3 - 2(2), 3(2)(-1)^2 \rangle$$

$$\langle -1-4, 6 \rangle$$

$$\boxed{\langle -5, 6 \rangle}$$

(b) (10 points) Calculate the **magnitude of the greatest rate of increase** at the point $(2, -1)$.

$$\sqrt{25+36} = \sqrt{61}$$

$$\boxed{\sqrt{61}}$$

(c) (10 points) Calculate the rate of change at the point $(2, -1)$ in the direction of the vector $\langle 3, -4 \rangle$.

$$\sqrt{9+16} = 5$$

$$\langle -5, 6 \rangle \cdot \langle \tfrac{3}{5}, -\tfrac{4}{5} \rangle$$

$$-\frac{15}{5} - \frac{24}{5}$$

$$\boxed{-\frac{39}{5}}$$

3. (30 points) Find the area of the surface, S, with vector equation

$$r(u,v) = \langle u, v, 3 + u^2 + v^2 \rangle$$

that lies above the disk $x^2 + y^2 \leq 1$ in the xy-plane.

4. (30 points) Evaluate $\oint_C \overset{P}{y^3}\, dx - \overset{Q}{x^3}\, dy$, where C is the positively-oriented cirlce of radius 2 centered at the origin.

$$\left(\frac{\partial Q}{\partial x} - \frac{dP}{dy}\right)$$

$$\int_0^{2\pi}\int_0^{2} \left(-3x^2 - 3y^2\right) r\, dr\, d\theta$$

$$-3\int_0^{2\pi}\int_0^{2} r^3\, dr\, d\theta$$

$$-3\int_0^{2\pi} d\theta \int_0^{2} r^3\, dr$$

$$-3(2\pi)\left(\frac{1}{4} r^4 \Big|_0^2\right)$$

$$-3(2\pi)(4) = -24\pi$$

$$\boxed{-24\pi}$$

5. (30 points) Compute the work done by the force field $\mathbf{F}$ in moving an object along the curve C.

$$\mathbf{F}(x,y,z) = \langle yz,\ x+z,\ y^2\rangle, \quad C: x = t+1,\ y = 2t,\ z = t+2,\ 0 \le t \le 2$$

$x' = 1 \quad y' = 2 \quad z' = 1$

$\langle (2t)(t+2),\ t+1+t+2,\ 4t^2 \rangle \cdot \langle 1, 2, 1 \rangle$

$\langle 2t^2+4t,\ 2t+3,\ 4t^2 \rangle \cdot \langle 1, 2, 1 \rangle$

$\cancel{2t^2} + \cancel{4t} + \cancel{4t} + 6 + \cancel{4t^2}$

$\int_0^2 6t^2 + 8t + 6 \, dt$

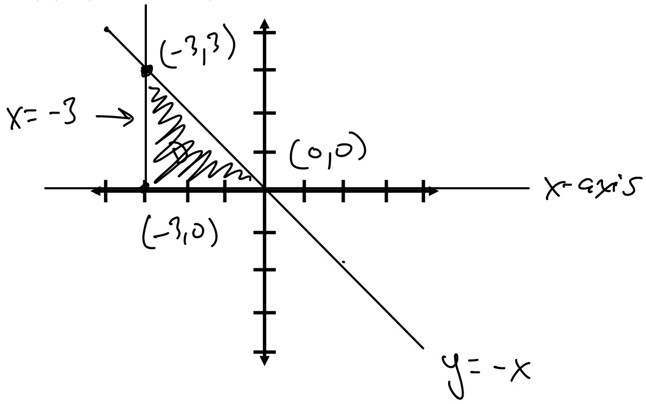
$2t^3 + 4t^2 + 6t \Big|_0^2$

$16 + 16 + 12$

$\boxed{44}$

6. The boundary of a lamina consists of the line $y = -x$, the x-axis, and $x = -3$. The lamina has a density function given by $\rho(x, y) = x^2 + y^2$.

   (a) (15 points) Draw and shade in the region; **label all lines, axes, and points of intersection**.



   (b) (15 points) Find the total mass of the lamina, $\iint_R \rho(x, y)\, dA$, where R is the bounded region above.

$$\int_{-3}^{0}\int_{0}^{-x} x^2+y^2 \, dy\, dx$$

$$x^2y+\frac{1}{3}y^3\Big|_0^{-x}$$

$$x^2(-x)+\frac{1}{3}(-x)^3$$

$$-1\int_{-3}^{0} x^3+\frac{1}{3}x^3 \quad dx$$

$$-1\left(\frac{1}{4}x^4+\frac{1}{12}x^4\Big|_{-3}^{0}\right)$$

$$-1\left(0-\frac{1}{4}(-3)^4-\frac{1}{12}(-3)^4\right)$$

$$-1\left(0-\frac{81}{4}-\frac{27}{4}\right)$$

$$\boxed{27}$$

7. (30 points) Use the method of Lagrange Multipliers to find the maximum and minimum values of

$$f(x, y) = x^2y \text{ subject to the constraint } x^2 + y^2 = 1.$$

$\nabla f = \lambda \nabla g$

$\langle 2xy, x^2 \rangle = \lambda \langle 2x, 2y \rangle$

$2xy = \lambda 2x$ $\quad$ $x^2 = \lambda 2y$ $\quad$ $x^2 + y^2 = 1$

$\lambda = y$ $\quad$ $\lambda = \frac{x^2}{2y}$

$2y^2 + y^2 = 1$

$x^2 = 2y^2$

$3y^2 = 1$

$x = \pm\sqrt{2y^2}$

$y = \pm\sqrt{\frac{1}{3}}$

$x^2 + \left(\sqrt{\frac{1}{3}}\right)^2 = 1$

$\left(\pm\sqrt{\frac{2}{3}}, \pm\sqrt{\frac{1}{3}}\right)$

$x^2 + \frac{1}{3} = 1$

$x^2 = \frac{2}{3}$

$x = \pm\sqrt{\frac{2}{3}}$

$\frac{2}{3\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} = \frac{2\sqrt{3}}{9}$

$$\boxed{\frac{2}{3\sqrt{3}} \text{ Max} \qquad -\frac{2}{3\sqrt{3}} \text{ Min}}$$

8. (30 points) Use a surface integral to calculate the flux of the field, $\mathbf{F}(x, y, z) = x\mathbf{i} + z\mathbf{j} + y\mathbf{k}$ across the surface, S, which is a cone defined parametrically by:

$$r(u, v) = \langle \overset{x}{u}, \overset{y}{u \sin v}, \overset{z}{u \cos v} \rangle$$

$\cancel{r_u = \langle 1, \sin v, \cos v \rangle \quad 1}$

$\cancel{r_v = \langle 0, u \cos v, -u \sin v \rangle \quad 0}$

where $0 \le u \le 3$ and $0 \le v \le \dfrac{3\pi}{2}$.

$$r_u \times r_v = \langle -u \sin^2 v - u \cos^2 v,\ 0 + u \sin v,\ u \cos v - 0 \rangle$$

$$\langle -u, u \sin v, u \cos v \rangle$$

$$\langle u, u \cos v, u \sin v \rangle \cdot \langle -u, u \sin v, u \cos v \rangle$$

$$-u^2 + u^2 \cos v \sin v + u^2 \sin v \cos v \, du$$

$u = \sin v$

$du = \cos v \, dv$

$$\int_0^3 -u^2 + 2u^2 \cos v \sin v \, du$$

$$-\frac{1}{3}u^3 + \frac{2}{3}u^3 \cos v \sin v \Big|_0^3$$

$$\int_0^{\frac{3\pi}{2}} -9 + 18 \cos v \sin v \, dv$$

$$-9v + \frac{18}{2}u^2 \Big|_0^{\frac{3\pi}{2}} \qquad 9 \sin^2\left(\frac{3\pi}{2}\right) - \cancel{9 \sin^2(0)}$$

$$-9\left(\frac{3\pi}{2}\right) + 9 = \boxed{9 - \frac{27\pi}{2}}$$

$\boxed{-33.4}$

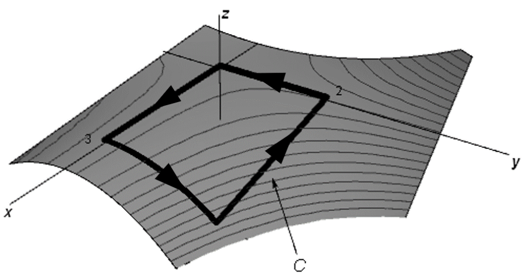
9. (30 points) Consider the vector field

$$\overset{P}{}\quad \overset{Q}{}\quad \overset{R}{}$$

$$\mathbf{F}(x, y, z) = x^2\mathbf{i} + z^2\mathbf{j} + 2y\mathbf{k}$$

and the curve, C, which is the boundary of the part of the surface $z = 1 - xy^2$ over the rectangular region $0 \le x \le 3$, $0 \le y \le 2$, with positive (counter-clockwise) orientation when viewed from above. (See the plot below.)

Use Stokes' Theorem to evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$. You may use page 4 for any additional workspace, but ensure your final answer is written inside the box.



$$-P\frac{\partial g}{\partial x} - Q\frac{\partial g}{\partial y} + R$$

$$\begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & z^2 & 2y \end{vmatrix} \begin{matrix} \frac{\partial}{\partial x} \\ x^2 \end{matrix}$$

$$\langle 2-2z, 0-0, 0-0 \rangle$$

$$\langle 2-2z, 0, 0 \rangle$$

$$-(2-2z)(-y^2) - (0)(-2xy) + 0$$

$$(-2+2z)(-y^2)$$

$$2y^2 - 2zy^2$$

$$2y^2 - 2(1-xy^2)y^2$$

$$2y^2 - 2y^2 + 2xy^4$$

$$\int_0^2 \int_0^3 2xy^4 \, dx\, dy$$

$$y^4 \quad x^2 \Big|_0^3$$

$$9\int_0^2 y^4\, dy$$

$$9\left(\frac{1}{5}y^5\Big|_0^2\right)$$

$$\boxed{\frac{288}{5}}$$

Use if necessary: extra page to show continued work for Question 9. Your answer must be given on the original page.

10. (30 points) Use the Divergence Theorem to calculate the flux of $\mathbf{F}$ across S where

$$\mathbf{F}(x, y, z) = ye^z \ \mathbf{i} + 3x^2y \ \mathbf{j} + z^3 \ \mathbf{k}$$

and where S is the surface of the solid bounded by the cylinder $x^2 + z^2 = 1$ and the planes $y = -3$ and $y = 1$.

$$\nabla \cdot F = 0 + 3x^2 + 3z^2$$

$$3 \int_0^{2\pi} \int_0^1 \int_{-3}^1 r^2 \, r \, dy \, dr \, d\theta$$

$$3 \int_{-3}^1 dy \int_0^1 r^3 \, dr \int_0^{2\pi} d\theta$$

$$3(1+3)\left(\frac{1}{4}\right)(2\pi) = 3(4)\left(\frac{1}{4}\right)(2\pi) = 6\pi$$

$$\boxed{6\pi}$$

SECTION III

ASSIGNMENTS

TAB L COURSE PROJECT

MA205 COURSE PROJECT

Mission Overview

You are a Sapper platoon leader tasked with selecting the optimal route to deliver construction materials to a bridge site under combat conditions. Enemy surveillance is active and time is limited. Your convoy must deliver supplies and secure a forward command post in 5 hours or less.

Three potential routes vary in terrain, exposure to risk, and construction feasibility. Intelligence reports indicate that enemy activity patterns vary with the **time of day**, so each route's risk profile reflects a different planned departure time and the corresponding risk density along that corridor. Your task is to evaluate these routes using multivariate calculus tools to determine which best balances speed, safety, and operational effectiveness.

Each member of the team will analyze one route. Your team must present a final course of action (COA) recommendation supported by models, risk analysis, and ethical considerations.

Tasks Overview

Task 1: Route Modeling & Arc Length

Each route is defined as a 3D vector function $\mathbf{r}(t)$, for $t \in [0, 5]$. Calculate the arc length:

$$L = \int_0^5 |\mathbf{r}'(t)| dt$$

Route Definitions:

- Route Black: $\mathbf{r}_1(t) = \langle t,\ 2\sin(0.5t),\ 0.5t^2 \rangle$
- Route Green: $\mathbf{r}_2(t) = \langle t,\ \ln(t+1),\ 3 - e^{-0.4t} \rangle$
- Route Red: $\mathbf{r}_3(t) = \langle t,\ \cos(0.6t),\ t\sin(0.5t) \rangle$

These routes simulate elevation profiles, changes in directions, and length of the route.

Use symbolic or numerical tools to calculate the arc length. Include:

- Full integral expression and tech-assisted evaluation
- 3D plot of route with labeled axes

Task 2: Risk Modeling via Risk Density Functions

Each route's region has varying risk modeled as a **risk density function** over (x, y), representing risk per square kilometer and capturing how risk changes with both terrain and the assigned departure time of day.

Evaluate:

$$R = \iint_D f(x, y)\, dA$$

Risk Density Functions:

- Black: $f_1(x, y) = \frac{2}{\pi} e^{-(x^2+y^2)},\ x^2 + y^2 \leq 4$
- Green: $f_2(x, y) = \frac{1}{2} e^{-0.5x} \cos^2(y),\ x \in [0, 4],\ y \in [-\pi/2, \pi/2]$
- Red: $f_3(x, y) = \frac{xy}{10} e^{-x-y},\ x, y \in [0, 4]$

Use technology to evaluate and visualize:

- Region of integration sketch
- Heatmap or contour plot of risk
- Integral expression and numeric result

Task 3: Berm Construction Optimization

Each route leads to a command post (CP) area to be secured by a berm. Each CP must be enclosed by a berm (rectangle + semicircle east side). Minimize total time using:

Constraint:

$$A = LW + \frac{1}{2}\pi W^2$$

Objective:

$$T = a(\pi W) + b(2L + W)$$

Route-Specific Data:

- Black (Riverbank): $A = 8200\ \text{m}^2,\ a = 1.5,\ b = 1.1$
- Green (Plateau): $A = 7700\ \text{m}^2,\ a = 0.7,\ b = 0.5$
- Red (Forest Gap): $A = 8000\ \text{m}^2,\ a = 1.2,\ b = 0.4$

Use Lagrange multipliers and include:

- Lagrangian setup and equations
- Tech-assisted solution for L, W that minimizes

$$T = \text{Time}_{\text{East}} + \text{Time}_{\text{Other Sides}}$$

- Diagram of shape and computed time

Task 4: Work in a vector field

Each route has a distinct force field to integrate against representing **Wind Resistance**, **Variable Terrain**, and **Slope Resistance.** Assume the force field models external resistance against movement, and the integral below gives the energy or effort required to move supplies along the route.

$$W = \int_0^5 \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt$$

Vector Fields:

- Black: $\mathbf{F}_1(x, y, z) = \langle -0.5x,\ -y,\ 0 \rangle$
- Green: $\mathbf{F}_2(x, y, z) = \langle 0,\ -0.3y,\ 0.2z \rangle$ push
- Red: $\mathbf{F}_3(x, y, z) = \langle -0.1y,\ 0.2x,\ 0.1z \rangle$

Use technology to evaluate numerically. Be sure to include:

- A setup of your line integral
- The numeric integral value
- Physical interpretation of result

Task 5: Final Recommendation & Ethical Assessment

Present a team recommendation based on:

- Arc length
- Risk exposure
- Berm construction time
- Work/energy required

Justify your COA with:

- Clear synthesis of results
- Ethical and operational reasoning
- Address safety vs. speed tradeoffs

Deliverables & Evaluation

1) Individual Progress Review (IPR) — 20 pts (Due: 26 Nov)

Each cadet submits one completed task, a short interpretation paragraph, and a collaboration plan.

Category	Description	Points
Technical Modeling	Correct symbolic setup and technology computation	10
Interpretation	Operational insight and clarity	5
Professionalism	Collaboration plan and formatting	5
	Total	**20**

2) Final Deliverables — 80 pts (Due: 9 Dec)

Team submission includes a slide deck (10 to 12-minute presentation + 5-minute Q&A), and code notebooks (.nb or .ipynb).

Category	Description	Points
Technical Accuracy	Correct modeling and computations	30
Synthesis & Ethics	Integrated COA and reflection	20
Visuals & Report Quality	Clarity, figures, organization	10
Presentation	Delivery, teamwork, professionalism	20
	Total	**80**

Timeline

Date	Milestone	Description
14 Nov	Project Assigned	Teams formed
26 Nov	**IPR Due (20 pts)**	Individual progress submission
9 Dec	Deliverables Due	Report, slides, and code submission
10–12 Dec	**Final Presentations (80 pts)**	10-12-min team briefing + 5-min Q&A

Note

You are encouraged to use any technology to complete your analysis, including Mathematica, Python, Wolfram Alpha, Desmos, or graphing calculators. This project synthesizes content from all blocks of MA205.

Generative AI Usage Policy

Intent: This project is the capstone experience for MA205 and is designed to assess your ability to apply multivariable calculus techniques, use technology responsibly, and communicate results clearly. Maintaining academic integrity prepares you for future coursework and professional expectations. To support your work on this project, you may use Generative AI tools (such as ChatGPT or similar systems) **only in limited ways**. These tools may assist you with coding, troubleshooting syntax, or **understanding** mathematical concepts.

Permitted Uses

You may use Generative AI to:

- Assist with writing or debugging Mathematica or Python code.
- Ask conceptual questions to better understand procedures or modeling steps.
- Help with formatting or error correction.

These uses are acceptable only when they support—but do not replace—your own mathematical work.

Prohibited Uses

You may **not** use Generative AI to:

- Produce solutions, computations, or written explanations for any part of the project.
- Generate code, plots, or symbolic expressions that you do not fully understand.
- Write interpretation paragraphs, analysis, or the final Course of Action (COA) recommendation.
- Create your IPR or final presentation content.

Any use of AI that replaces your own modeling or reasoning is strictly prohibited. If you are unsure if your use is prohibited, please see your instructor before submitting your work.

Consequences for Misuse

If Generative AI is used to **perform the work of the project itself**, you will receive **no credit** for the assignment—including the IPR and the final deliverables—even if the assistance is acknowledged.

Documentation Requirement

If you use Generative AI within the permitted boundaries, you must document assistance in accordance with the DAAW.

Outstanding Student Work

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Route and Risk Analysis

By: CDT Kellen Daly, CDT Lucas Silva, CDT Luke Johnson

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Agenda

- Task 1 Data
- Task 2 Data
- Task 3 Data
- Task 4 Data
- Final recommendation
- Ethical Considerations
- Assumptions
- Works Cited

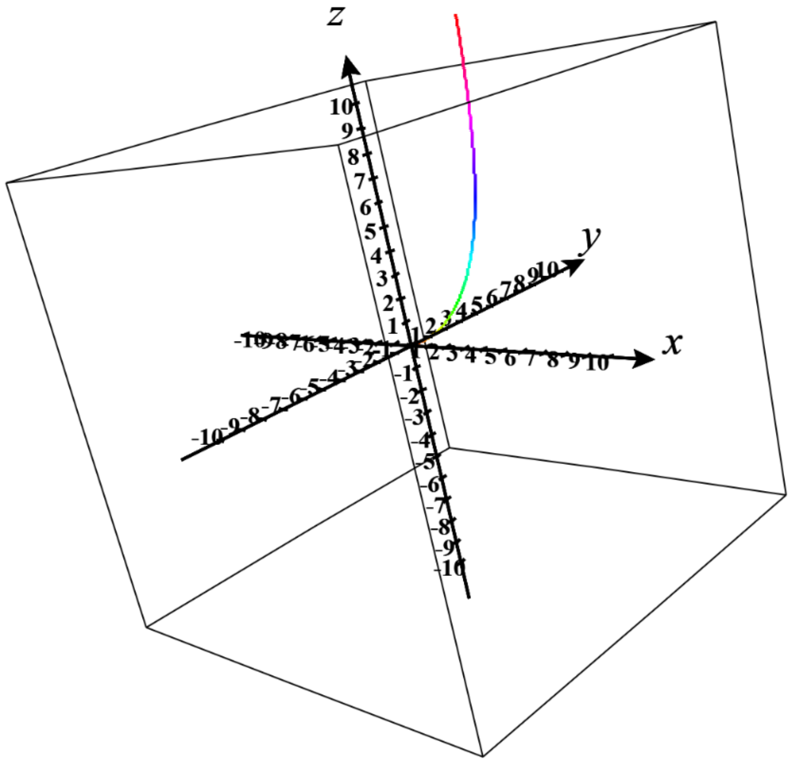
Task 1: Route Length and Difficulty

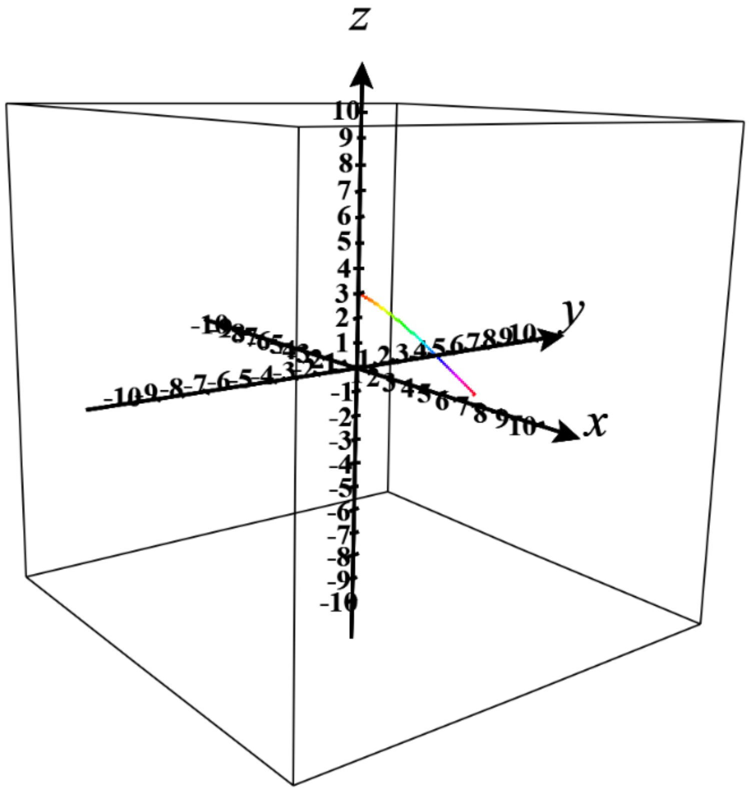
- Each route is defined as a 3D vector function r(t), for t ∈ [0, 5]. Calculate the arc length:
- $L = \int_0^5 |r'(t)|\, dt$

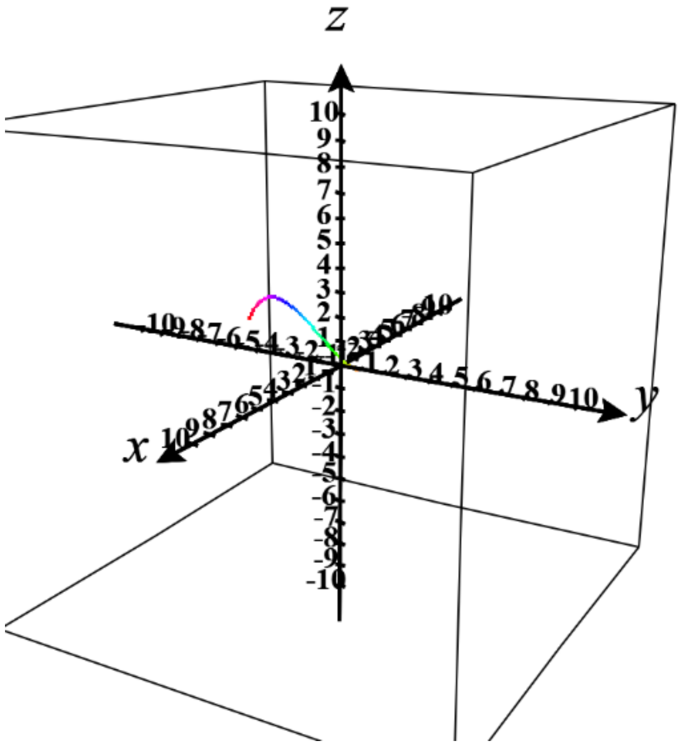
Route Definitions:

- **Route Black:** r(t) = ⟨t, 2sin(0.5t), $0.5t^2$⟩
- $L = \int_0^5 \sqrt{1 + \left(cos(0.5t)\right)\text{^}2 + t^2}\, dt$
- = 14.4752
- **Route Green**: r(t) = ⟨t, ln(t + 1), 3 – $e^{-0.4t}$⟩
- $L = \int_0^5 \sqrt{1 + (\frac{1}{t+1})^2 + (0.4e^{-0.4t})\text{^}2}\, dt$
- = 5.46807
- **Route Red:** r(t) = ⟨t, cos(0.6t), t sin(0.5t)⟩
- $L = \int_0^5 \sqrt{1 + ((-0.6\sin(0.6t))^2 + (0.5tcos(0.5t) + \sin(0.5t))^2}\, dt$
- = 7.13491

Task 1: Route Length and Difficulty (MODELS)







ROUTE: BLACK

ROUTE: GREEN

ROUTE: RED

Task 2: Route Risk

- Each route is modeled as a risk density function representing the risk per square kilometer
- $R = \iint f(x,y)dA$

Route Definitions:

- **Route Black:** f(x,y) = $\frac{2}{\pi}e^{-(x^2+y^2)}, D = \{(x,y)|x^2 + y^2 \leq 4\}$
- $R = \int_0^{2\pi}\int_0^2 \frac{2}{\pi}e^{-(r^2)}r\ dr\ d\theta$
- = 1.9634
- **Route Green**: f(x,y) = $\frac{1}{2}e^{-0.5x}\cos^2(y)\,, D = \left\{(x,y)|x \in\ [0,4], y \in [-\frac{\pi}{2}, \frac{\pi}{2}]\right\}$
- $R = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\int_0^4 \frac{1}{2}e^{-0.5x}\cos^2(y)\ dx\ dy$
- = 1.3582
- **Route Red:** f(x,y) = $\frac{xy}{10}e^{-x-y}, D = \{(x,y)|x,y \in [0,4]\}$
- $R = \int_0^4\int_0^4 \frac{xy}{10}e^{-x-y}\ dx\ dy$
- = 0.082523

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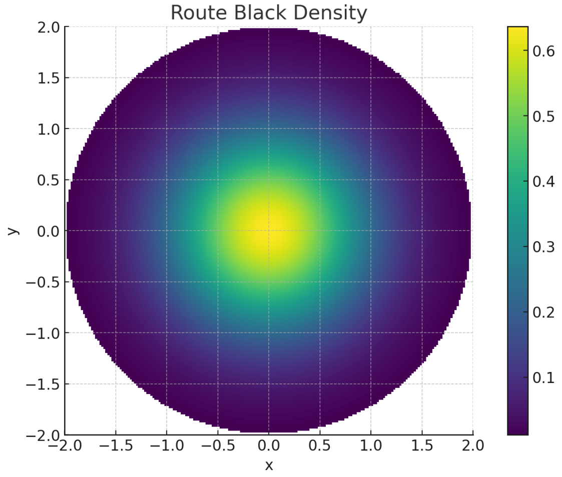
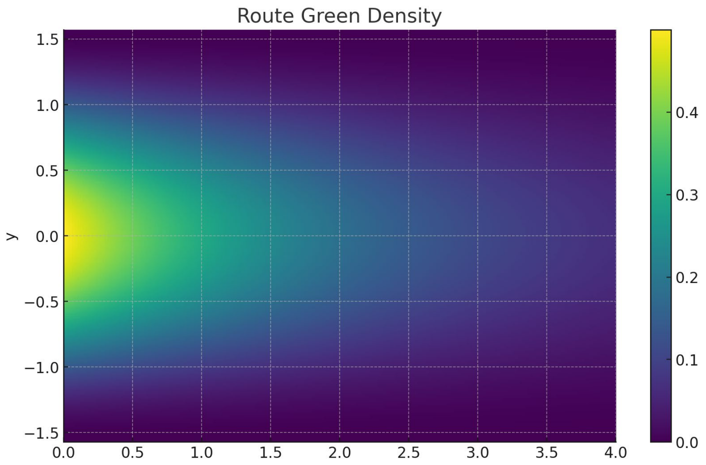
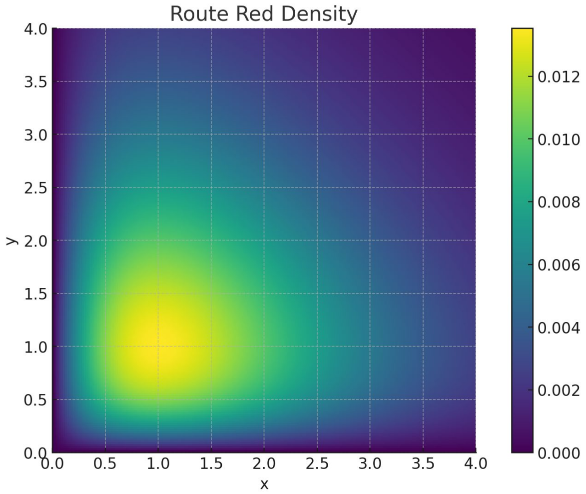
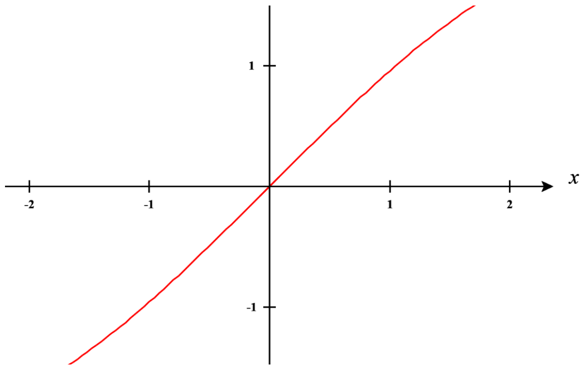
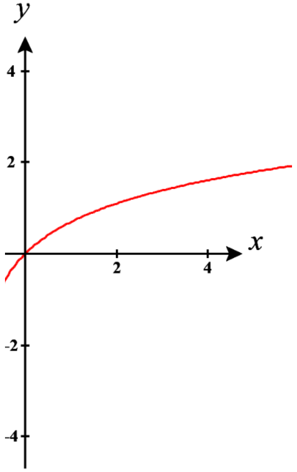
Task 2: Route Risk Heatmap

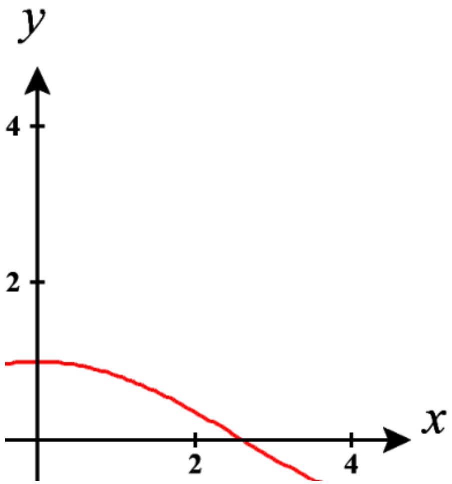












Task 3: Optimizing Berm Construction

Each Command Post is to be secured by a berm while minimizing time and meeting the required area.

Objective Function: $T = a(\pi W) + b(2L + W)$

Area Constraint: $A = LW + \frac{1}{8}\pi W^2$

Lagrange Set Up $\nabla T = \lambda \nabla g$

$$\frac{dT}{dL} = \lambda \frac{dg}{dL}$$

$$2b = \lambda W$$

$$\frac{dT}{dW} = \lambda \frac{dg}{dW}$$

$$a\pi + b = \lambda(L + \frac{\pi W}{4})$$

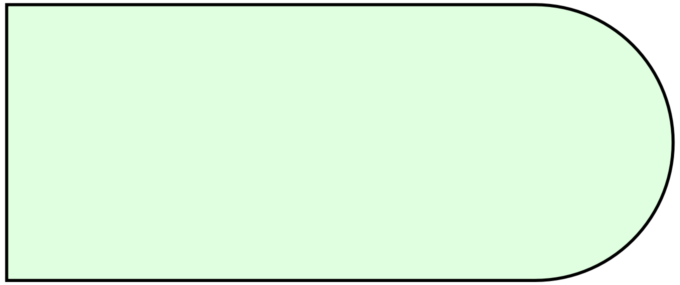
Task 3: Berm Optimization Times and Shapes

Route Black (Riverbank) $A = 8200\ m^2, a = 1.5, b = 1.1$

Length = 112.1, Width = 60.4, Time = 597.6 minutes

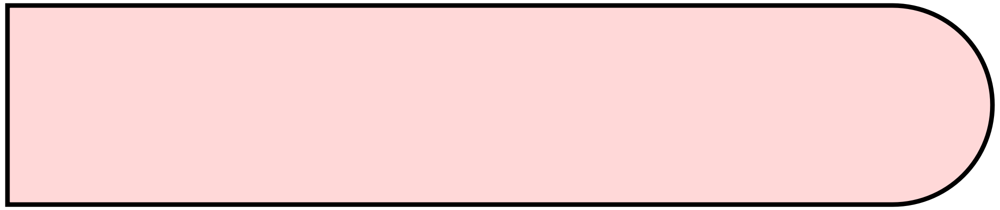
Route Green (Plateau) $A = 7700\ m^2, a = .7, b = 0.5$

Length = 110.6, Width = 57.8, Time = 266.5 minutes



Route Red (Forest Gap) $A = 8000\ m^2, a = 1.2, b = 0.4$

Length = 180.4, Width = 40.7, Time = 314.2 minutes



Task 4 : Work in a Vector Field

- Each route passes through a distinct vector field representing Wind Resistance, Variable Terrain, and Slope Resistance.
- $W = \int_0^5 F(r(t)) \cdot r'(t)dt$
- F is the function of the vector field each route passes through, and r(t) is the path itself given from task 1

Route Definitions:

- **Route Black:** F(x,y,z) = ⟨-0.5x, -y, 0⟩
- $W = \int_0^5 \langle -0.5t, -2\sin(0.5t), 0\rangle \cdot \langle 1, \cos(0.5t), t\rangle dt$
- = -6.966 J
- **Route Green**: F(x,y,z) = ⟨0, -0.3y, 0.2z⟩
- $W = \int_0^5 \langle 0, -0.3\ln(t+1), 0.2(3-e^{-0.4t})\rangle \cdot \langle 1, \frac{1}{t+1}, 0.4e^{(-0.4t)^2}\rangle dt$
- = 0.789 J
- **Route Red:** F(x,y,z) = ⟨-0.1y, 0.2x, 0.1z⟩
- $L = \int_0^5 \langle -0.1\cos(0.6t), 0.2t, 0.1t\sin(0.5)\rangle \cdot \langle 1, -0.6\sin(0.6t), 0.5t\cos(0.5t) + \sin(0.5t)\rangle dt$
- = 1.842 J

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Real World meaning

- The result of the integral is the amount of work that is done to move supplies along the route. A large positive value means it takes a lot of energy to overcome the terrain while a large negative value means the terrain helps with the movement of supplies.

Final Recommendation

	Route Black	Route Green	Route Red
Task 1	14.4752 km	5.46807 km	7.13491 km
Task 2	1.9634 overall risk	1.3582 overall risk	0.082523 overall risk
Task 3	597.6 minutes	266.5 minutes	314.2 minutes
Task 4	-6.96634 J	0.789 J	1.842 J

Our course of action is to take Route Green to deliver the construction materials to the bridge site.

Key Points:

- Lowest overall distance and time to construct the berm
- Not the largest risk
- In our needed time range of under 300 minutes
- Not a lot of extraneous energy being used along the route

Ethical Considerations

- While Route Green does have a chance of enemy interference, it completes the task
- Shorter walk means the soldier can be more alert the whole time
- There is not a lot of outside forces making the trip harder or making the soldiers more tired

Ethical Considerations

- We are going to do jobs that are dangerous and have risk
- The forward command post we are supplying could desperately need the supplies

Assumptions

- We assume that the value we got for Task 1 is in km
- We assume that the risk value in Task 2 is how many times we are likely to see an enemy
- We assume the value we receive in Task 3 is in minutes

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Works Cited and CODE

- Arctic Warrior. Sapper. Flickr. Airborne Sappers “Clear The Line” | Combat engineers assigne… | Flickr. Accessed 12/5/2025
- Wolfram Alpha. Wolfram|Alpha: Computational Intelligence. Accessed 11/21/2025. We utilized this for calculating the values of the integrals. This included Tasks 1, 2, and 4.
- CalcPlot3D. CalcPlot3D. Accessed 11/21/2025. We utilized this for providing 3-D models of the routes.
- Assitance given to author, generative AI ChatGPT. https://chatgpt.com/share/6935d879-4dc8-800f-a3f5-5d2852cbfb93 and https://chatgpt.com/share/6935d887-4e94-800f-a334-0d6dea59d47a. We utilized this as well as Wolfram Apha to solve the double integrals in task 2. We also used this to create the heatmaps for task 2. This helped us immensely because it helped us to generate difficult heat maps that we couldn’t have otherwise. West Point, NY. 061700DEC2025.
- Assistance given to author. Generative AI. ChatGPT. https://chatgpt.com/share/69379238-5f14-8000-80fc-d6f3fbb73ca8. I used chat GPT not to code, but to help me code. I used my talk with MAJ Fisher in class to help me know what I need to do on paper then I asked chat gpt what phrases I need for the code. These were the questions I asked Chat. “dont give me a ton of code just give me the code words for lagrange in mathematica, Now that I have the lagrange help me set up a system of equations for the derivative with respect to L W and the g(L,W), why is my lag blue, and I pasted my code and asked why wont it run.” I used these questions and my work of taking derivatives with the lagrange to solve the problem. Chat GPT helped a lot, but I used it as a tool not a crutch. West Point, NY 082210DEC2025.
- Assistance Given to author. Generative AI. ChatGPT. https://chatgpt.com/share/69379999-4e3c-8000-a527-3bc69ba8a2bc. I had no clue how to generate a picture of my berm shapes so I asked chat gpt how to do it in mathematica as well as many other questions when the code didn’t run. This helped me so much and I wouldn’t have been able to do it without chat gpt. West Point, NY 082210DEC2025.
- Wolfram Alpha. https://www.wolframcloud.com/obj/ec07173a-efba-4a31-8e4c-1f1f5978b8e5. Accessed 12/8/2025. We utilized this for calculating the values of the integrals. This included Tasks 3.

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